

MODULE 8:- Jenkins

Continious Integration Using Jenkins

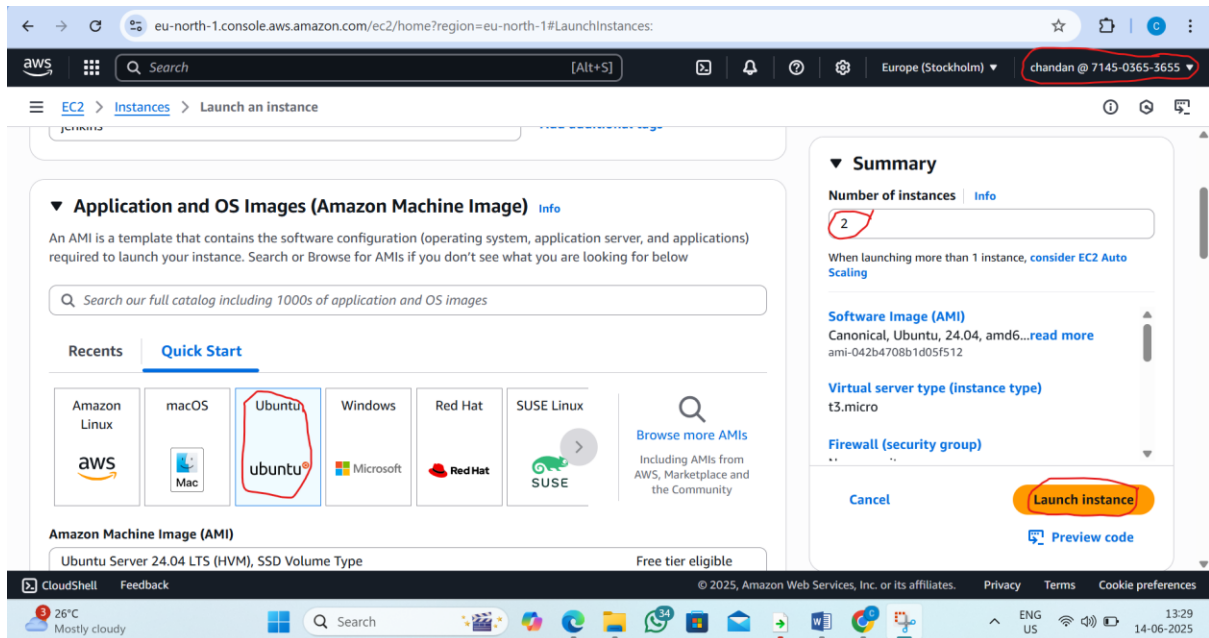
SUBMITTED BY :- CHANDAN YADAV KS

SUBMITTED TO :- VIKUL

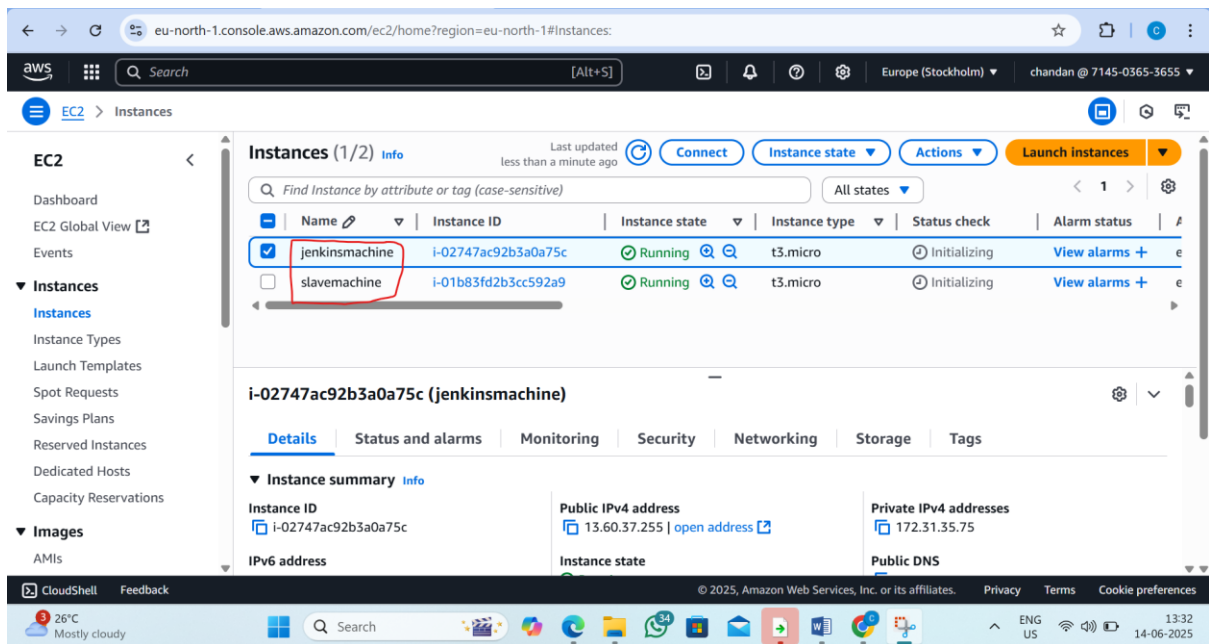
DATE OF SUBMISSION :- 27-06-2025

1. L1 - Create Jenkins Master-Slave Configurations and Create a workspace in Jenkins Slave Node using Jenkins Free-style project

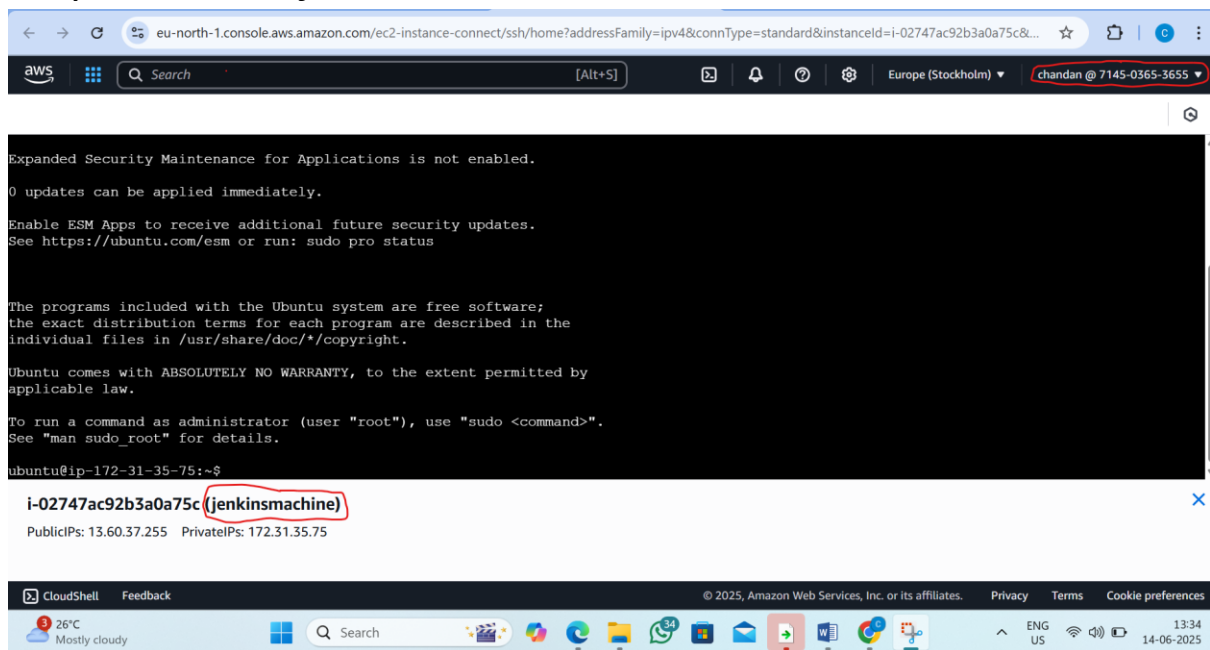
Step1: Launch 2 ubuntu machines in aws



Step2: now name 1 machine as **jenkinsmachine** and another as **Slave machine**



Step3:connect jenkinsmachine



Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-35-75:~\$

i-02747ac92b3a0a75c (jenkinsmachine)

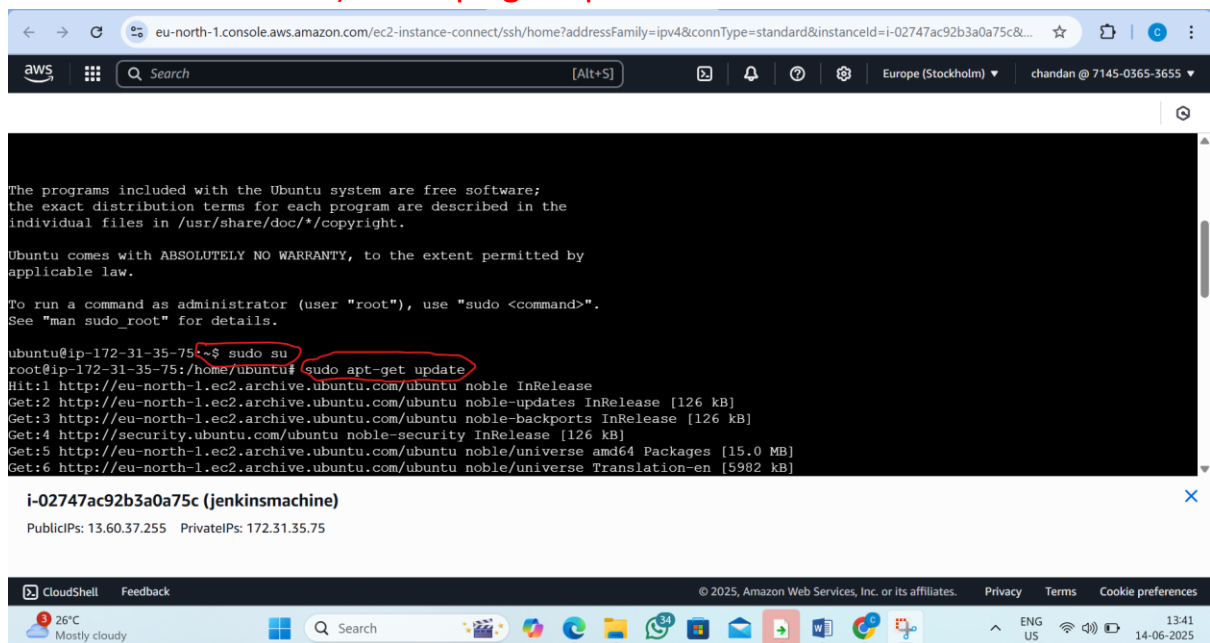
PublicIPs: 13.60.37.255 PrivateIPs: 172.31.35.75

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26°C Mostly cloudy Search 13:34 14-06-2025

Step4:now go to the root user and update all packages

By the commands: 1) `sudo su`
2) `sudo apt-get update`



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-35-75:~\$ **sudo su**
root@ip-172-31-35-75:/home/ubuntu# **sudo apt-get update**
Hit:1 <http://eu-north-1.ec2.archive.ubuntu.com/ubuntu> noble InRelease
Get:2 <http://eu-north-1.ec2.archive.ubuntu.com/ubuntu> noble-updates InRelease [126 kB]
Get:3 <http://eu-north-1.ec2.archive.ubuntu.com/ubuntu> noble-backports InRelease [126 kB]
Get:4 <http://security.ubuntu.com/ubuntu> noble-security InRelease [126 kB]
Get:5 <http://eu-north-1.ec2.archive.ubuntu.com/ubuntu> noble/universe amd64 Packages [15.0 MB]
Get:6 <http://eu-north-1.ec2.archive.ubuntu.com/ubuntu> noble/universe Translation-en [5982 kB]

i-02747ac92b3a0a75c (jenkinsmachine)

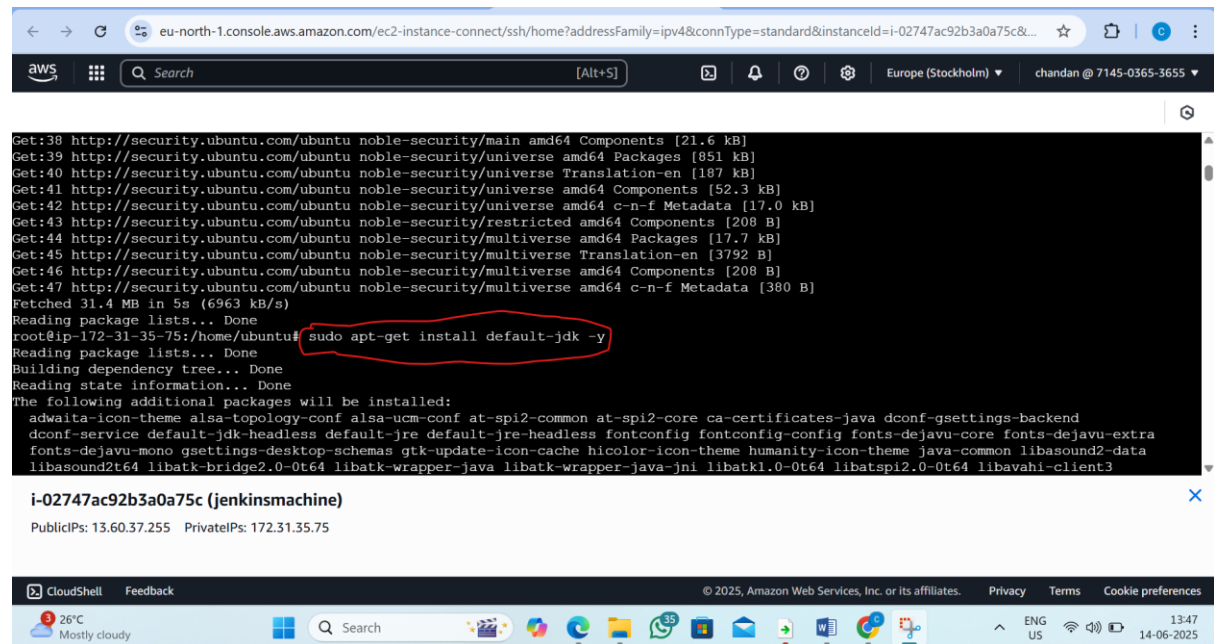
PublicIPs: 13.60.37.255 PrivateIPs: 172.31.35.75

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Step5 :install the java 11

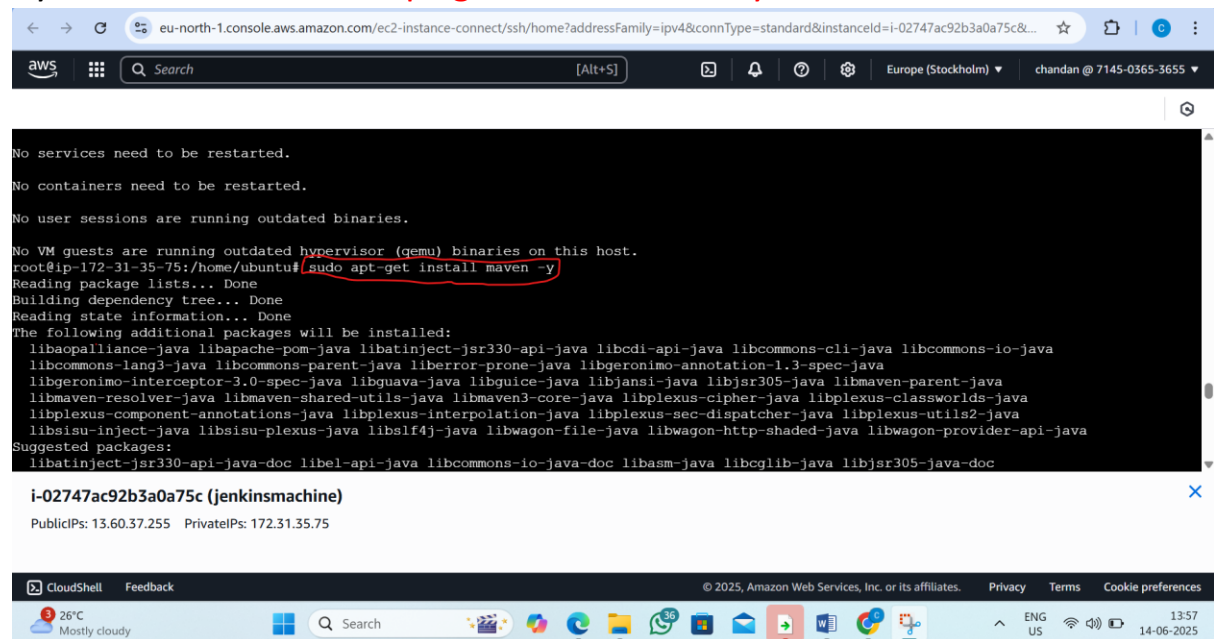
By the command : 1)sudo apt-get install default-jdk -y



```
Get:38 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:39 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [851 kB]
Get:40 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [187 kB]
Get:41 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:42 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [17.0 kB]
Get:43 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:44 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [17.7 kB]
Get:45 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [3792 B]
Get:46 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:47 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [380 B]
Fetched 31.4 MB in 5s (6963 kB/s)
Reading package lists... Done
root@ip-172-31-35-75:/home/ubuntu# sudo apt-get install default-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gsettings-backend
  dconf-service default-jdk-headless default-jre default-jre-headless fontconfig fontconfig-config fonts-dejavu-core fonts-dejavu-extra
  fonts-dejavu-mono gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme java-common libasound2-data
  libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64 libatspi2.0-0t64 libavahi-client3
  libpam-systemd
Suggested packages:
  libasound2-plugins alsa-utils cups-common default-jdk-headless | default-jdk | default-jre | default-jre-headless | fonts-dejavu-extra | fonts-dejavu-mono | fonts-dejavu-core | fonts-dejavu-extra | fonts-dejavu-mono | gsettings-desktop-schemas | gtk-update-icon-cache | hicolor-icon-theme | humanity-icon-theme | java-common | libasound2-data | libasound2t64 | libatk-bridge2.0-0t64 | libatk-wrapper-java | libatk-wrapper-java-jni | libatk1.0-0t64 | libatspi2.0-0t64 | libavahi-client3 | libpam-systemd
```

Step6 :install maven

By the command:sudo apt-get install maven -y

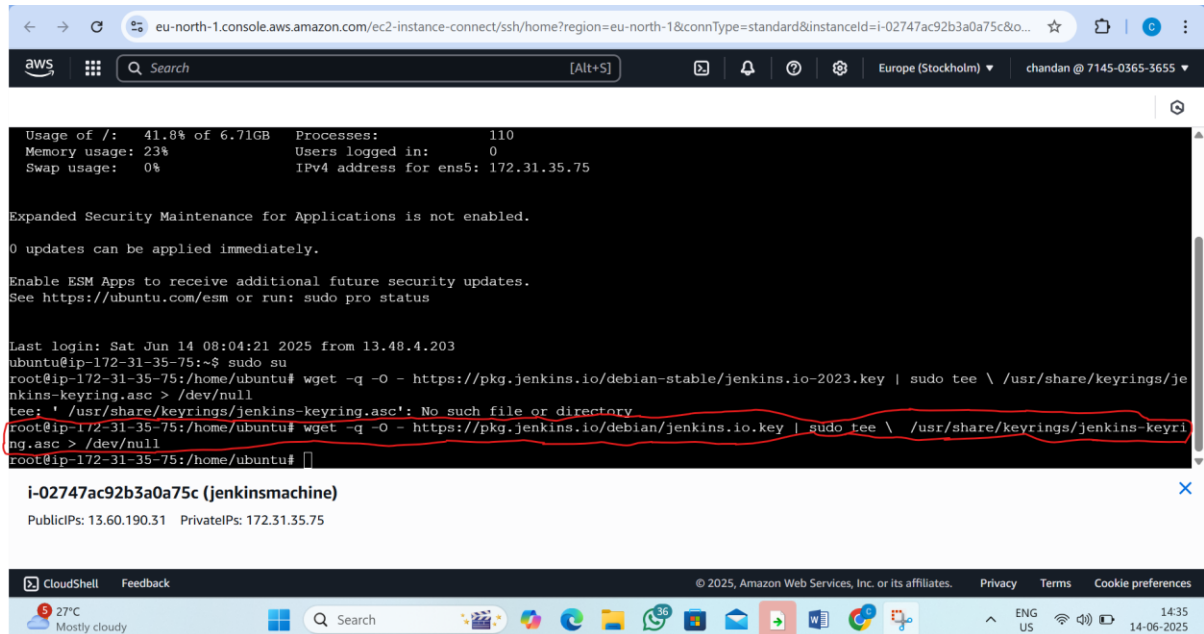


```
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-35-75:/home/ubuntu# sudo apt-get install maven -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libaopalliance-java libapache-pom-java libatinject-jer330-api-java libcdi-api-java libcommons-cli-java libcommons-io-java
  libcommons-lang3-java libcommons-parent-java liberror-prone-java libgeronimo-annotation-1.3-spec-java
  libgeronimo-interceptor-3.0-spec-java libguava-java libguice-java libjansi-java libjer305-java libmaven-parent-java
  libmaven-resolver-java libmaven-shared-utils-java libmaven3-core-java libplexus-cipher-java libplexus-classworlds-java
  libplexus-component-annotations-java libplexus-interpolation-java libplexus-sec-dispatcher-java libplexus-utils2-java
  libsisu-inject-java libsisu-plexus-java libslf4j-java libwagon-file-java libwagon-http-shaded-java libwagon-provider-api-java
Suggested packages:
  libatinject-jer330-api-java-doc libel-api-java libcommons-io-java-doc libasm-java libcglib-java libjer305-java-doc
```


Step7:now install the Jenkins

1)add jenkins key to the system

By the command: `wget -q -O - https://pkg.jenkins.io/debian/jenkins.io.key | sudo tee \`
`/usr/share/keyrings/jenkins-keyring.asc > /dev/null`



The screenshot shows the AWS CloudShell interface with a terminal window. The terminal output includes system statistics, security notices, and the execution of the command to add the Jenkins key. The command is highlighted with a red box. The terminal shows the key being downloaded and added to the keyring successfully.

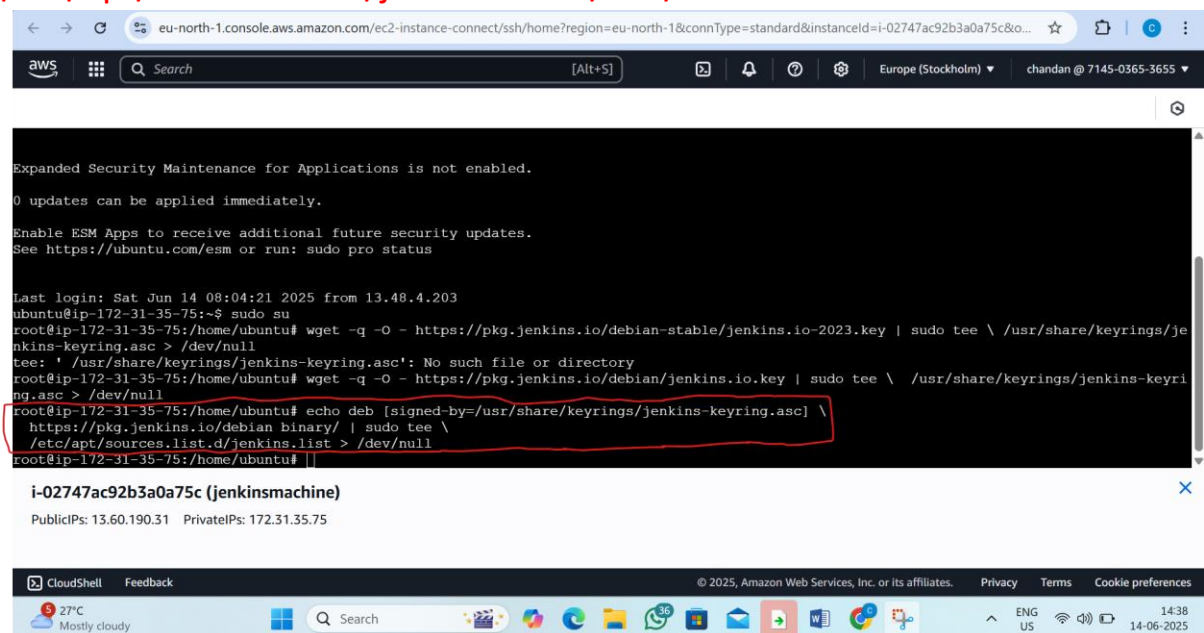
```
Usage of /: 41.8% of 6.71GB Processes: 110
Memory usage: 23% Users logged in: 0
Swap usage: 0% IPv4 address for ens5: 172.31.35.75

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Jun 14 08:04:21 2025 from 13.48.4.203
ubuntu@ip-172-31-35-75:~$ sudo su
root@ip-172-31-35-75:/home/ubuntu# wget -q -O - https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee \
/usr/share/keyrings/jenkins-keyring.asc > /dev/null
tee: '/usr/share/keyrings/jenkins-keyring.asc': No such file or directory
root@ip-172-31-35-75:/home/ubuntu# wget -q -O - https://pkg.jenkins.io/debian/jenkins.io.key | sudo tee \
/usr/share/keyrings/jenkins-keyring.asc > /dev/null
root@ip-172-31-35-75:/home/ubuntu#
```

2) Add the Jenkins repository

By the command: `echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \`
`https://pkg.jenkins.io/debian binary/ | sudo tee \`
`/etc/apt/sources.list.d/jenkins.list > /dev/null`



The screenshot shows the AWS CloudShell interface with a terminal window. The terminal output includes the same system statistics and security notices as the previous screenshot. The command to add the Jenkins repository is highlighted with a red box. The terminal shows the repository being added to the sources list successfully.

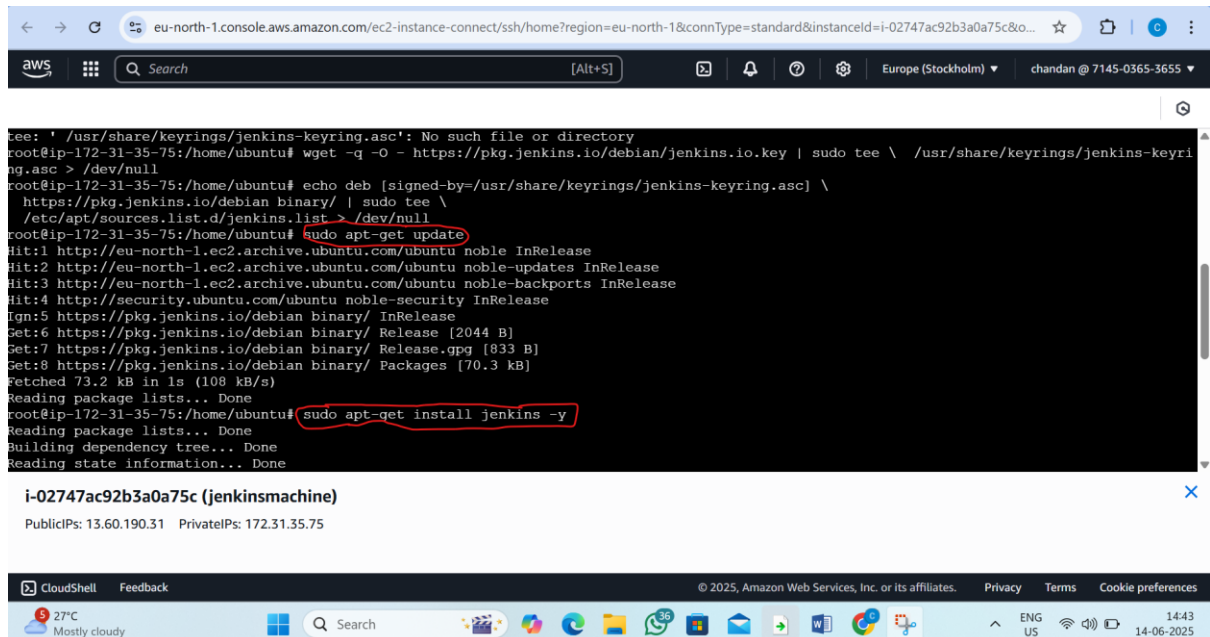
```
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Jun 14 08:04:21 2025 from 13.48.4.203
ubuntu@ip-172-31-35-75:~$ sudo su
root@ip-172-31-35-75:/home/ubuntu# wget -q -O - https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee \
/usr/share/keyrings/jenkins-keyring.asc > /dev/null
tee: '/usr/share/keyrings/jenkins-keyring.asc': No such file or directory
root@ip-172-31-35-75:/home/ubuntu# wget -q -O - https://pkg.jenkins.io/debian/jenkins.io.key | sudo tee \
/usr/share/keyrings/jenkins-keyring.asc > /dev/null
root@ip-172-31-35-75:/home/ubuntu# echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
root@ip-172-31-35-75:/home/ubuntu#
```

3)now update the pakages and Install Jenkins

By the command:1) **sudo apt-get update**

2)**sudo apt-get install Jenkins -y**



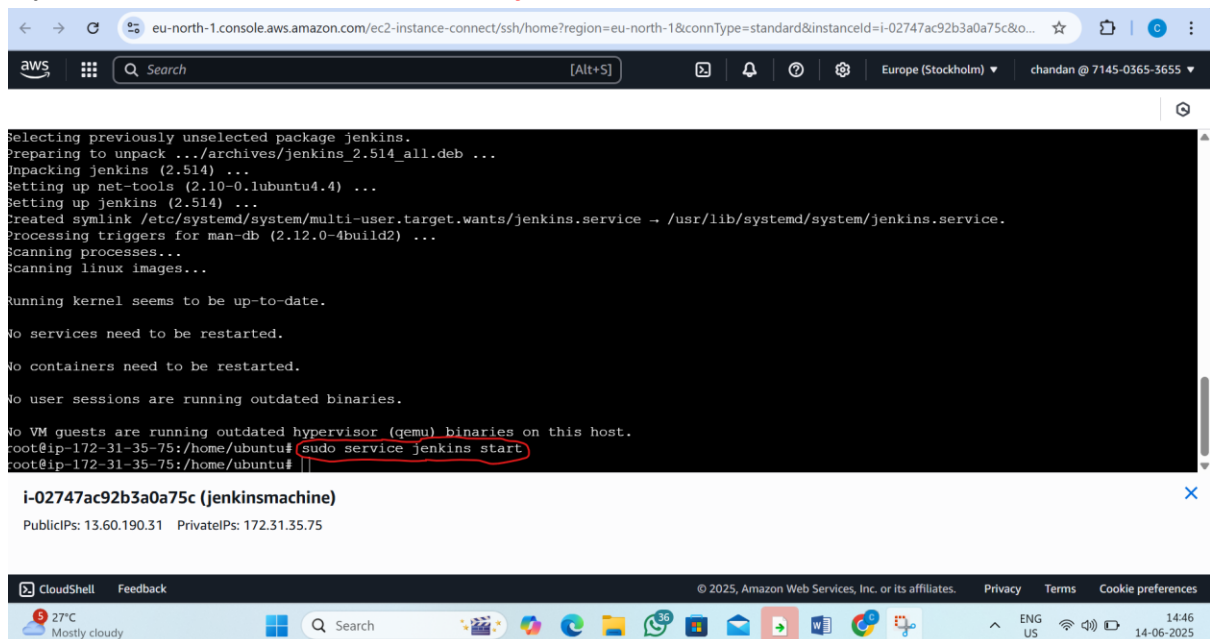
The screenshot shows the AWS CloudShell interface. The terminal output displays the execution of two commands: `sudo apt-get update` and `sudo apt-get install jenkins -y`. The first command updates the package lists, and the second command installs Jenkins. The output for the second command shows the download of Jenkins 2.514 and its dependencies. Below the terminal output, the instance details for 'i-02747ac92b3a0a75c (jenkinsmachine)' are shown, including public and private IP addresses. The CloudShell toolbar at the bottom includes a search bar, a taskbar with various application icons, and system status information like temperature and time.

```
root@ip-172-31-35-75:/home/ubuntu# wget -q -O - https://pkg.jenkins.io/debian/jenkins.io.key | sudo tee \ /usr/share/keyrings/jenkins-keyr
root@ip-172-31-35-75:/home/ubuntu# echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
root@ip-172-31-35-75:/home/ubuntu# sudo apt-get update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:5 https://pkg.jenkins.io/debian binary/ InRelease
Get:6 https://pkg.jenkins.io/debian binary/ Release [2044 B]
Get:7 https://pkg.jenkins.io/debian binary/ Release.gpg [833 B]
Get:8 https://pkg.jenkins.io/debian binary/ Packages [70.3 kB]
Fetched 73.2 kB in 1s (108 kB/s)
Reading package lists... Done
root@ip-172-31-35-75:/home/ubuntu# sudo apt-get install jenkins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

i-02747ac92b3a0a75c (jenkinsmachine)
PublicIPs: 13.60.190.31 PrivateIPs: 172.31.35.75

4)now start the Jenkins

by the command:-**sudo service jenkins start**



The screenshot shows the AWS CloudShell interface. The terminal output displays the execution of the command `sudo service jenkins start`. The output shows the service starting successfully, including downloading the Jenkins 2.514 package and setting up the service. Below the terminal output, the instance details for 'i-02747ac92b3a0a75c (jenkinsmachine)' are shown, including public and private IP addresses. The CloudShell toolbar at the bottom includes a search bar, a taskbar with various application icons, and system status information like temperature and time.

```
selecting previously unselected package jenkins.
Preparing to unpack .../archives/jenkins_2.514_all.deb ...
Unpacking jenkins (2.514) ...
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Setting up jenkins (2.514) ...
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service -> /usr/lib/systemd/system/jenkins.service.
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-35-75:/home/ubuntu# sudo service jenkins start
root@ip-172-31-35-75:/home/ubuntu#
```

i-02747ac92b3a0a75c (jenkinsmachine)
PublicIPs: 13.60.190.31 PrivateIPs: 172.31.35.75

Step 8: now got to the instance **security group** and **edit the inbound rule** and add enable **8080 port**

The screenshot shows the AWS Management Console interface for editing inbound rules on a security group. The breadcrumb navigation indicates the path: EC2 > Security Groups > sg-0f5eb9bdd06a0e8d4 - launch-wizard-3 > Edit inbound rules. The main heading is 'Edit inbound rules' with a sub-note: 'Inbound rules control the incoming traffic that's allowed to reach the instance.'

The 'Inbound rules' table is as follows:

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-038303e6a7753ccd4	SSH	TCP	22	Cu...	
-	Custom TCP	TCP	8080	An...	

Below the table, there is an 'Add rule' button. A yellow warning banner states: 'Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.'

At the bottom right, there are three buttons: 'Cancel', 'Preview changes', and 'Save rules' (which is highlighted with an orange circle).

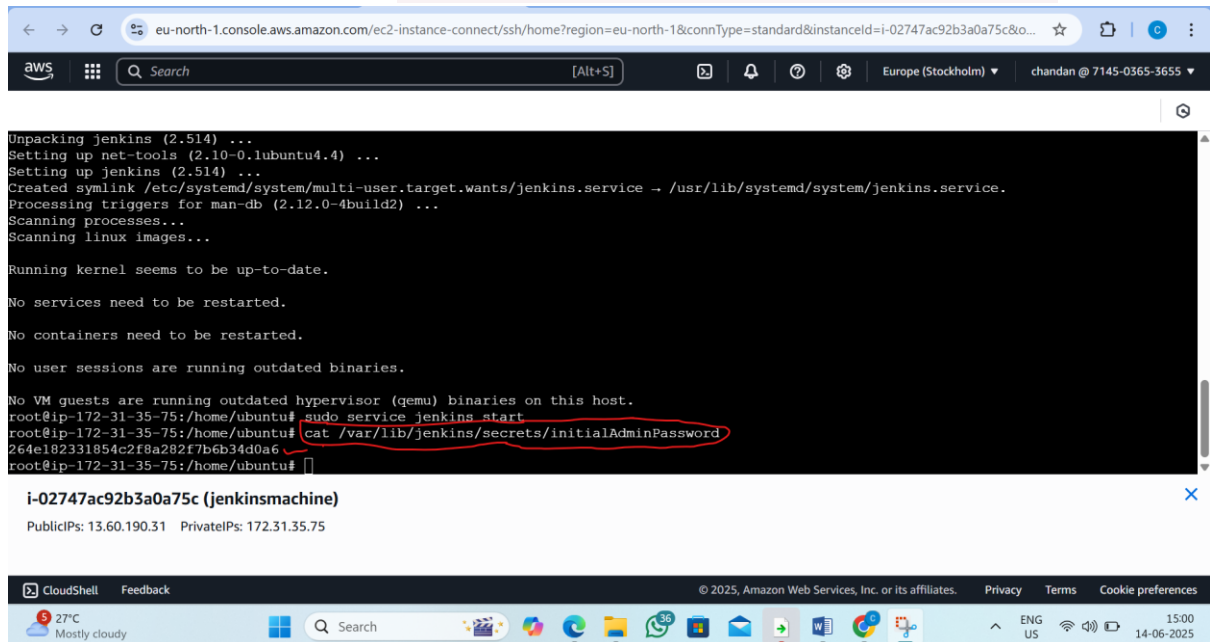
Step 9: now to to browser and type public ip of the instance and port 8080
Type: **13.60.190.31:8080/**

The screenshot shows a web browser window with the URL `13.60.190.31:8080/login?from=%2F`. The page is titled 'Getting Started' and 'Unlock Jenkins'. The text on the page reads: 'To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server: `/var/lib/jenkins/secrets/initialAdminPassword`'. Below this, it says 'Please copy the password from either location and paste it below.' and 'Administrator password'.

At the bottom of the page, there is a 'Continue' button.

Step10:now copy the **administrator pass** then the **command** in the ubuntu then then the password will be shown

By the command:**cat /var/lib/jenkins/secrets/initialAdminPassword**



```
eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-02747ac92b3a0a75c&o...
aws
Search [Alt+S] Europe (Stockholm) chandan @ 7145-0365-3655

Unpacking jenkins (2.514) ...
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Setting up jenkins (2.514) ...
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

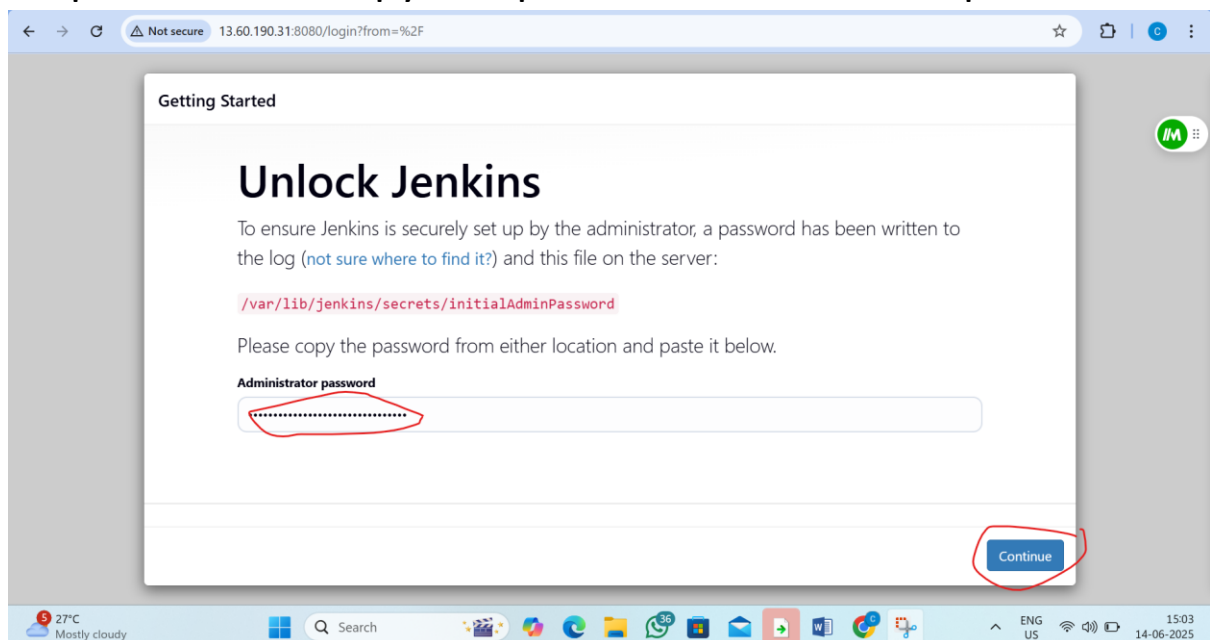
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-35-75:/home/ubuntu# sudo service jenkins start
root@ip-172-31-35-75:/home/ubuntu# cat /var/lib/jenkins/secrets/initialAdminPassword
264e182331854c2f8a282f7b6b34d0a6
root@ip-172-31-35-75:/home/ubuntu#

i-02747ac92b3a0a75c (jenkinsmachine)
PublicIPs: 13.60.190.31 PrivateIPs: 172.31.35.75

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27°C Mostly cloudy Search 15:00 14-06-2025
```

Step11:now the copy and paste in Administrator password



Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

`/var/lib/jenkins/secrets/initialAdminPassword`

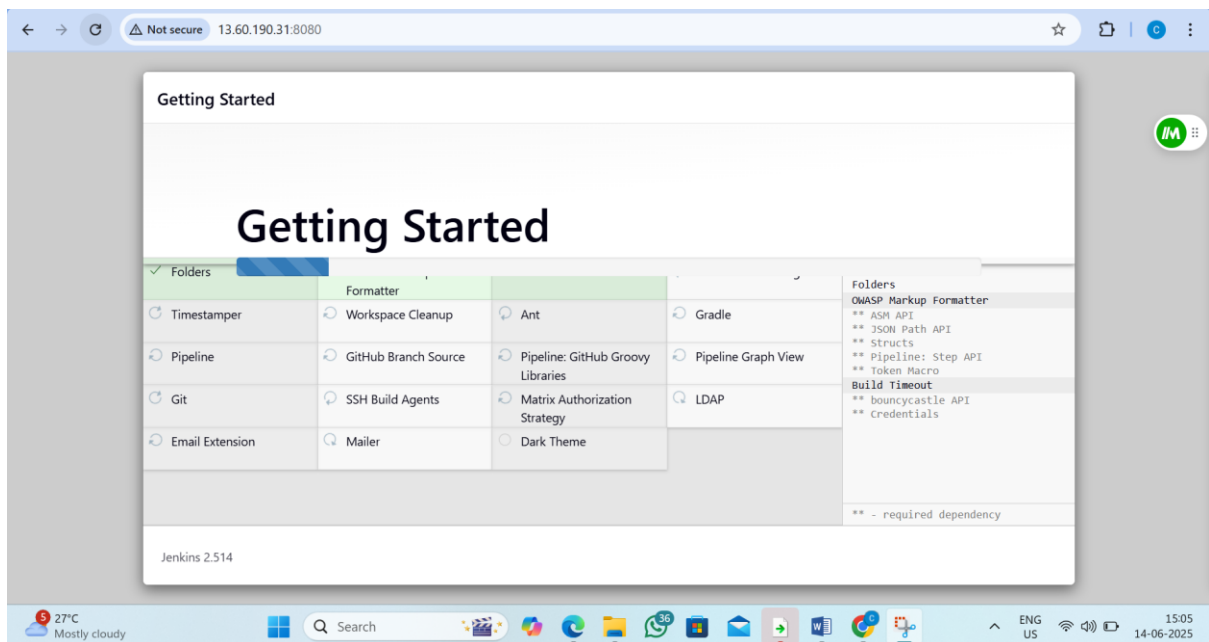
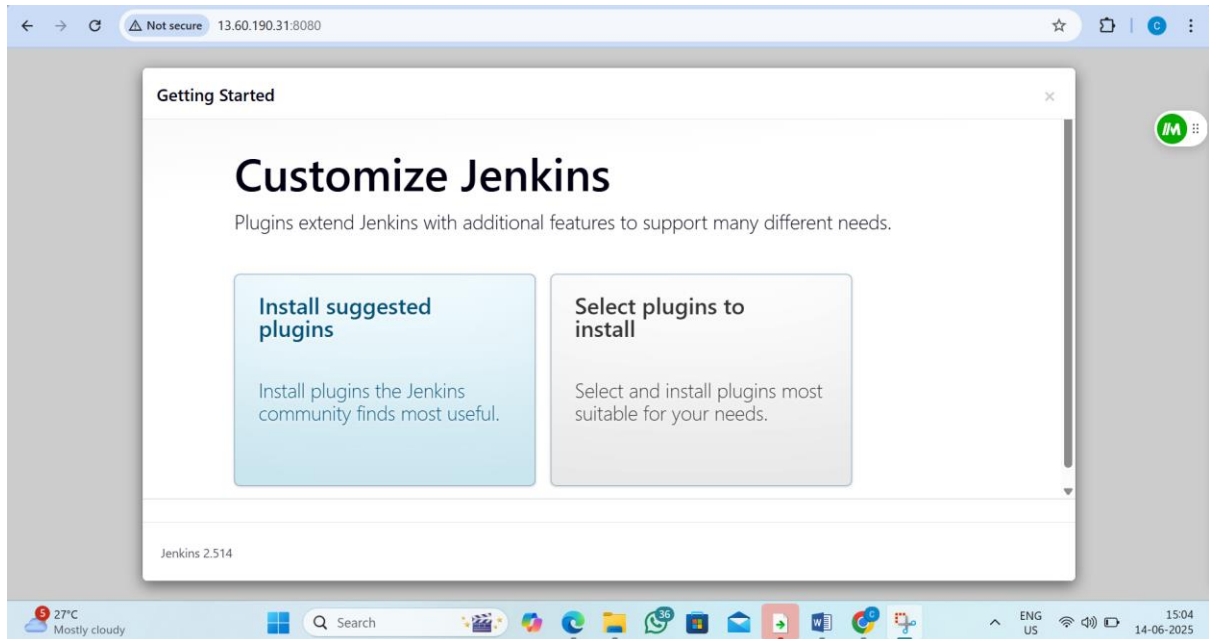
Please copy the password from either location and paste it below.

Administrator password

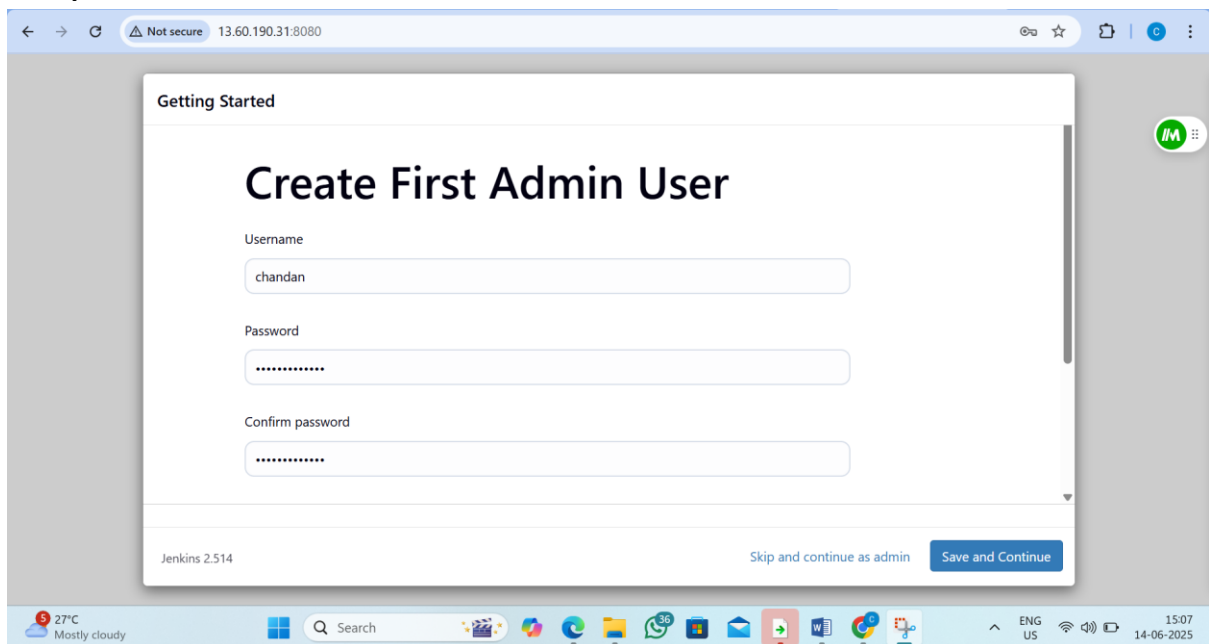
Continue

step12:getting started Jenkins

1)install pulgins



Step13:Create Admin user



A screenshot of a web browser showing the Jenkins 'Getting Started' page. The main heading is 'Create First Admin User'. There are three input fields: 'Username' with the value 'chandan', 'Password' with masked characters, and 'Confirm password' also with masked characters. At the bottom right, there are two buttons: 'Skip and continue as admin' and 'Save and Continue'. The bottom of the browser window shows a Windows taskbar with various icons and a system tray displaying '27°C Mostly cloudy' and the date '14-06-2025'.

Getting Started

Create First Admin User

Username
chandan

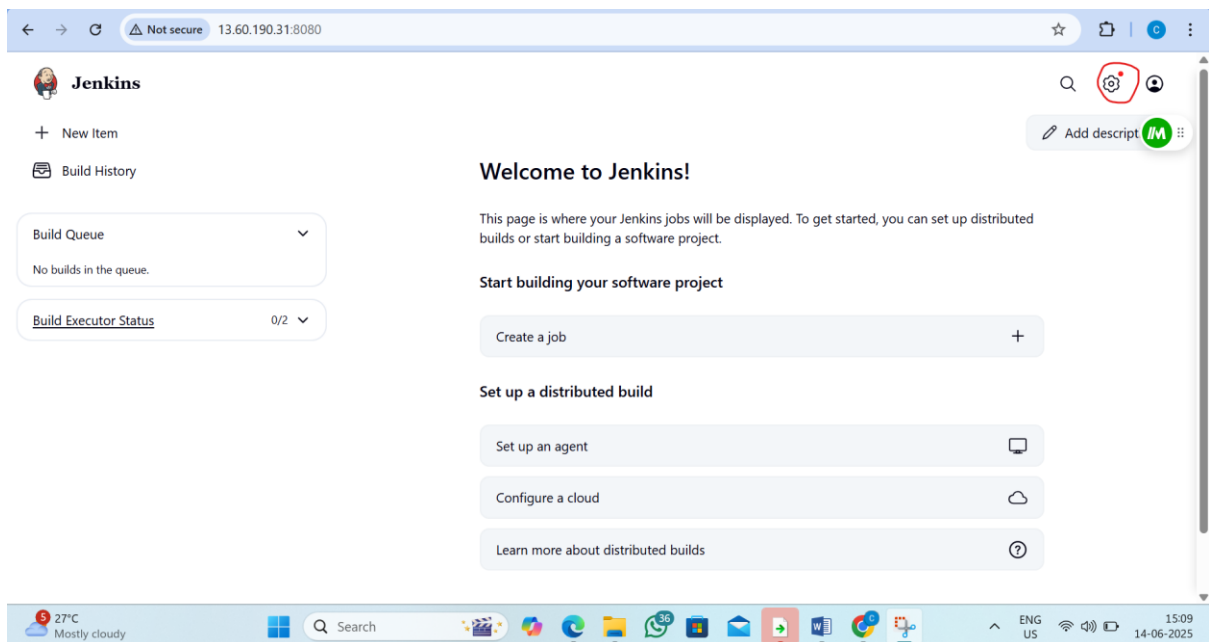
Password
.....

Confirm password
.....

Jenkins 2.514

[Skip and continue as admin](#) [Save and Continue](#)

Step14:go to manage jenkins



A screenshot of the Jenkins 'Welcome to Jenkins!' dashboard. The left sidebar contains links for 'New Item', 'Build History', 'Build Queue' (showing 'No builds in the queue.'), and 'Build Executor Status' (showing '0/2'). The main content area has a 'Welcome to Jenkins!' message, followed by instructions on how to get started. Below this, there are three sections: 'Start building your software project' with a 'Create a job' button, 'Set up a distributed build' with buttons for 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'. The top right corner shows a search icon, a settings icon (circled in red), and a user profile icon. The bottom of the browser window shows a Windows taskbar with various icons and a system tray displaying '27°C Mostly cloudy' and the date '14-06-2025'.

Jenkins

+ New Item

Build History

Build Queue
No builds in the queue.

Build Executor Status
0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

Step15:click on node

The screenshot shows the Jenkins 'Manage Jenkins' interface. At the top, there's a navigation bar with 'Jenkins / Manage Jenkins'. Below it, a search bar and a 'Search settings' button are visible. A warning banner states: 'Building on the built-in node can be a security issue. You should set up distributed builds. See the documentation.' Below this, there are buttons for 'Set up agent', 'Set up cloud', and 'Dismiss'. The 'System Configuration' section contains several links: 'System', 'Plugins', 'Clouds', 'Tools', 'Nodes', and 'Appearance'. The 'Nodes' link is highlighted with a red circle. The Windows taskbar at the bottom shows the date as 14-06-2025 and time as 15:13.

step16:create a new node

The screenshot shows the Jenkins 'Nodes' page. At the top, there's a navigation bar with 'Jenkins / Manage Jenkins / Nodes'. Below it, there's a search bar and a 'Search settings' button. A 'New Node' button is highlighted with a red circle. Below this, there's a table with columns: 'S', 'Name', 'Architecture', 'Clock Difference', 'Free Disk Space', 'Free Swap Space', 'Free Temp Space', and 'Response Time'. The table contains one row for the 'Built-In Node'. Below the table, there's a legend with 'Icon: S M L'. The Windows taskbar at the bottom shows the date as 14-06-2025 and time as 15:14.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	3.57 GiB	0 B	3.57 GiB	0ms
	Data obtained	30 min	30 min	30 min	30 min	30 min	30 min

← → ↻ Not secure 13.60.190.31:8080/manage/computer/new

Jenkins / Manage Jenkins / Nodes / New node

New node

Node name
agent

Type
☒ Permanent Agent
Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

Create

REST API Jenkins 2.514

13.60.190.31:8080

27°C Mostly cloudy

Search

ENG US 15:15 14-06-2025

step17:setting up the node

← → ↻ Not secure 13.60.190.31:8080/manage/computer/createItem

Jenkins / Manage Jenkins / Nodes

Number of executors ?
1

Remote root directory ?
/opt/build

Labels ?
agent

Usage ?
Use this node as much as possible

Launch method ?
Launch agent by connecting it to the controller

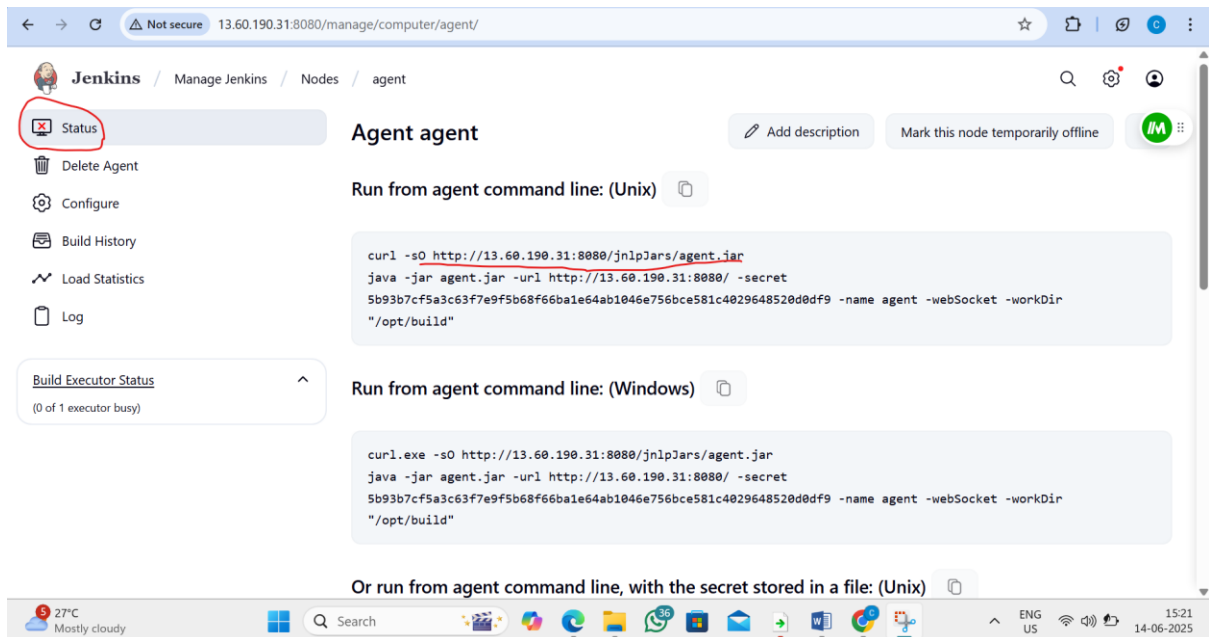
Save

27°C Mostly cloudy

Search

ENG US 15:20 14-06-2025

step18 : the status of the node



The screenshot shows the Jenkins web interface for managing an agent. The browser address bar displays '13.60.190.31:8080/manage/computer/agent/'. The left sidebar contains navigation links: 'Status' (highlighted with a red circle), 'Delete Agent', 'Configure', 'Build History', 'Load Statistics', and 'Log'. The main content area is titled 'Agent agent' and includes buttons for 'Add description' and 'Mark this node temporarily offline'. It provides command-line instructions for running the agent on Unix and Windows. The Unix commands are:

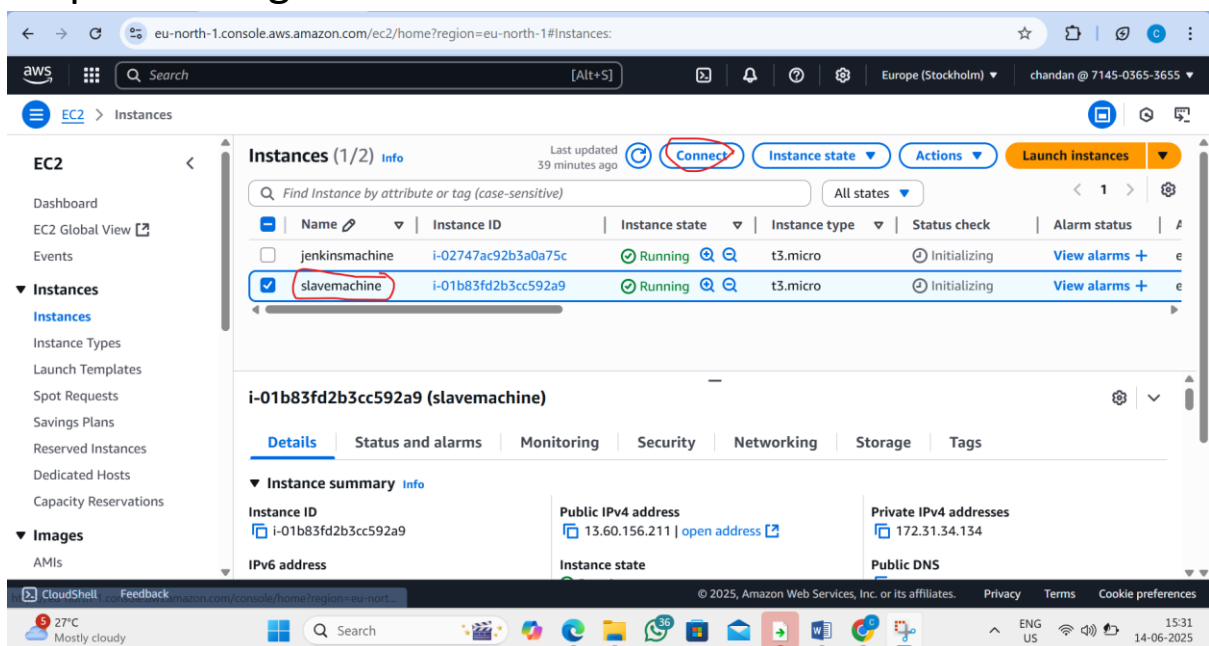
```
curl -sO http://13.60.190.31:8080/jnlpJars/agent.jar
java -jar agent.jar -url http://13.60.190.31:8080/ -secret
5b93b7cf5a3c63f7e9f5b68f66ba1e64ab1046e756bce581c4029648520d0df9 -name agent -webSocket -workDir
"/opt/build"
```

 The Windows commands are:

```
curl.exe -sO http://13.60.190.31:8080/jnlpJars/agent.jar
java -jar agent.jar -url http://13.60.190.31:8080/ -secret
5b93b7cf5a3c63f7e9f5b68f66ba1e64ab1046e756bce581c4029648520d0df9 -name agent -webSocket -workDir
"/opt/build"
```

 A 'Build Executor Status' section shows '(0 of 1 executor busy)'. The bottom status bar indicates a temperature of 27°C and a date of 14-06-2025.

Step19: Now go to instance ubuntu slave machine



The screenshot shows the AWS Management Console for the 'eu-north-1' region. The 'Instances' page is active, displaying a table of EC2 instances. The 'Connect' button is highlighted with a red circle. The table lists two instances: 'jenkinsmachine' and 'slavemachine'. The 'slavemachine' instance is highlighted with a red circle. Below the table, the details for the 'slavemachine' instance (ID: i-01b83fd2b3cc592a9) are shown, including its state (Running), type (t3.micro), and public IP address (13.60.156.211). The bottom status bar shows a temperature of 27°C and a date of 14-06-2025.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
jenkinsmachine	i-02747ac92b3a0a75c	Running	t3.micro	Initializing	View alarms +
slavemachine	i-01b83fd2b3cc592a9	Running	t3.micro	Initializing	View alarms +

Step20:now connect to slave machine instance

The screenshot shows the AWS Management Console interface for connecting to an EC2 instance. The breadcrumb navigation is EC2 > Instances > i-01b83fd2b3cc592a9 > Connect to instance. The instance ID is i-01b83fd2b3cc592a9 (slavemachine). There are two main options: 'Connect using a Public IP' (selected) and 'Connect using a Private IP'. Under 'Public IP', the 'Public IPv4 address' is 13.60.156.211. The 'Username' field is set to 'ubuntu'. A note states: 'Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.' At the bottom right, there are 'Cancel' and 'Connect' buttons. The bottom of the screen shows a Windows taskbar with a search bar and various application icons.

The screenshot shows the AWS CloudShell terminal window. The terminal output displays the Ubuntu login screen, including the message 'Expanded Security Maintenance for Applications is not enabled.' and the prompt 'ubuntu@ip-172-31-34-134:~\$'. Below the terminal output, the instance ID 'i-01b83fd2b3cc592a9 (slavemachine)' and its public IP address '13.60.156.211' are shown. The bottom of the screen shows a Windows taskbar with a search bar and various application icons.

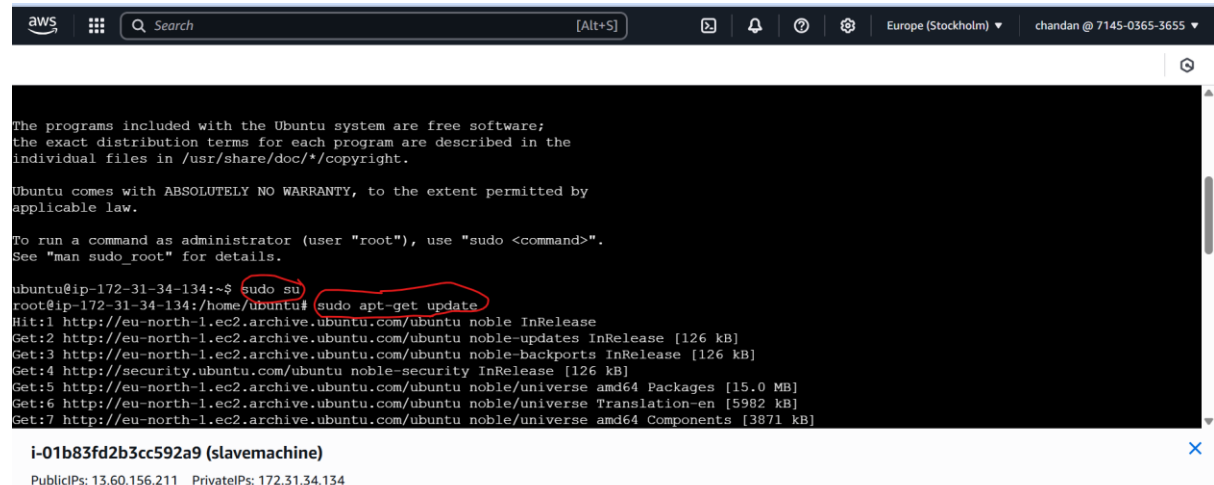
Step21:now we will put the agent in the slave machine

1)install java in slave machine

By the command:1)sudo su

2)sudo apt-get update

3)sudo apt-get install default-jdk-y



```
aws
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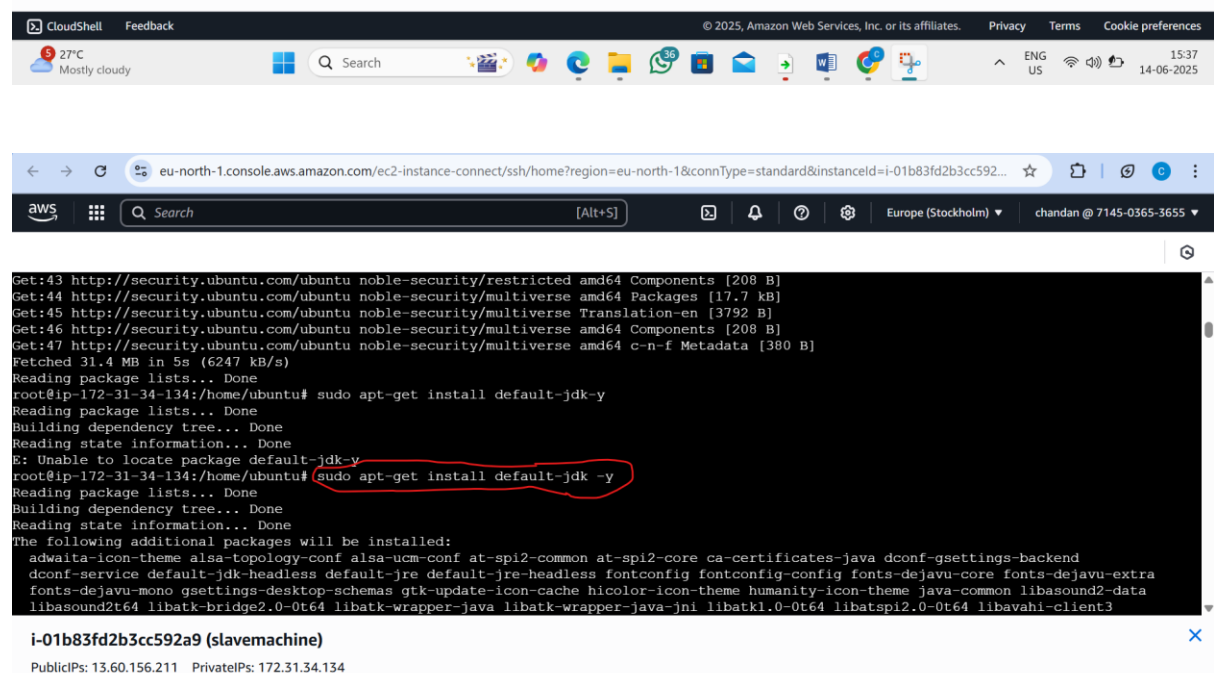
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-34-134:~$ sudo su
root@ip-172-31-34-134:/home/ubuntu# sudo apt-get update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
```

i-01b83fd2b3cc592a9 (slavemachine)
PublicIPs: 13.60.156.211 PrivateIPs: 172.31.34.134



```
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27°C Mostly cloudy
Search

eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-01b83fd2b3cc592...

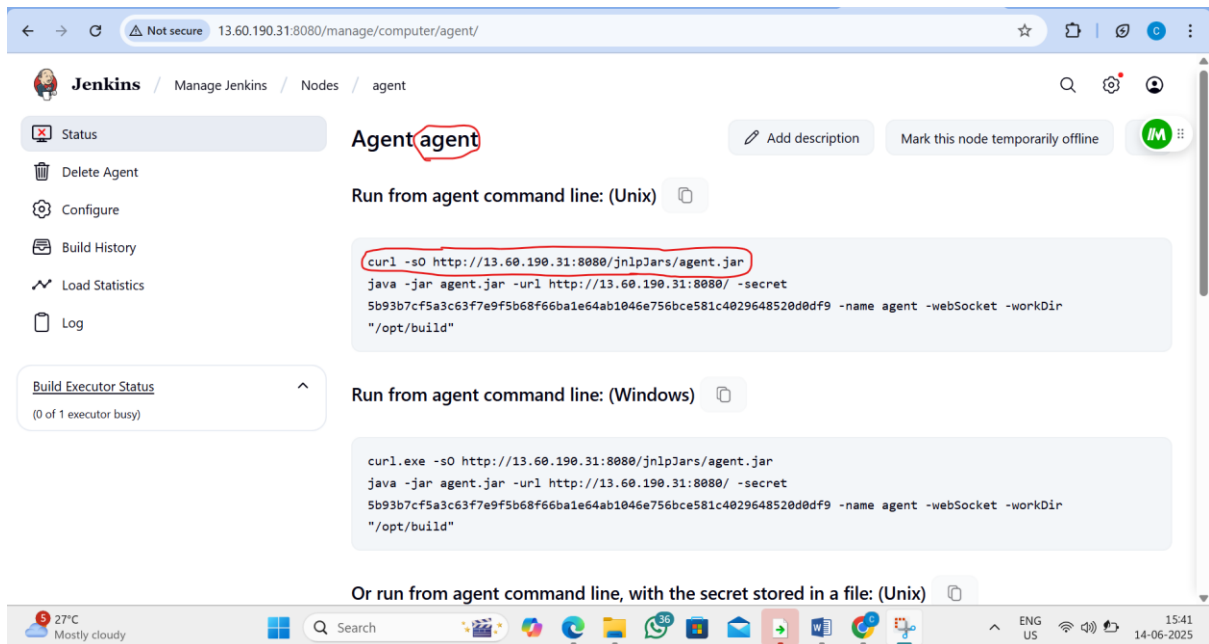
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Get:43 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:44 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [17.7 kB]
Get:45 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [3792 B]
Get:46 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:47 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [380 B]
Fetched 31.4 MB in 5s (6247 kB/s)
Reading package lists... Done
root@ip-172-31-34-134:/home/ubuntu# sudo apt-get install default-jdk-y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package default-jdk-y
root@ip-172-31-34-134:/home/ubuntu# sudo apt-get install default-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gsettings-backend
  dconf-service default-jdk-headless default-jre default-jre-headless fontconfig fontconfig-config fonts-dejavu-core fonts-dejavu-extra
  fonts-dejavu-mono gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme java-common libasound2-data
  libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64 libatspi2.0-0t64 libavahi-client3
```

i-01b83fd2b3cc592a9 (slavemachine)
PublicIPs: 13.60.156.211 PrivateIPs: 172.31.34.134



Step22:go to the agent node in jenkins and copy the command and paste in the slave machine



The screenshot shows the Jenkins web interface for an agent node named 'agent'. The left sidebar contains links for Status, Delete Agent, Configure, Build History, Load Statistics, and Log. The main content area has a title 'Agent agent' with a red circle around the word 'agent'. Below the title are two sections for commands to run from the agent command line: 'Run from agent command line: (Unix)' and 'Run from agent command line: (Windows)'. Both sections contain a red box around the 'curl' command. At the bottom, there is a section 'Or run from agent command line, with the secret stored in a file: (Unix)'.

Jenkins / Manage Jenkins / Nodes / agent

Status

Delete Agent

Configure

Build History

Load Statistics

Log

Build Executor Status

(0 of 1 executor busy)

Agent agent

Add description

Mark this node temporarily offline

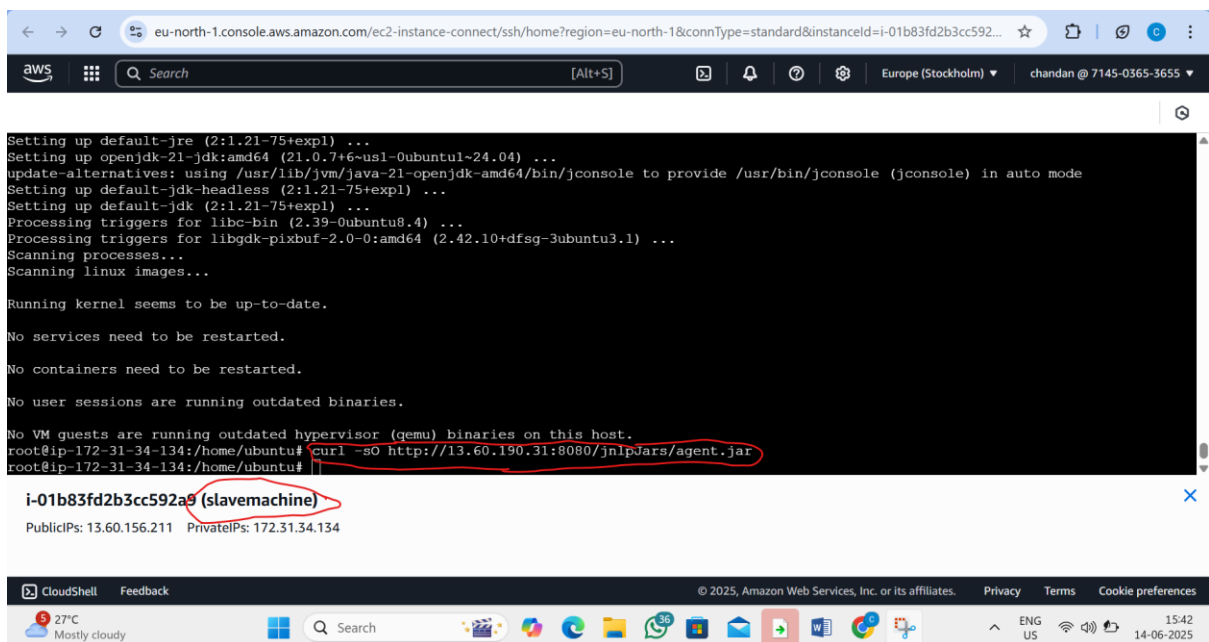
Run from agent command line: (Unix)

```
curl -sO http://13.60.190.31:8080/jnlpJars/agent.jar
java -jar agent.jar -url http://13.60.190.31:8080/ -secret
5b93b7cf5a3c63f7e9f5b68f66ba1e64ab1046e756bce581c4029648520d0df9 -name agent -webSocket -workDir
"/opt/build"
```

Run from agent command line: (Windows)

```
curl.exe -sO http://13.60.190.31:8080/jnlpJars/agent.jar
java -jar agent.jar -url http://13.60.190.31:8080/ -secret
5b93b7cf5a3c63f7e9f5b68f66ba1e64ab1046e756bce581c4029648520d0df9 -name agent -webSocket -workDir
"/opt/build"
```

Or run from agent command line, with the secret stored in a file: (Unix)



The screenshot shows the AWS CloudShell terminal interface. The terminal output displays the results of running the 'curl' command from the previous screenshot. The output shows the download of the Jenkins agent.jar file and the execution of the Java command. The terminal prompt is 'root@ip-172-31-34-134:/home/ubuntu#'. Below the terminal output, the instance ID 'i-01b83fd2b3cc592a9' is highlighted with a red circle and labeled '(slavemachine)'. The Public IP is 13.60.156.211 and the Private IP is 172.31.34.134.

eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-01b83fd2b3cc592...

aws

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Setting up default-jre (2:1.21-75+exp1) ...

Setting up openjdk-21-jdk:amd64 (21.0.7+6-us1-0ubuntu1~24.04) ...

update-alternatives: using /usr/lib/jvm/java-21-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode

Setting up default-jdk-headless (2:1.21-75+exp1) ...

Setting up default-jdk (2:1.21-75+exp1) ...

Processing triggers for libc-bin (2.39-0ubuntu8.4) ...

Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.10+dfsg-3ubuntu3.1) ...

Scanning processes...

Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

root@ip-172-31-34-134:/home/ubuntu# curl -sO http://13.60.190.31:8080/jnlpJars/agent.jar

root@ip-172-31-34-134:/home/ubuntu#

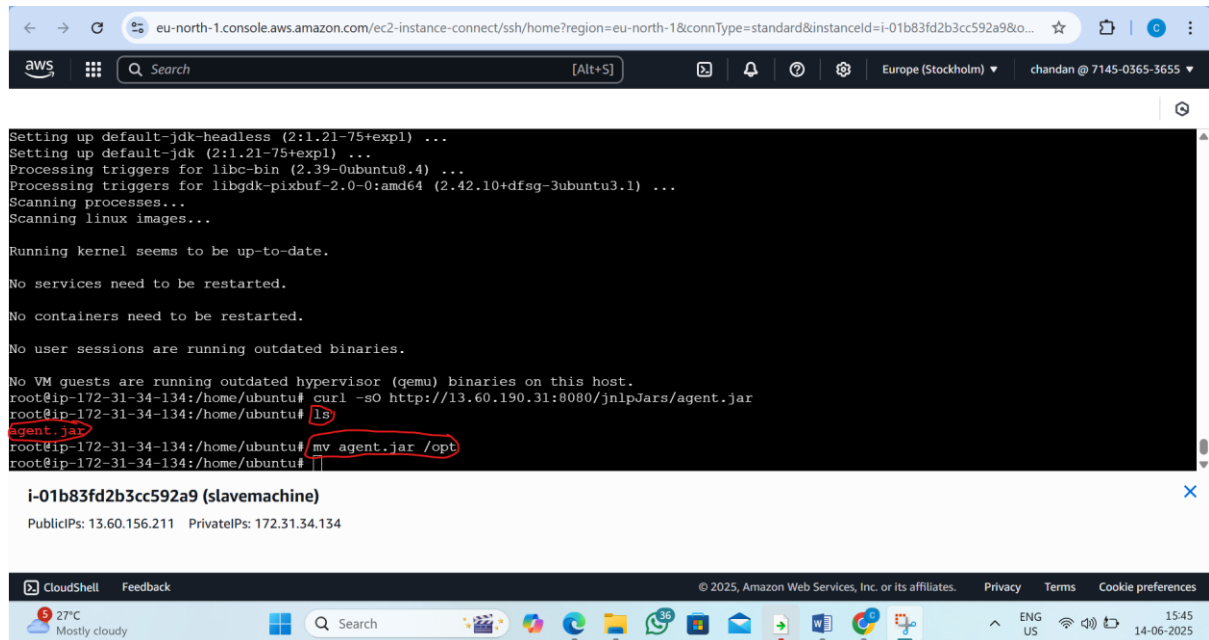
i-01b83fd2b3cc592a9 (slavemachine)

PublicIPs: 13.60.156.211 PrivateIPs: 172.31.34.134

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Step23:follow the command
by the command:1)ls
2)mv agent.jar /opt

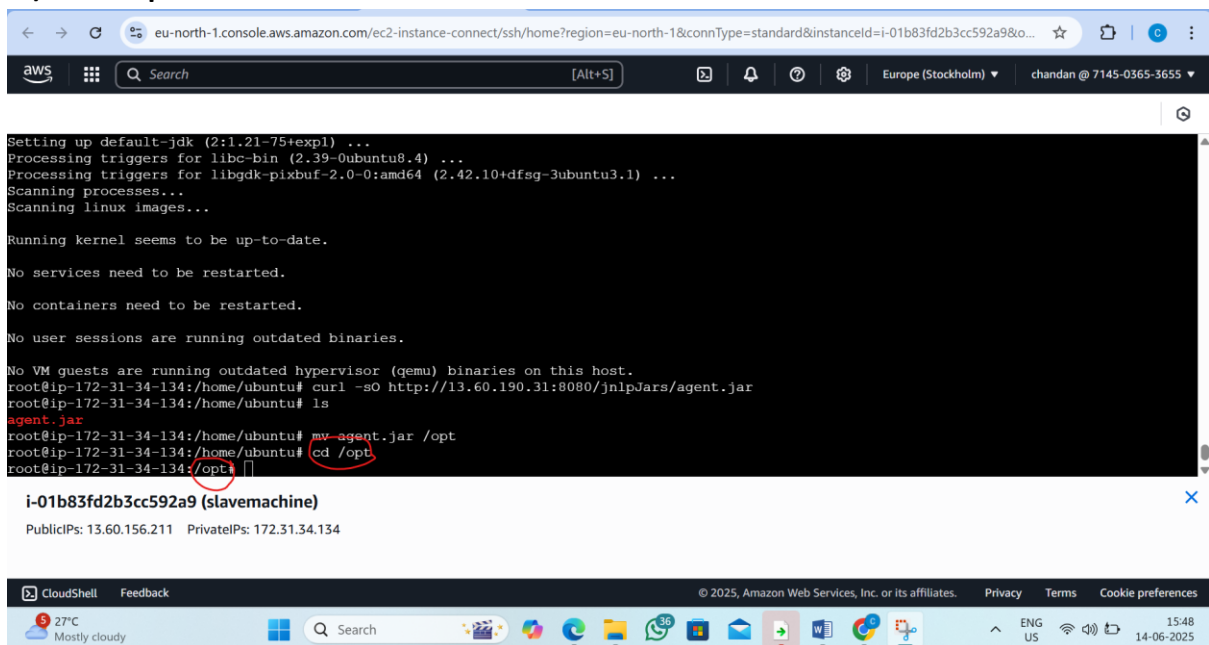


The screenshot shows the AWS CloudShell interface. The terminal output displays system updates and then the user's commands. The 'ls' command shows 'agent.jar' in the current directory. The 'mv agent.jar /opt' command is executed successfully. The instance name 'i-01b83fd2b3cc592a9 (slavemachine)' and its IP addresses are listed below the terminal. The bottom of the screen shows a Windows taskbar with various application icons and the system clock.

```
Setting up default-jdk-headless (2:1.21-75+expl) ...  
Setting up default-jdk (2:1.21-75+expl) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...  
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.10+dfsg-3ubuntu3.1) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
root@ip-172-31-34-134:/home/ubuntu# curl -sO http://13.60.190.31:8080/jnlpJars/agent.jar  
root@ip-172-31-34-134:/home/ubuntu# ls  
agent.jar  
root@ip-172-31-34-134:/home/ubuntu# mv agent.jar /opt  
root@ip-172-31-34-134:/home/ubuntu#
```

i-01b83fd2b3cc592a9 (slavemachine)
PublicIPs: 13.60.156.211 PrivateIPs: 172.31.34.134

3)cd /opt



This screenshot is similar to the previous one, showing the same system updates and the 'ls' command. The 'cd /opt' command is now executed, changing the directory to '/opt'. The instance details and the Windows taskbar are also visible.

```
Setting up default-jdk (2:1.21-75+expl) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...  
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.10+dfsg-3ubuntu3.1) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
root@ip-172-31-34-134:/home/ubuntu# curl -sO http://13.60.190.31:8080/jnlpJars/agent.jar  
root@ip-172-31-34-134:/home/ubuntu# ls  
agent.jar  
root@ip-172-31-34-134:/home/ubuntu# mv agent.jar /opt  
root@ip-172-31-34-134:/home/ubuntu# cd /opt  
root@ip-172-31-34-134:/opt#
```

i-01b83fd2b3cc592a9 (slavemachine)
PublicIPs: 13.60.156.211 PrivateIPs: 172.31.34.134

Step24:now copy the command in node agent in Jenkins and paste in the slave master

Jenkins / Manage Jenkins / Nodes / agent

Status

Delete Agent

Configure

Build History

Load Statistics

Log

Build Executor Status
(0 of 1 executor busy)

Agent agent

Add description

Mark this node temporarily offline

Run from agent command line: (Unix)

```
curl -sO http://13.60.190.31:8080/jnlpJars/agent.jar
java -jar agent.jar -url http://13.60.190.31:8080/ -secret
5b93b7cf5a3c63f7e9f5b68f66ba1e64ab1046e756bce581c4029648520d0df9 -name agent -webSocket -workDir
"/opt/build"
```

Run from agent command line: (Windows)

```
curl.exe -sO http://13.60.190.31:8080/jnlpJars/agent.jar
java -jar agent.jar -url http://13.60.190.31:8080/ -secret
5b93b7cf5a3c63f7e9f5b68f66ba1e64ab1046e756bce581c4029648520d0df9 -name agent -webSocket -workDir
"/opt/build"
```

Or run from agent command line, with the secret stored in a file: (Unix)

eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-01b83fd2b3cc592a9&o...

aws

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```
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-34-134:/home/ubuntu# curl -sO http://13.60.190.31:8080/jnlpJars/agent.jar
root@ip-172-31-34-134:/home/ubuntu# ls
agent.jar
root@ip-172-31-34-134:/home/ubuntu# mv agent.jar /opt
root@ip-172-31-34-134:/home/ubuntu# cd /opt
root@ip-172-31-34-134:/opt# java -jar agent.jar -url http://13.60.190.31:8080/ -secret 5b93b7cf5a3c63f7e9f5b68f66ba1e64ab1046e756bce581c40
29648520d0df9 -name agent -webSocket -workDir "/opt/build"
Jun 14, 2025 10:21:26 AM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /opt/build/remoting as a remoting work directory
Jun 14, 2025 10:21:27 AM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /opt/build/remoting
Jun 14, 2025 10:21:27 AM hudson.remoting.Launcher createEngine
INFO: Setting up agent: agent
Jun 14, 2025 10:21:27 AM hudson.remoting.Engine startEngine
INFO: Using Remoting version: 3309.v27b_9314fdla_4
Jun 14, 2025 10:21:27 AM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /opt/build/remoting as a remoting work directory
```

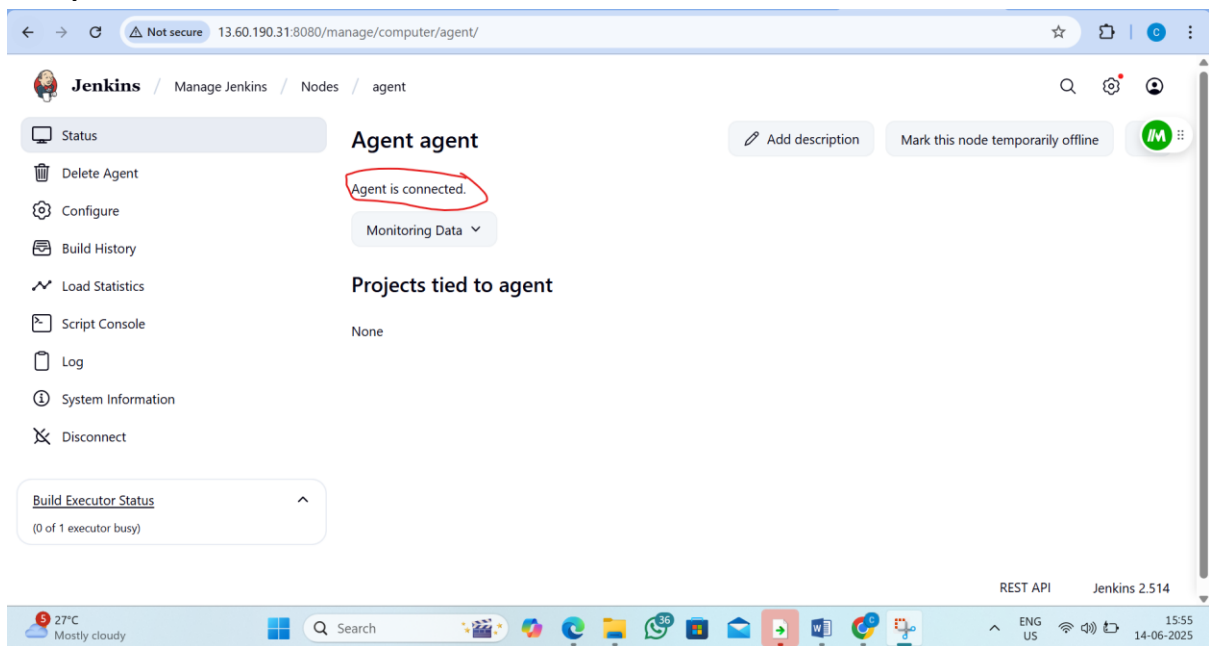
i-01b83fd2b3cc592a9 (slavemachine)

PublicIPs: 13.60.156.211 PrivateIPs: 172.31.34.134

CloudShell Feedback

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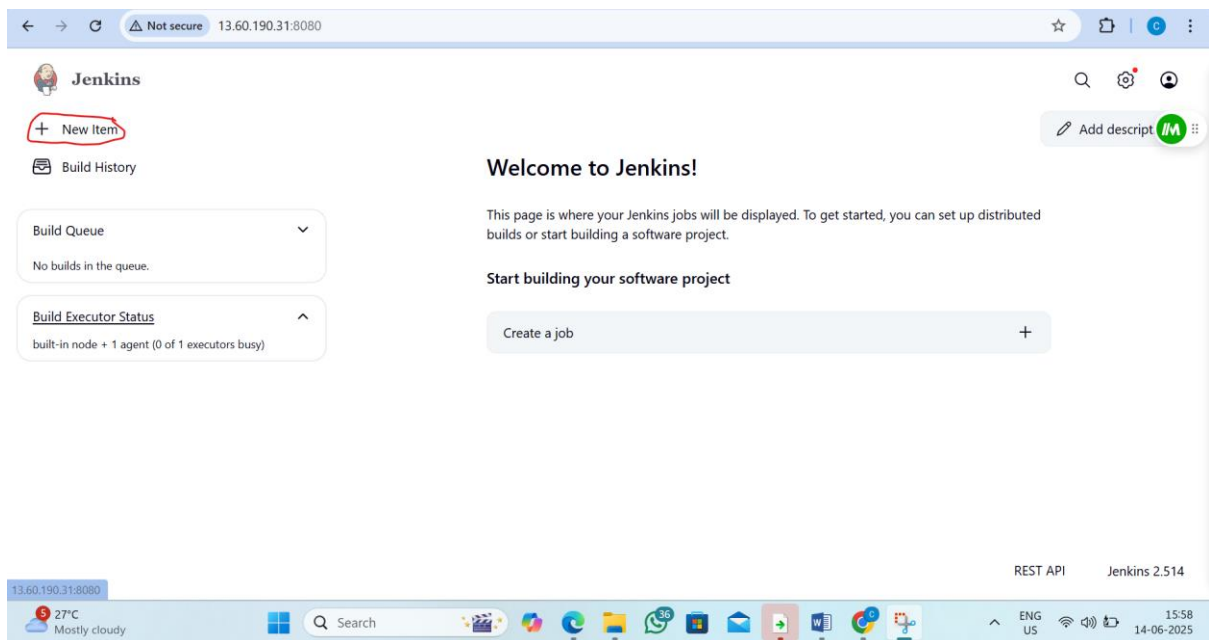
Step25:it shows connected



The screenshot shows the Jenkins web interface at the URL `13.60.190.31:8080/manage/computer/agent/`. The left sidebar contains a list of actions: Status, Delete Agent, Configure, Build History, Load Statistics, Script Console, Log, System Information, and Disconnect. The main content area is titled "Agent agent" and displays the status "Agent is connected." which is circled in red. Below this, there is a "Monitoring Data" dropdown and a section titled "Projects tied to agent" showing "None". At the bottom left, a "Build Executor Status" box indicates "(0 of 1 executor busy)". The bottom of the page shows a Windows taskbar with various application icons and a system tray with weather and time information.

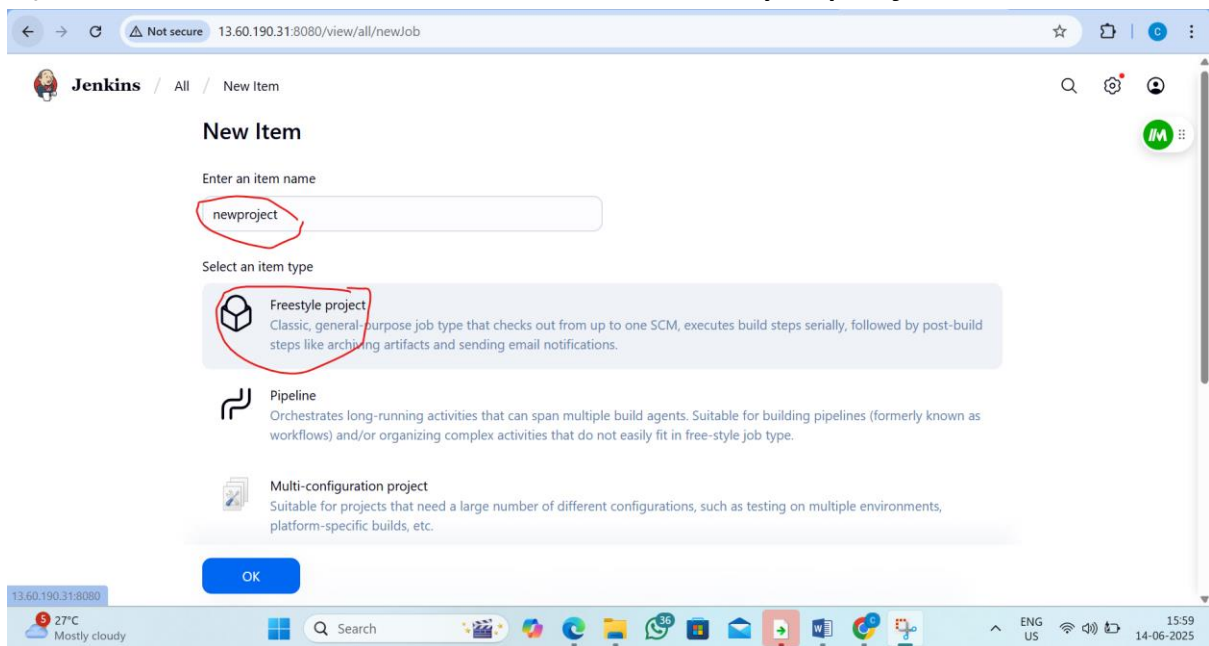
step26:now create a freestyle project

1)create new item



The screenshot shows the Jenkins main dashboard at the URL `13.60.190.31:8080`. The left sidebar features a "+ New Item" button, which is circled in red, and a "Build History" link. Below these are two status boxes: "Build Queue" showing "No builds in the queue." and "Build Executor Status" showing "built-in node + 1 agent (0 of 1 executors busy)". The main content area displays a "Welcome to Jenkins!" message and instructions on how to get started. A "Start building your software project" section includes a "Create a job" button with a plus icon. The bottom of the page shows a Windows taskbar with various application icons and a system tray with weather and time information.

2) Enter the name and select the freestyle project



The screenshot shows the Jenkins 'New Item' configuration page. The browser address bar displays '13.60.190.31:8080/view/all/newJob'. The page title is 'New Item'. Under 'Enter an item name', the text 'newproject' is entered in a text box. Under 'Select an item type', three options are listed: 'Freestyle project' (selected with a red circle), 'Pipeline', and 'Multi-configuration project'. The 'Freestyle project' description reads: 'Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.' The 'Pipeline' description reads: 'Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.' The 'Multi-configuration project' description reads: 'Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.' An 'OK' button is at the bottom.

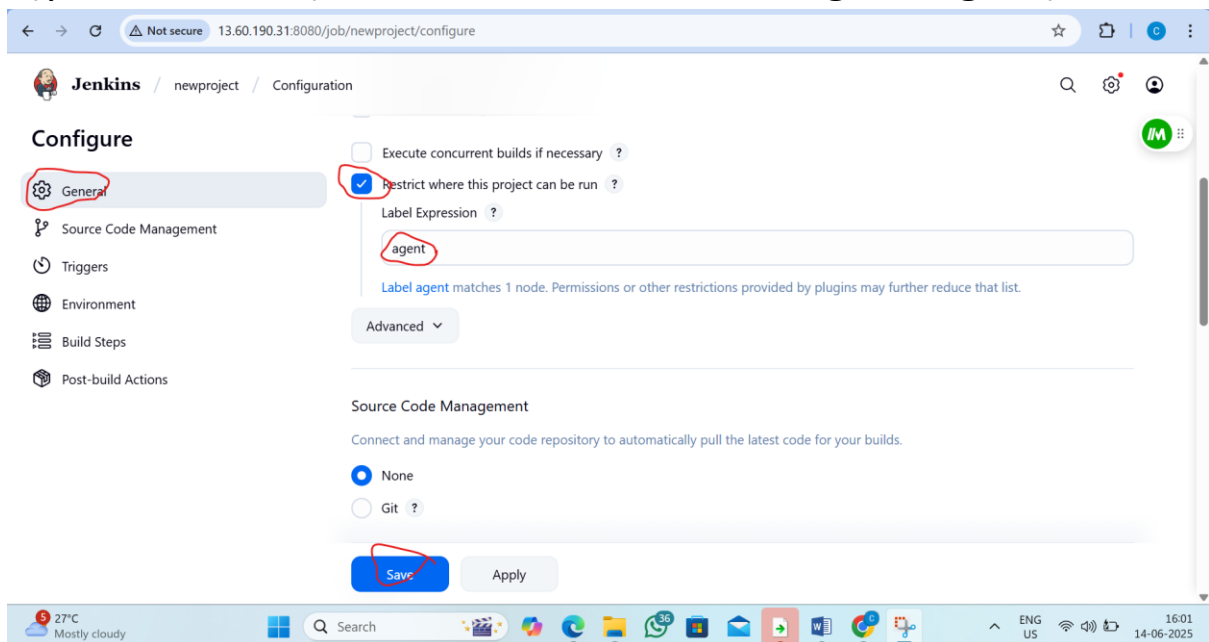
13.60.190.31:8080

27°C Mostly cloudy

Search

ENG US 15:59 14-06-2025

3) put the label (which I used while creating the agent)



The screenshot shows the Jenkins 'Configure' page for the 'newproject' job. The browser address bar displays '13.60.190.31:8080/job/newproject/configure'. The page title is 'Configure'. On the left sidebar, 'General' is selected (circled in red). The 'General' section contains the following options: 'Execute concurrent builds if necessary' (unchecked), 'Restrict where this project can be run' (checked with a red circle), and 'Label Expression' (text box containing 'agent', circled in red). Below the 'Label Expression' text box, it says 'Label agent matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.' There is an 'Advanced' dropdown menu. Below the 'General' section, the 'Source Code Management' section is visible, with 'None' selected. At the bottom, there are 'Save' and 'Apply' buttons, with 'Save' circled in red.

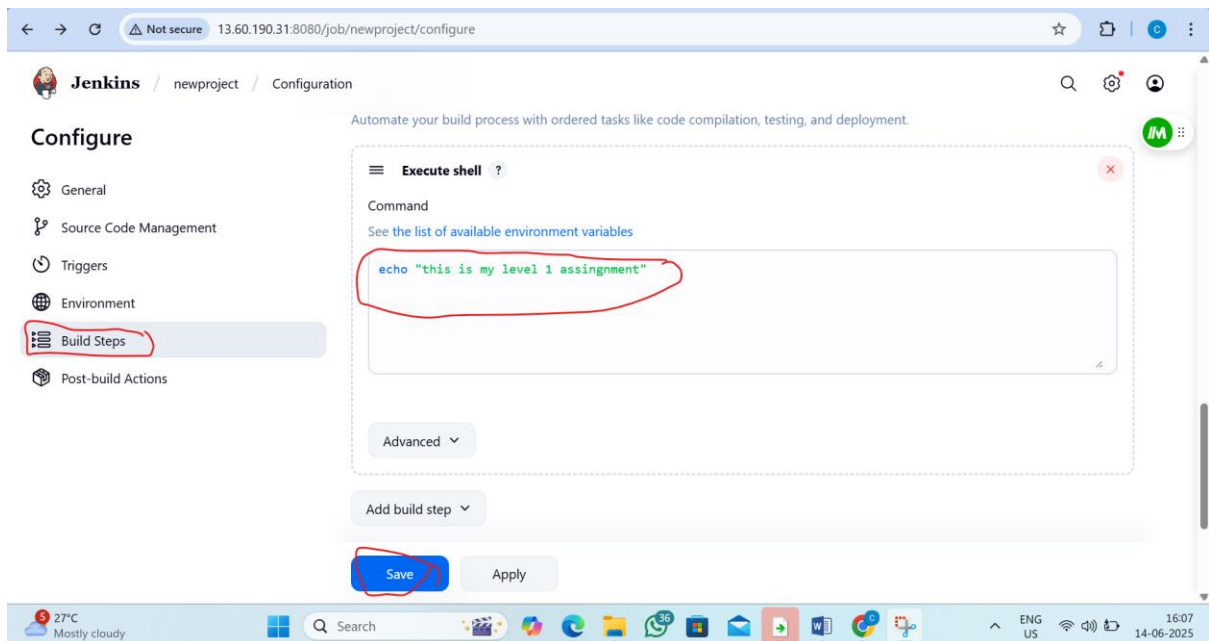
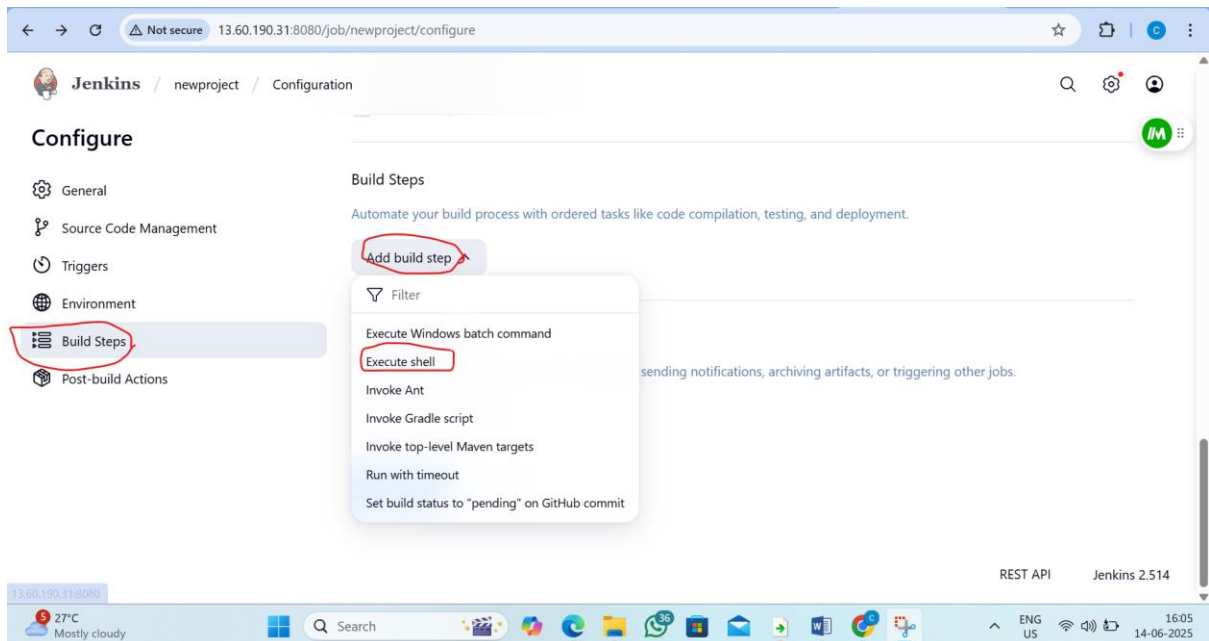
13.60.190.31:8080/job/newproject/configure

27°C Mostly cloudy

Search

ENG US 16:01 14-06-2025

Step27:go to build steps and select the execute shell because iam using ubuntu



Step28: click on build now

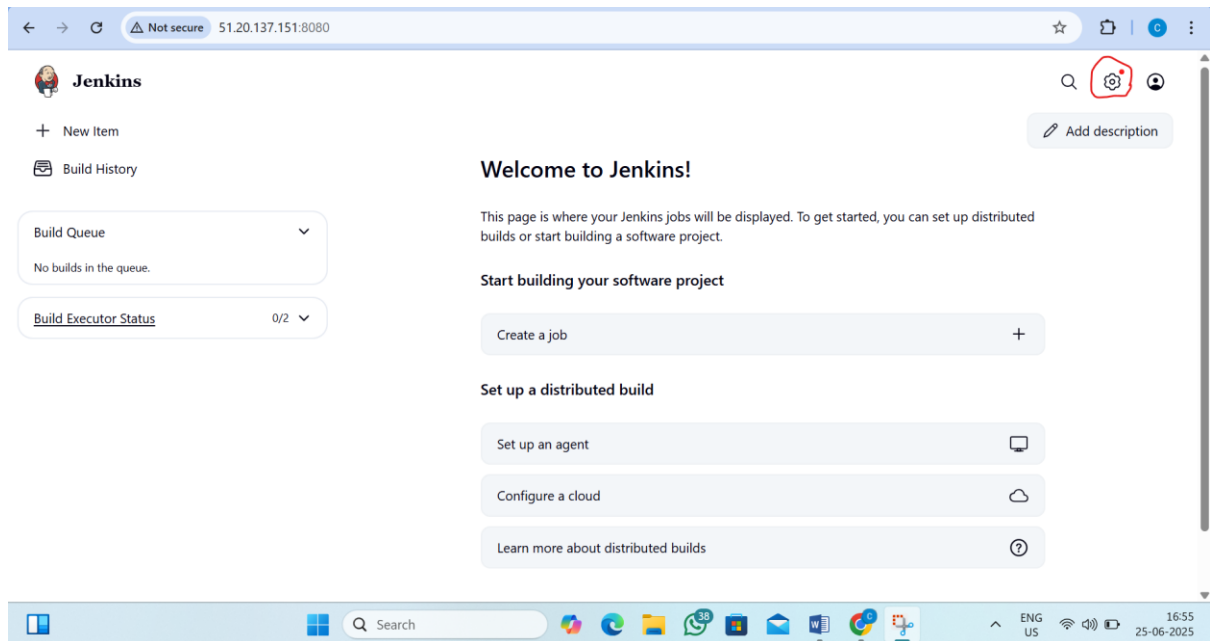
The screenshot shows the Jenkins web interface for a job named 'newproject'. The left sidebar contains a menu with 'Status', 'Changes', 'Workspace', 'Build Now' (circled in red), 'Configure', 'Delete Project', and 'Rename'. The main area displays the job status as 'newproject' with a green checkmark. Below this, there are 'Permalinks' for the last build, stable build, successful build, and completed build, all indicating they occurred 3 minutes and 58 seconds ago. At the bottom left, a 'Builds' section shows a list of builds for 'Today', with build #2 at 10:38 AM and build #1 at 10:34 AM. Both are marked with green checkmarks. The browser's address bar shows '13.60.190.31:8080/job/newproject/'. The Windows taskbar at the bottom shows the system clock as 16:08 on 14-06-2025.

The screenshot shows the Jenkins web interface for the 'newproject' job, specifically the details for build #2. The left sidebar menu includes 'Status', 'Changes', 'Console Output', 'Edit Build Information', 'Delete build '#2'', 'Timings', and 'Previous Build'. The main area displays the build status as '#2 (Jun 14, 2025, 10:38:17 AM)' with a green checkmark. To the right of the status, there are buttons for 'Add description' and 'Keep this build forever'. Below the status, it indicates the build was 'Started by user chandan' and 'Started 2 min 17 sec ago'. A section titled 'This run spent:' lists the following durations: '6 ms waiting;', '0.46 sec build duration;', and '0.47 sec total from scheduled to completion.'. There is also a 'No changes.' message. The browser's address bar shows '13.60.190.31:8080/job/newproject/2/'. The Windows taskbar at the bottom shows the system clock as 16:10 on 14-06-2025.

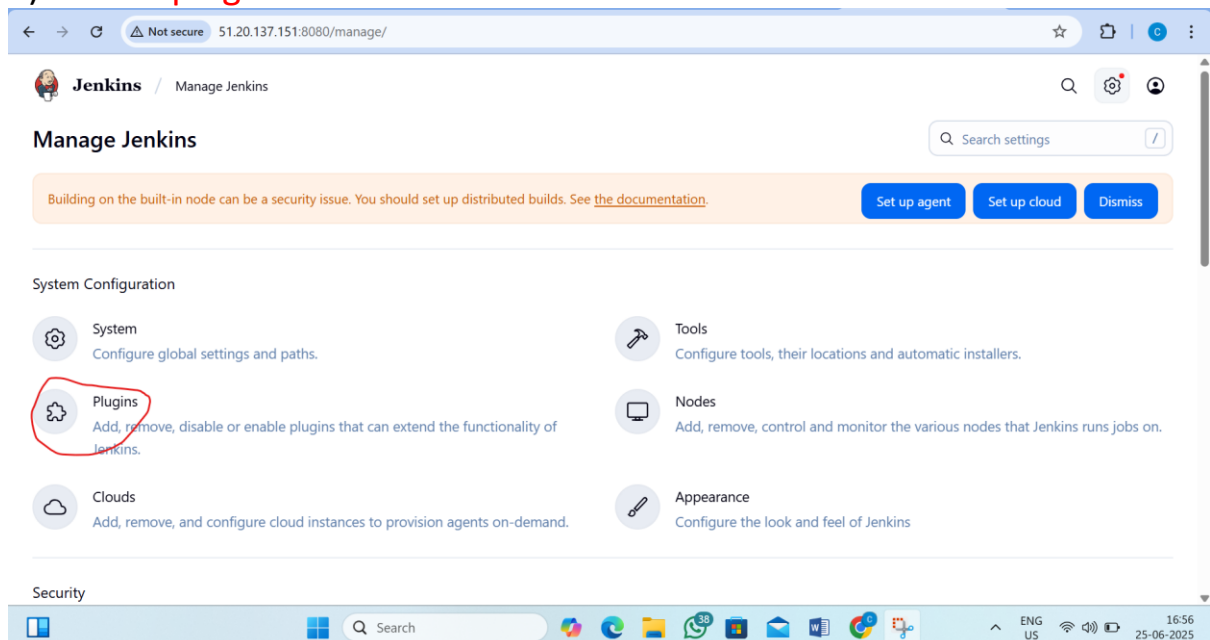
2. L2 - Create CI/CD Pipeline to Clone and Build Java Maven Web Application

Step 1: Install Required Plugins

1) go to Jenkins dashboard and click on **manage Jenkins**



2) click on **plugins**



3) click on available plugins and install git plugin and Maven Integration plugin

The screenshot shows the Jenkins 'Manage Jenkins' > 'Plugins' page. The 'Available plugins' tab is selected. A search bar contains the text 'maven'. The 'Install' button is circled in red. The table below lists available plugins:

Install	Name	Released	Health
☒	Git Push v4.vd474e0fe7b_ec git Source Code Management This plugin allows to perform a git push as a post build step via Git Plugin	3 yr 4 mo ago	86
☒	Maven Integration v3.26 Build Tools This plugin provides a deep integration between Jenkins and Maven. It adds support for automatic triggers between projects depending on SNAPSHOTS as well as the automated configuration of various Jenkins publishers such as Junit.	2 mo 2 days ago	100
☐	Config File Provider 988.v0461fcc2b_9d1 Groovy-related External Site/Tool Integrations Maven Ability to provide configuration files (e.g. settings.xml for maven, XML, groovy, custom files,...) loaded through the UI which will be copied to the job workspace.	1 mo 27 days ago	100

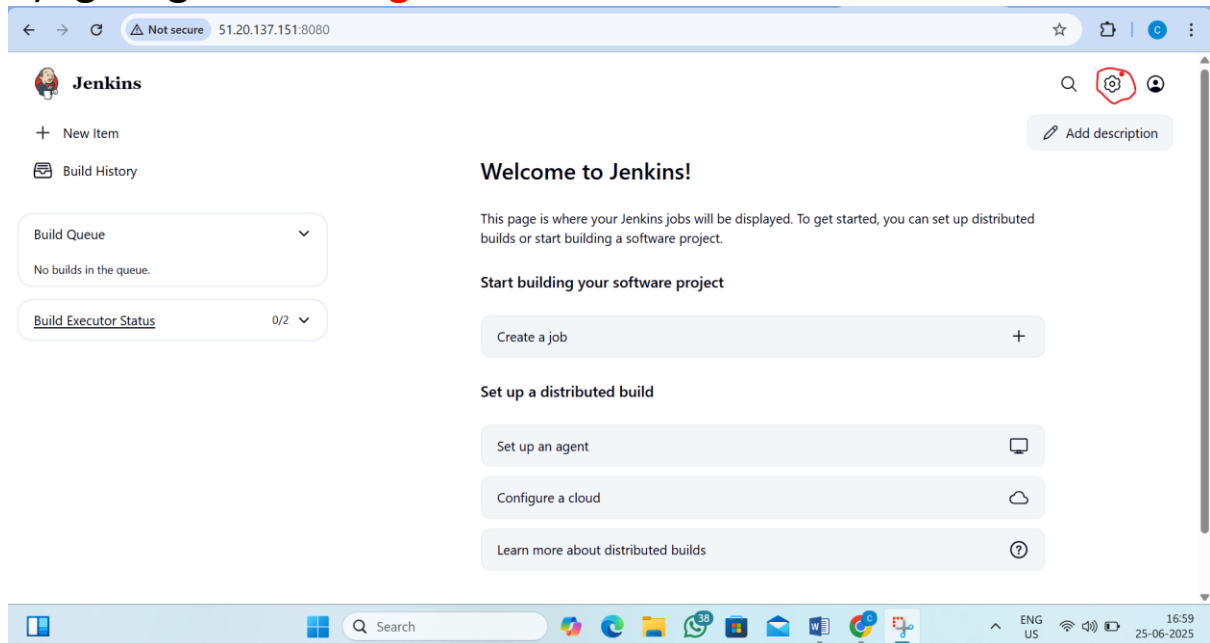
The screenshot shows the Jenkins 'Manage Jenkins' > 'Plugins' page. The 'Available plugins' tab is selected. The table below lists installed plugins:

Plugin	Status
LDAP	Success
jsoup API	Success
Email Extension	Success
Mailer	Success
Theme Manager	Success
Dark Theme	Success
Loading plugin extensions	Success
JavaMail API	Success
SSH server	Success
Git Push	Success
Javadoc	Success
Dev Tools Symbols API	Success
JSch dependency	Success
Maven Integration	Success
Loading plugin extensions	Success

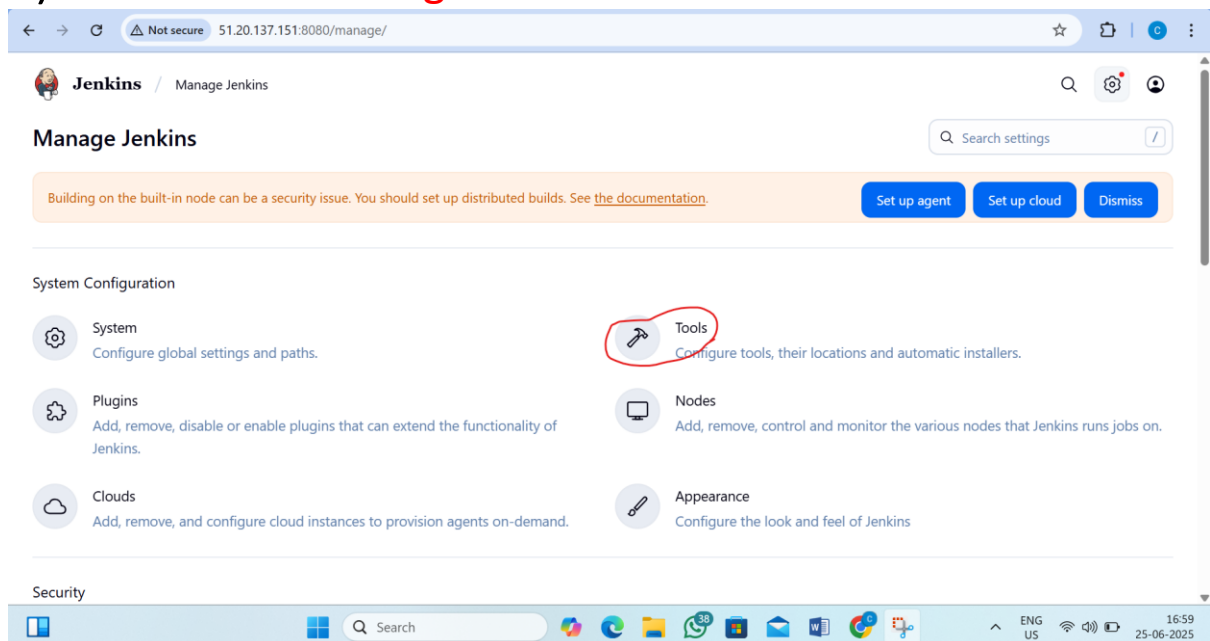
→ [Go back to the top page](#)
(you can start using the installed plugins right away)

Step 2: Create Maven Tool in Jenkins

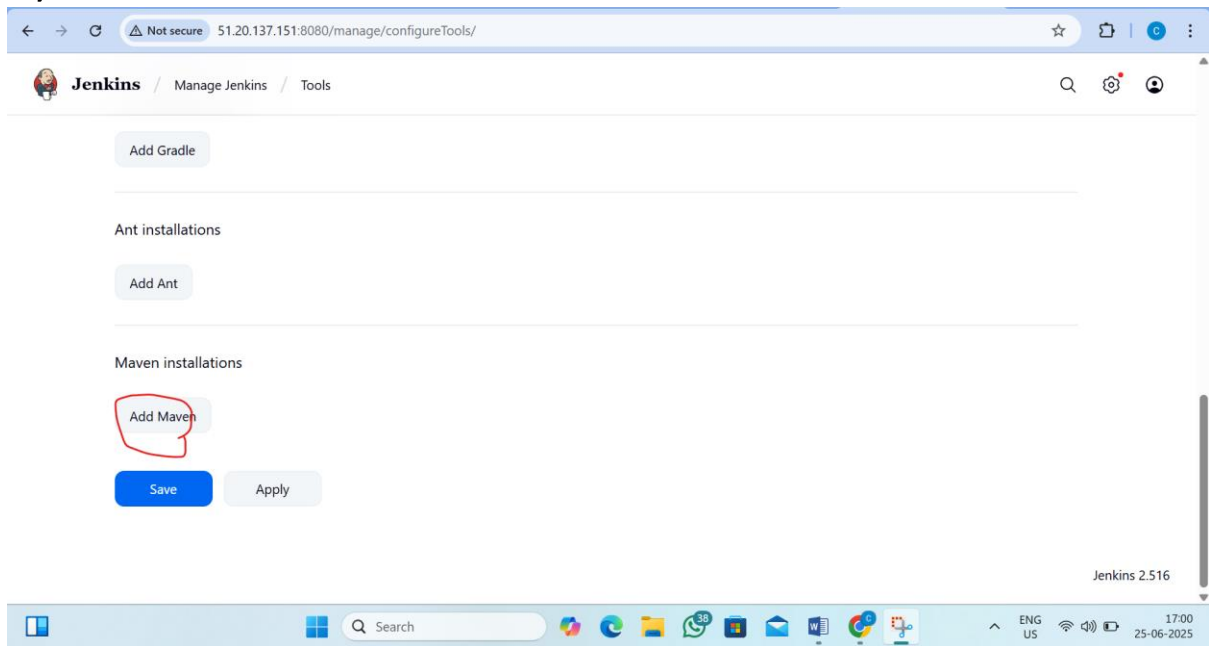
1) Again go to **manage Jenkins**



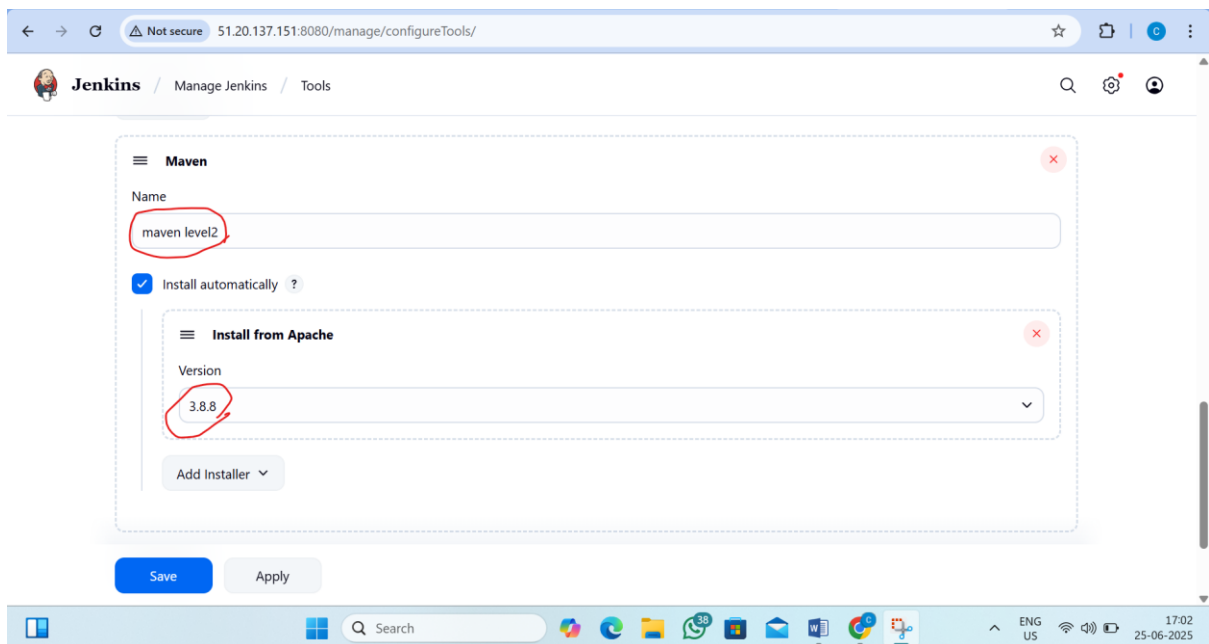
2) click on **Tool Configuration**



3) Scroll to **Maven** section → Click **Add Maven**



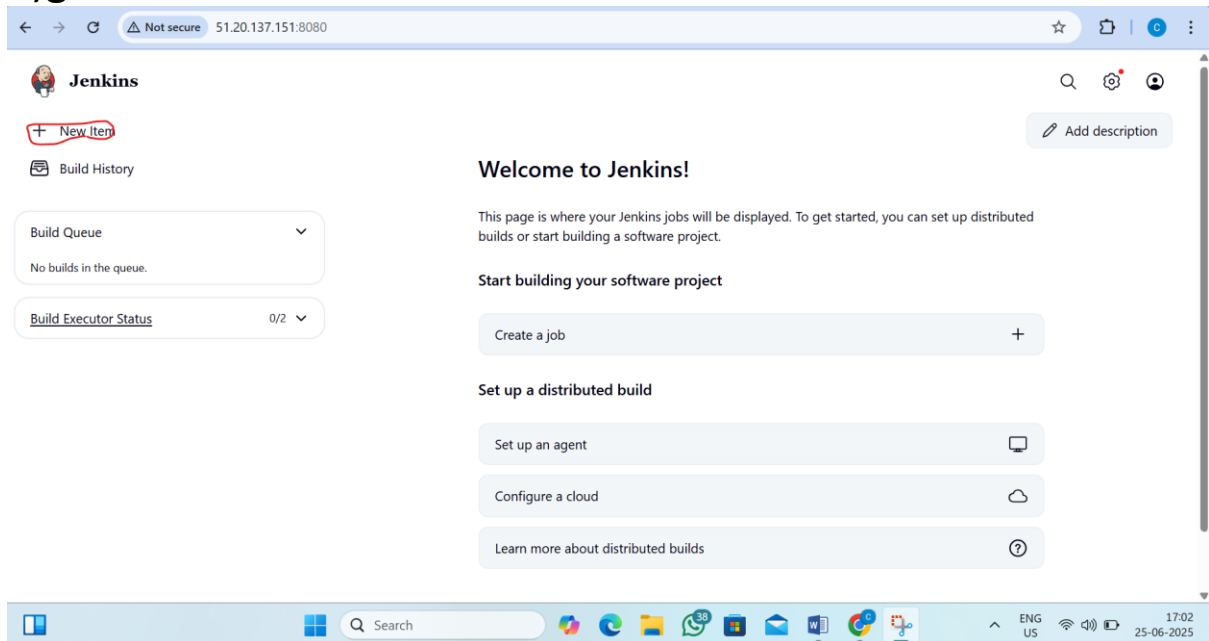
This screenshot shows the Jenkins 'Tools' configuration page. The browser address bar indicates the URL '51.20.137.151:8080/manage/configureTools/'. The page title is 'Jenkins / Manage Jenkins / Tools'. Under the 'Maven installations' section, the 'Add Maven' button is circled in red. Below this button are 'Save' and 'Apply' buttons. The Jenkins version '2.516' is noted in the bottom right corner of the page content.



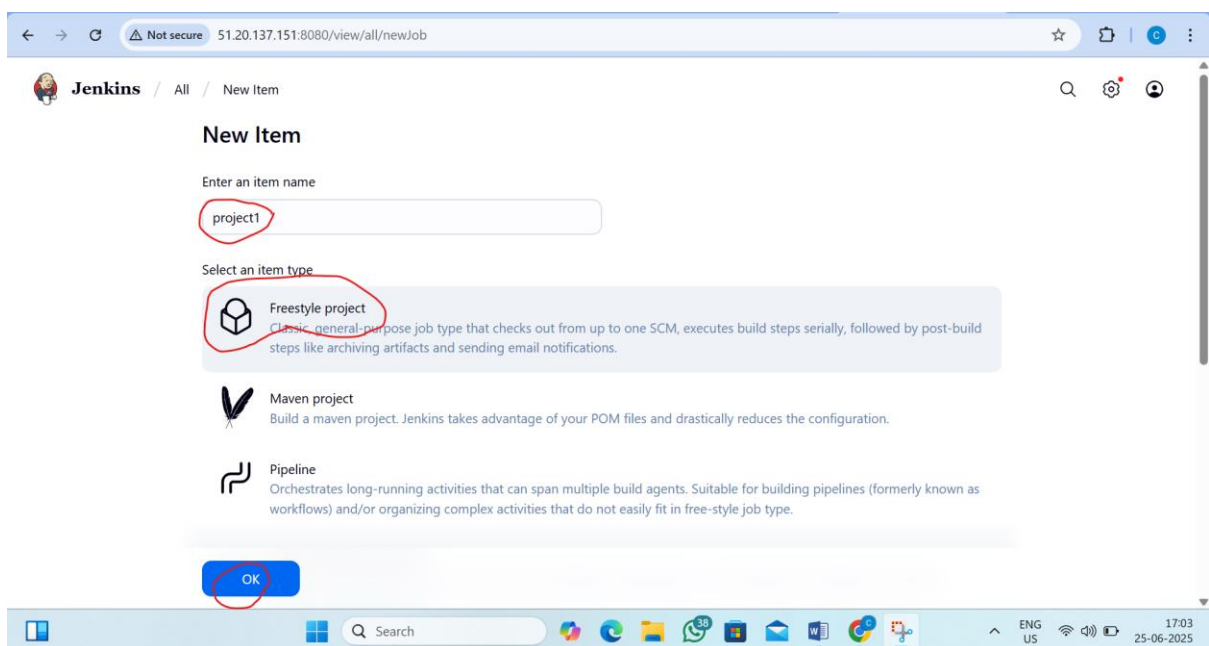
This screenshot shows the 'Maven' configuration dialog box. The 'Name' field contains 'maven level2' and is circled in red. The 'Install automatically' checkbox is checked. Within the 'Install from Apache' section, the 'Version' dropdown menu is set to '3.8.8' and is also circled in red. At the bottom of the dialog are 'Save' and 'Apply' buttons. The system taskbar at the bottom shows the time as 17:02 on 25-06-2025.

Step 3: Create a New Freestyle Project

1) go to Jenkins dashboard and click on **new item**

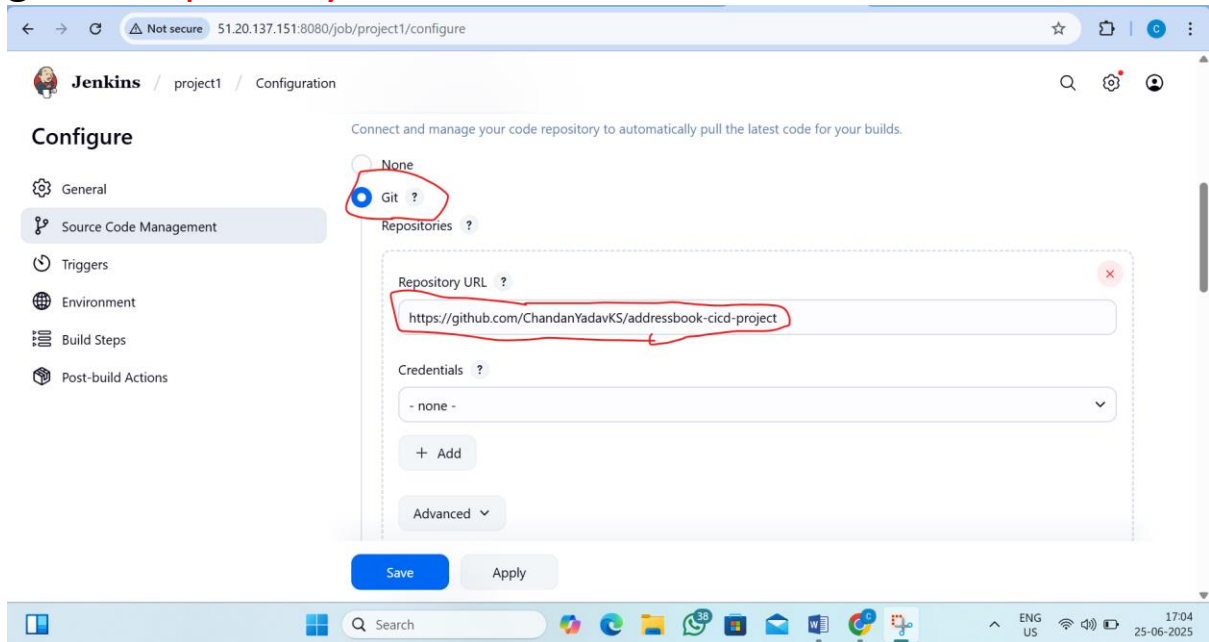


2) Enter the project name and select the freestyle project



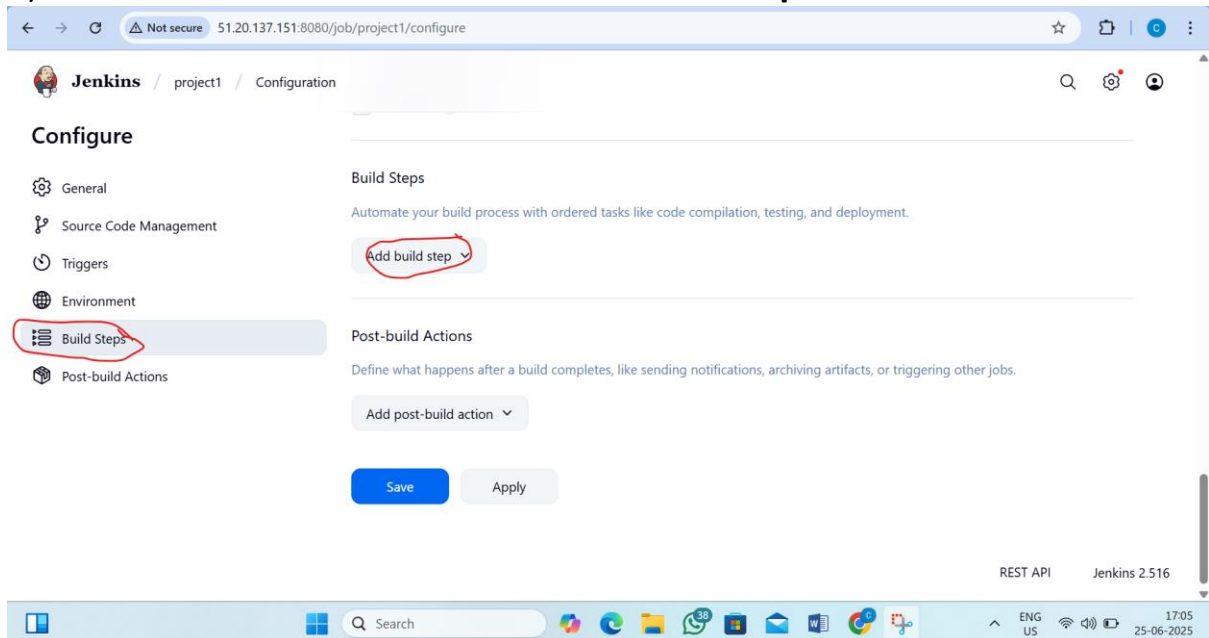
Step 4: Configure GitHub Repository

1) In the project → **Source Code Management** tab and add github repository url

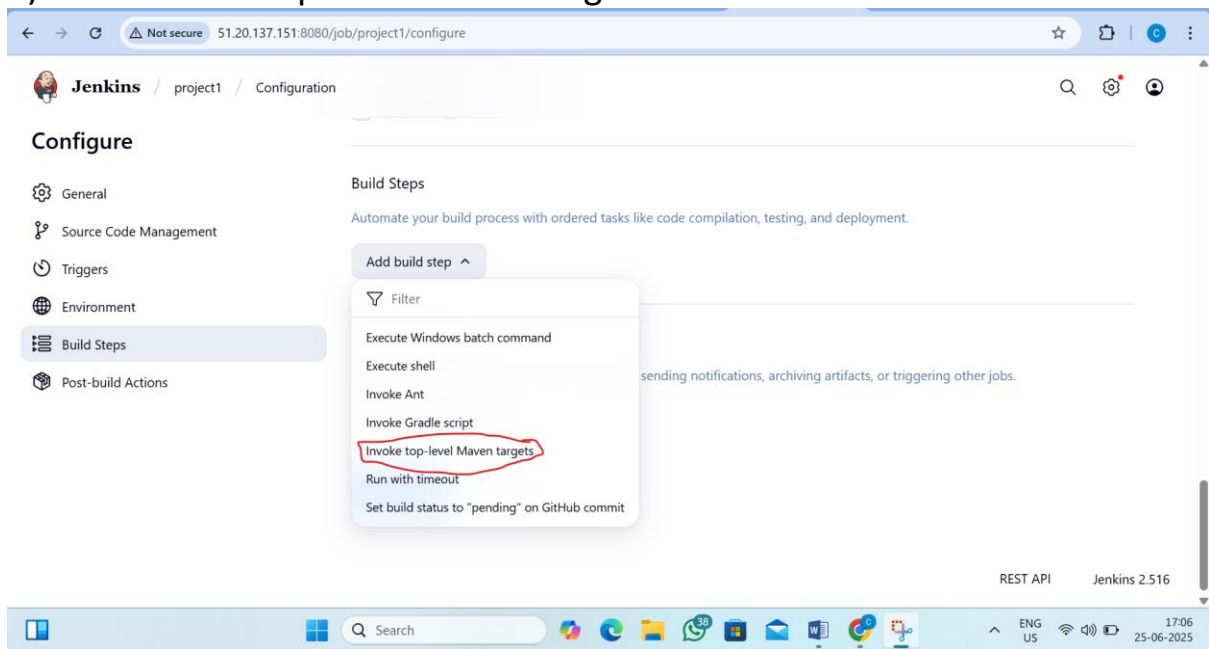


Step 5: Configure Build Environment

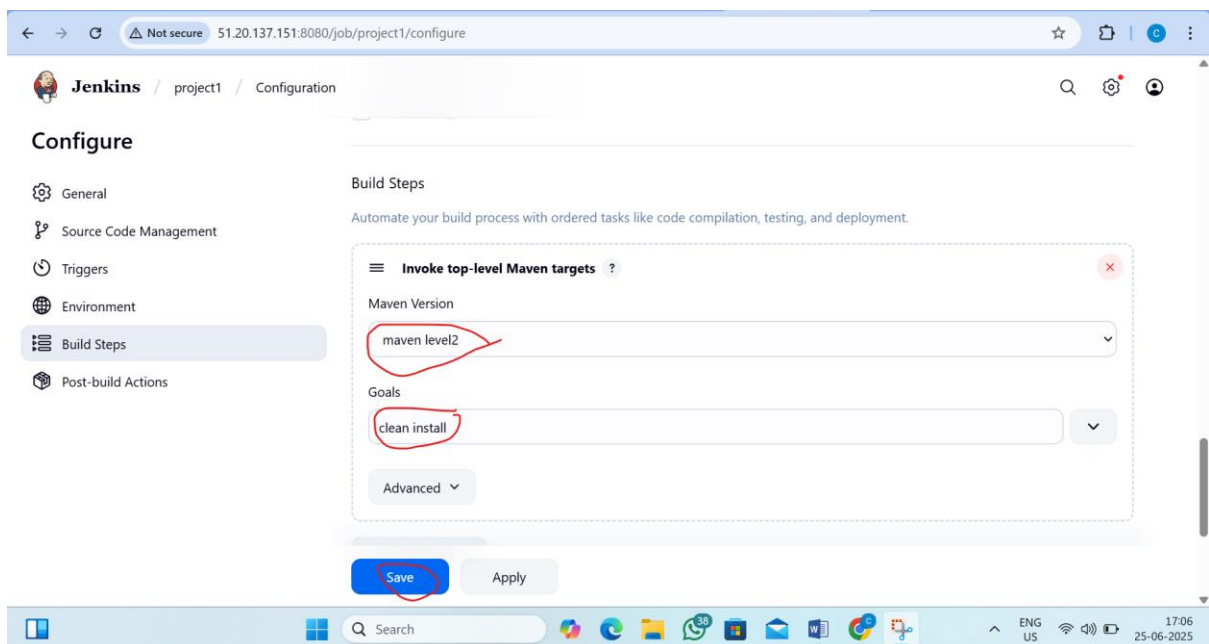
1) Go to the **Build** section → Click **Add build step**



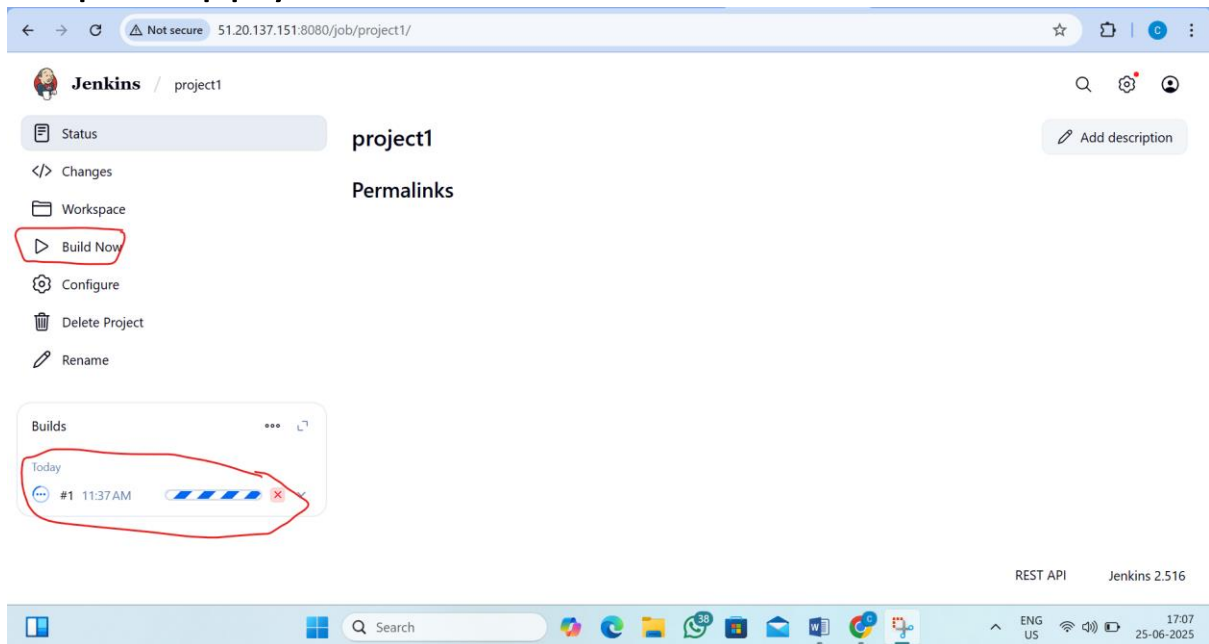
2) Select invoke top-level Maven targets



3) now select **maven version** which u have created and
In goals type **clean install**

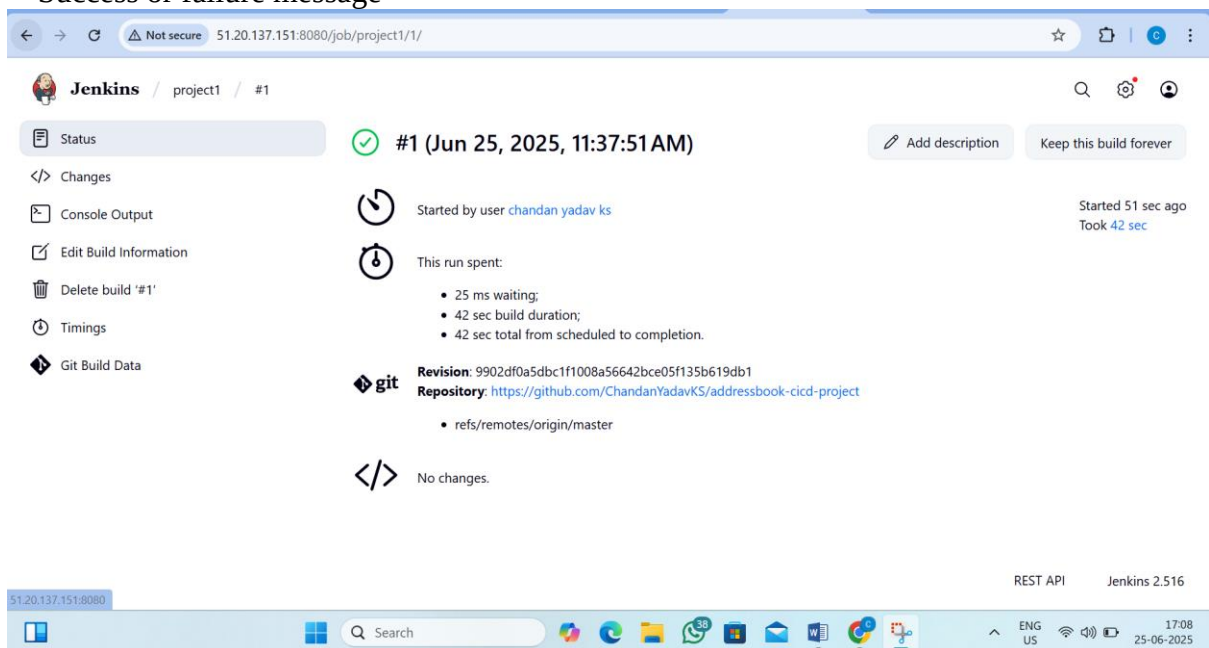


Step 6: Apply & Build



Step 7: Check Output by clicking the console output

- Project being cloned
- Maven build running
- Success or failure message



← → ↻ ⚠ Not secure 51.20.137.151:8080/job/project1/1/console ☆ 📄 🔍

Jenkins / project1 / #1 / Console Output 🔍 ⚙️ 👤

📄 Status </> Changes 📄 **Console Output** 📄 Edit Build Information 🗑 Delete build '#1' ⌚ Timings 📦 Git Build Data

📄 **Console Output** 📄 Download 📄 Copy 📄 View as plain text

🕒 This log is too long to show here, 487 KB has been skipped — click to see the complete log

```
Progress (4): 205/293 kB | 228 kB | 46 kB | 81/315 kB

Downloading from central: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-core/1.3/hamcrest-core-1.3.jar
Progress (4): 209/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 213/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 217/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 221/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 225/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 229/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 233/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 238/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 242/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 246/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 250/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 254/293 kB | 228 kB | 46 kB | 81/315 kB
Progress (4): 258/293 kB | 228 kB | 46 kB | 81/315 kB
```

📄 Search ENG US 📄 17:12 25-06-2025

← → ↻ ⚠ Not secure 51.20.137.151:8080/job/project1/1/console ☆ 📄 🔍

Jenkins / project1 / #1 / Console Output 🔍 ⚙️ 👤

Progress (2): 12 kB | 230 kB

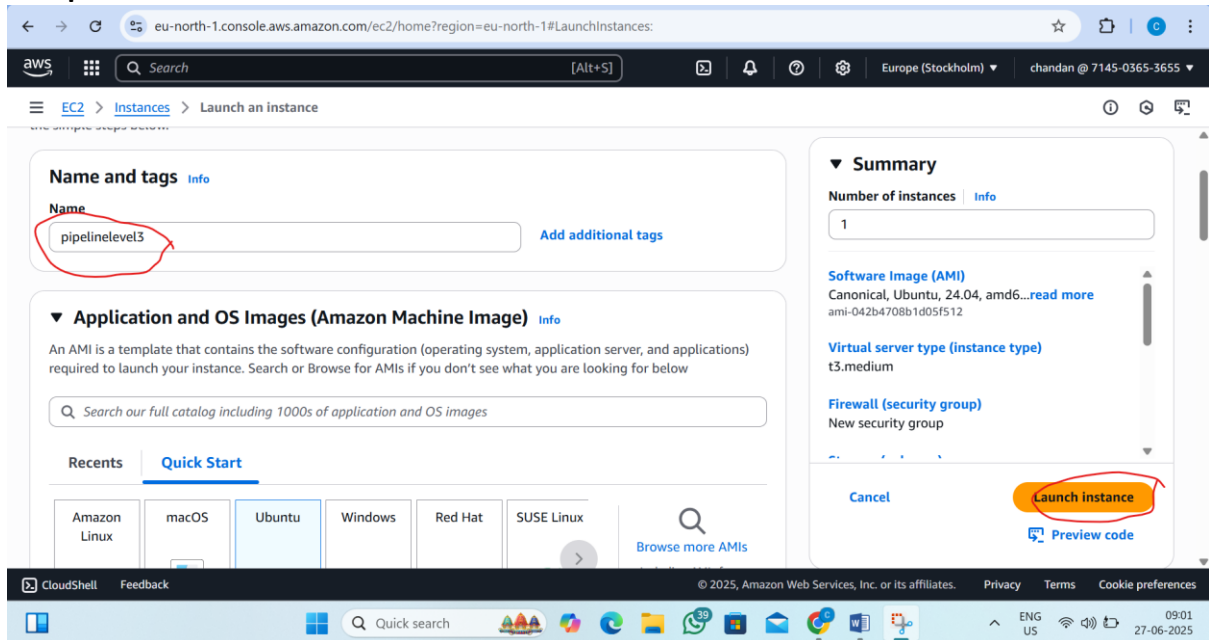
```
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar (12 kB at 1.5 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0.5/plexus-utils-3.0.5.jar (230 kB at 19 MB/s)
[INFO] Installing /var/lib/jenkins/workspace/project1/target/addressbook.war to
/var/lib/jenkins/.m2/repository/com/edurekademo/tutorial/addressbook/2.0/addressbook-2.0.war
[INFO] Installing /var/lib/jenkins/workspace/project1/pom.xml to
/var/lib/jenkins/.m2/repository/com/edurekademo/tutorial/addressbook/2.0/addressbook-2.0.pom
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 39.525 s
[INFO] Finished at: 2025-06-25T11:38:33Z
[INFO] -----
Finished: SUCCESS
```

REST API Jenkins 2.516

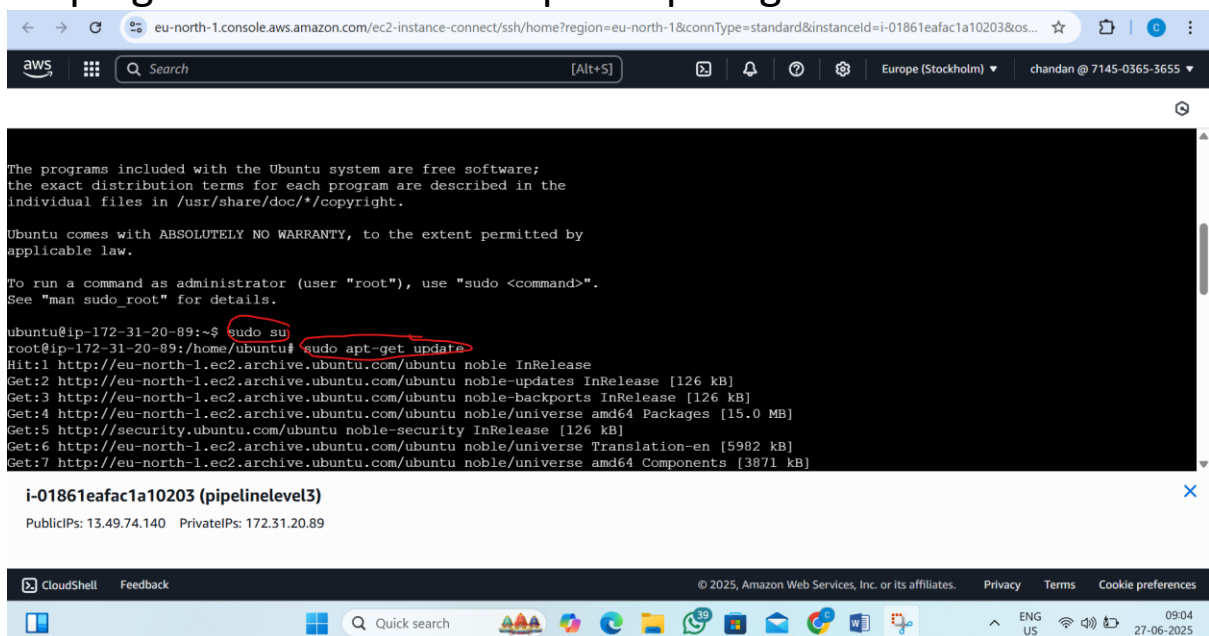
📄 Search ENG US 📄 17:12 25-06-2025

3. L3 - Create CI/CD Pipeline to Deploy the Maven Web Application in Tomcat Server

step1: launch instance and connect



Step2: go to root user and update packages



Step3:install java

```
eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-01861eafac1a10203&os=...

aws Search [Alt+S] Europe (Stockholm) chandan @ 7145-0365-3655

Reading package lists... Done
root@ip-172-31-20-89:/home/ubuntu# java --version
Command 'java' not found, but can be installed with:
apt install default-jre # version 2:1.17-75, or
apt install openjdk-17-jre-headless # version 17.0.14+7-1~24.04
apt install openjdk-21-jre-headless # version 21.0.6+7-1~24.04.1
apt install openjdk-11-jre-headless # version 11.0.26+4-1ubuntu1~24.04
apt install openjdk-8-jre-headless # version 8u442-b06-us1-0ubuntu1~24.04
apt install openjdk-19-jre-headless # version 19.0.2+7-4
apt install openjdk-20-jre-headless # version 20.0.2+9-1
apt install openjdk-22-jre-headless # version 22~22ea-1
root@ip-172-31-20-89:/home/ubuntu# sudo apt install openjdk-17-jre-headless
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
 alsa-topology-conf alsa-ucm-conf ca-certificates-java fontconfig-config fonts-dejavu-core fonts-dejavu-mono java-common libasound2-data
  libasound2t64 libavahi-client3 libavahi-common-data libavahi-common3 libcups2t64 libfontconfig1 libgraphite2-3 libharfbuzz0b
  libjpeg-turbo8 libjpeg8 liblcms2-2 libpcsclite1
Suggested packages:
  default-jre alsa-utils libasound2-plugins cups-common liblcms2-utils pcsclib libnss-mdns fonts-dejavu-extra fonts-ipafont-gothic
i-01861eafac1a10203 (pipelinelevel3)
PublicIPs: 13.49.74.140 PrivateIPs: 172.31.20.89

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Quick search ENG US 09:06 27-06-2025
```

Step4:-now install Jenkins

by the command:- **wget**

<https://raw.githubusercontent.com/akshu20791/Deployment-script/refs/heads/main/jenkins.sh>

```
eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-01861eafac1a10203&os=...

aws Search [Alt+S] Europe (Stockholm) chandan @ 7145-0365-3655

Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-20-89:/home/ubuntu# wget https://raw.githubusercontent.com/akshu20791/Deployment-script/refs/heads/main/jenkins.sh
--2025-06-27 03:40:57-- https://raw.githubusercontent.com/akshu20791/Deployment-script/refs/heads/main/jenkins.sh
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 642 [text/plain]
Saving to: 'jenkins.sh'

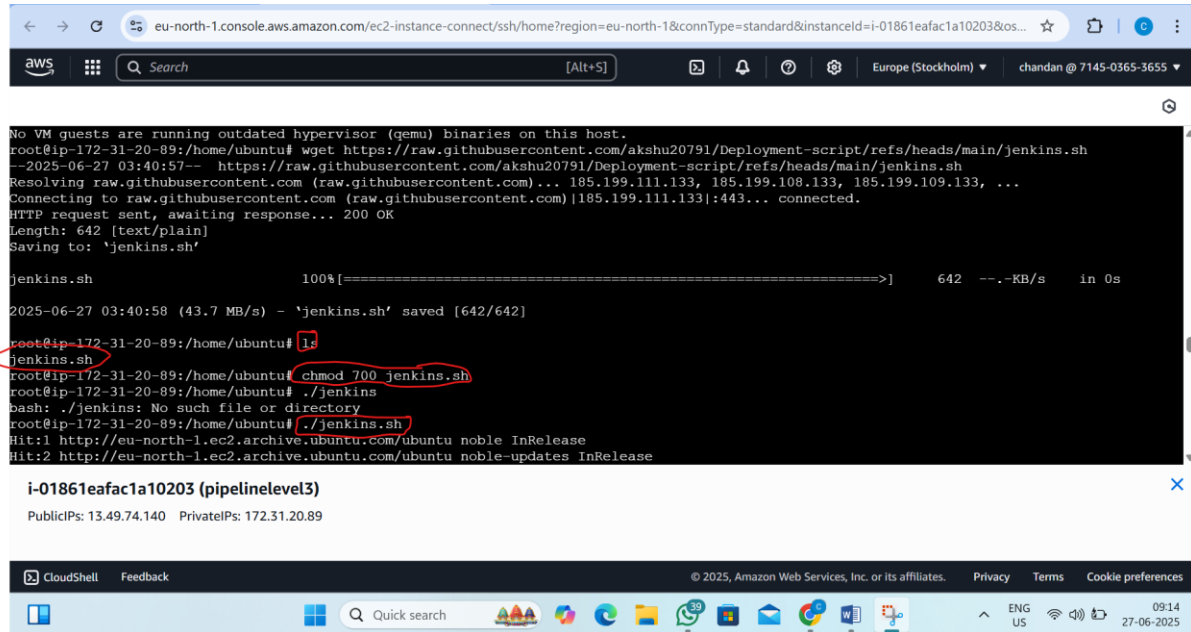
jenkins.sh          100%[=====>] 642 --.-KB/s  in 0s

2025-06-27 03:40:58 (43.7 MB/s) - 'jenkins.sh' saved [642/642]

i-01861eafac1a10203 (pipelinelevel3)
PublicIPs: 13.49.74.140 PrivateIPs: 172.31.20.89

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Quick search ENG US 09:11 27-06-2025
```


Step4:-give the permission to access and start the Jenkins by the command:1)chmod 700 jenkins.sh
2)./Jenkins.sh



```
eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-01861eafac1a10203&os=...

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-20-89:/home/ubuntu# wget https://raw.githubusercontent.com/akshu20791/Deployment-script/refs/heads/main/jenkins.sh
--2025-06-27 03:40:57-- https://raw.githubusercontent.com/akshu20791/Deployment-script/refs/heads/main/jenkins.sh
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 642 [text/plain]
Saving to: 'jenkins.sh'

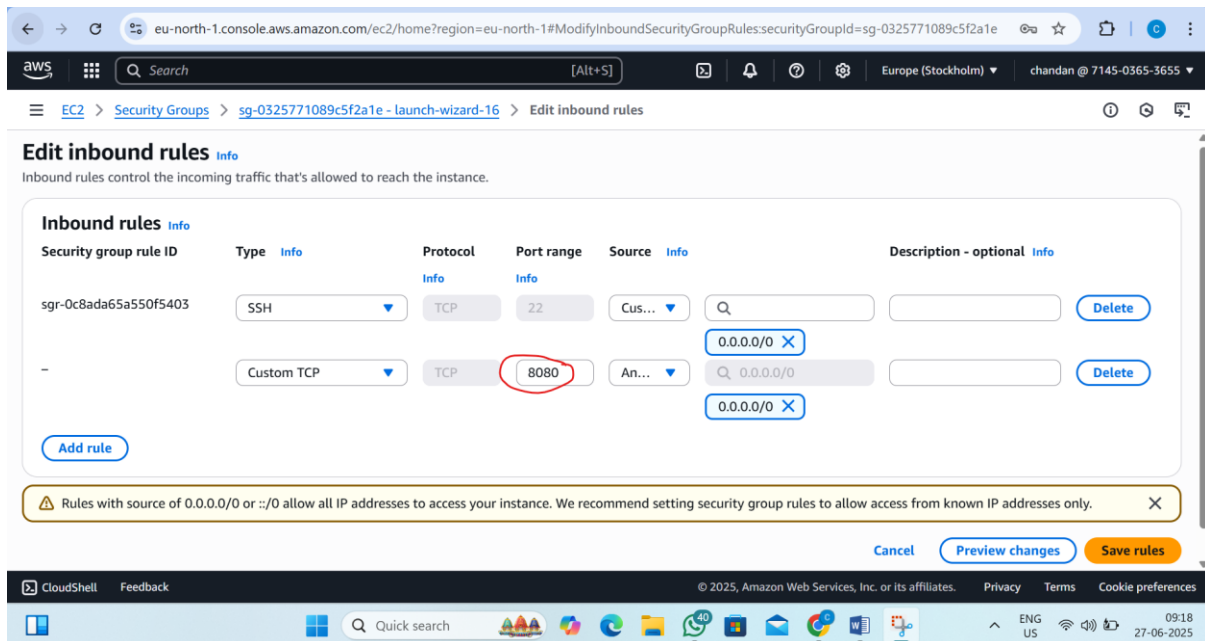
jenkins.sh                               100%[=====>]          642  --.-KB/s  in 0s

2025-06-27 03:40:58 (43.7 MB/s) - 'jenkins.sh' saved [642/642]

root@ip-172-31-20-89:/home/ubuntu# ls
jenkins.sh
root@ip-172-31-20-89:/home/ubuntu# chmod 700 jenkins.sh
root@ip-172-31-20-89:/home/ubuntu# ./jenkins
bash: ./jenkins: No such file or directory
root@ip-172-31-20-89:/home/ubuntu# ./jenkins.sh
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease

i-01861eafac1a10203 (pipelinelevel3)
PublicIPs: 13.49.74.140  PrivateIPs: 172.31.20.89
```

Step5:go to security group and edit the inbound rule
add the 8080 port in inbound rule



eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#ModifyInboundSecurityGroupRules:securityGroupId=sg-0325771089c5f2a1e

EC2 > Security Groups > sg-0325771089c5f2a1e - launch-wizard-16 > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

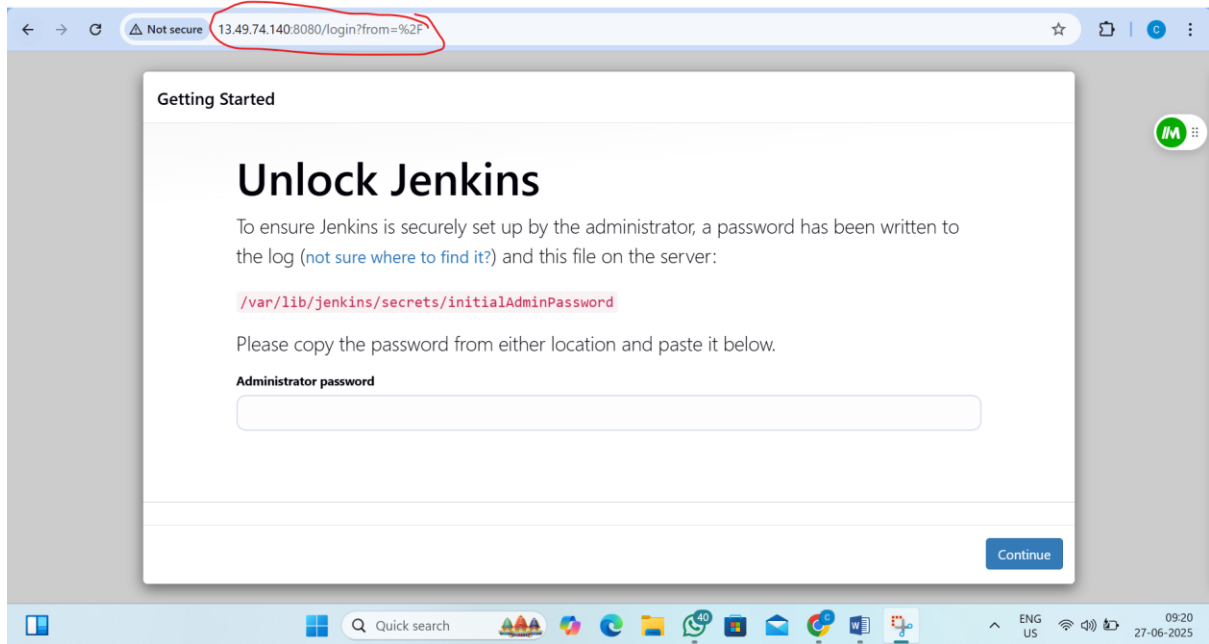
Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sg-0c8ada65a550f5403	SSH	TCP	22	Cus...		Delete
-	Custom TCP	TCP	8080	An...	0.0.0.0/0	Delete

[Add rule](#)

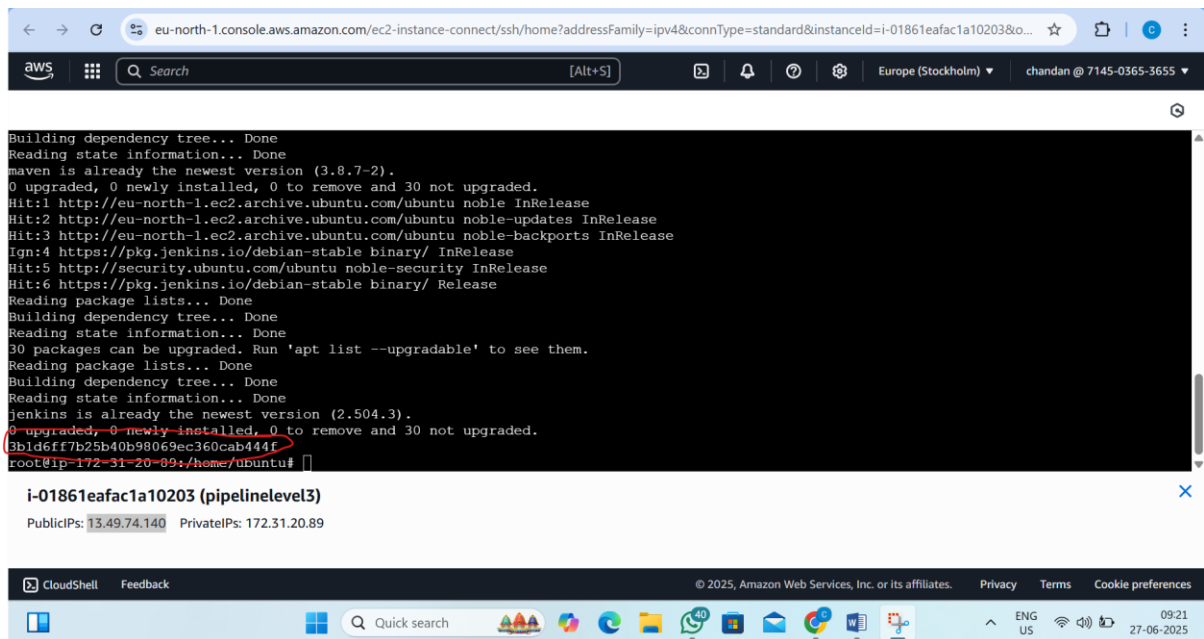
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#) [Preview changes](#) [Save rules](#)

Step6:now open the Browser and open the Jenkins

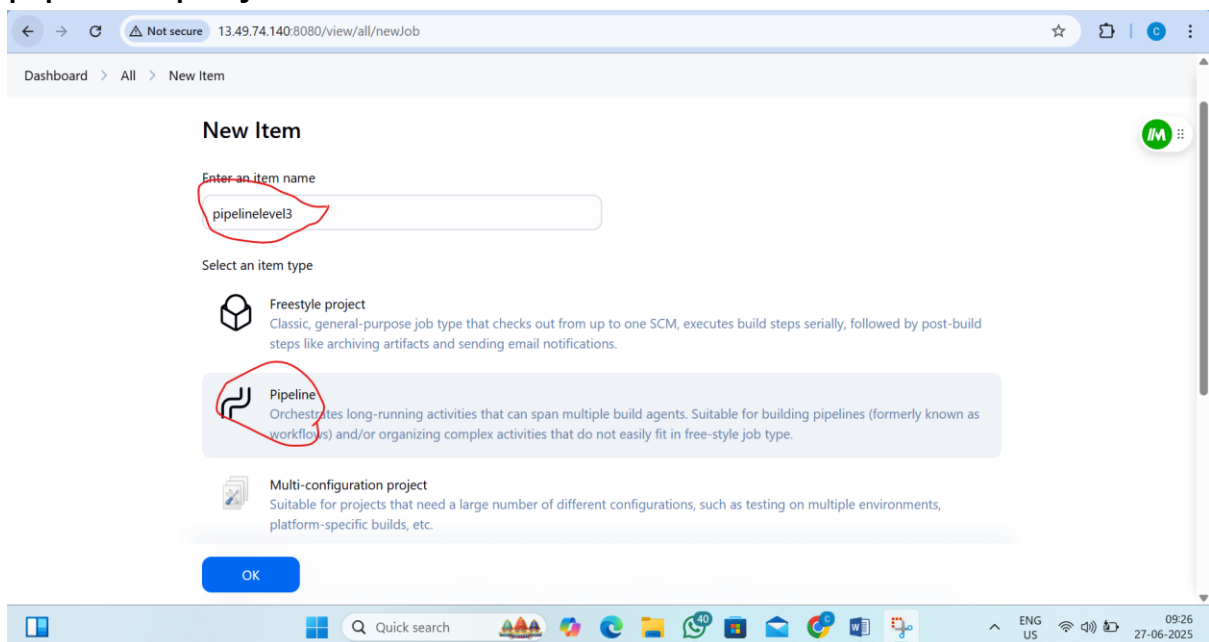


Step7:now copy the password and paste





Step8:now create a new item in Jenkins and select the pipeline project



Step9:now go to pipeline and write the script
script is :-

```
pipeline{
  agent any
  stages{
    stage("1st stage: checkout the code from github"){
      steps{
        git url: "https://github.com/ChandanYadavKS/addressbook-cicd-project"
      }
    }
    stage("2nd stage: compile the code"){
      steps{
        sh "mvn compile"
      }
    }
    stage("3rd stage: testing the code"){
      steps{
        sh "mvn test"
      }
    }
    stage("4th stage: quality assurance of the code"){
      steps{
        sh "mvn pmd:pmd"
      }
    }
    stage("5th stage: package the project"){
      steps{
        sh "mvn package"
      }
    }
  }
}
```

← → ↻ Not secure 13.49.74.140:8080/job/pipelinelevel3/configure ☆ 📄 ⓘ

Dashboard > pipelinelevel3 > Configuration

Configure

- General
- Triggers
- Pipeline
- Advanced

Script ?

```
17 }
18 }
19 stage("4th stage: quality assurance of the code"){
20 steps{
21 sh "mvn pmd:pmd"
22 }
23 }
24 stage("5th stage: package the project"){
25 steps{
26 sh "mvn package"
27 }
28 }
29 }
30 }
31
```

try sample Pipeline...

☒ Use Groovy Sandbox ?

Save Apply

Quick search ENG US 09:30 27-06-2025

← → ↻ Not secure 13.49.74.140:8080/job/pipelinelevel3/configure ☆ 📄 ⓘ

Dashboard > pipelinelevel3 > Configuration

Configure

- General
- Triggers
- Pipeline
- Advanced

Script ?

```
1 pipeline{
2 agent any
3 stages{
4 stage("1st stage: checkout the code from github"){
5 steps{
6 git url: "https://github.com/ChandanYadavKS/addressbook-cicd-project"
7 }
8 }
9 stage("2nd stage: compile the code"){
10 steps{
11 sh "mvn compile"
12 }
13 }
14 stage("3rd stage: testing the code"){
15 steps{
```

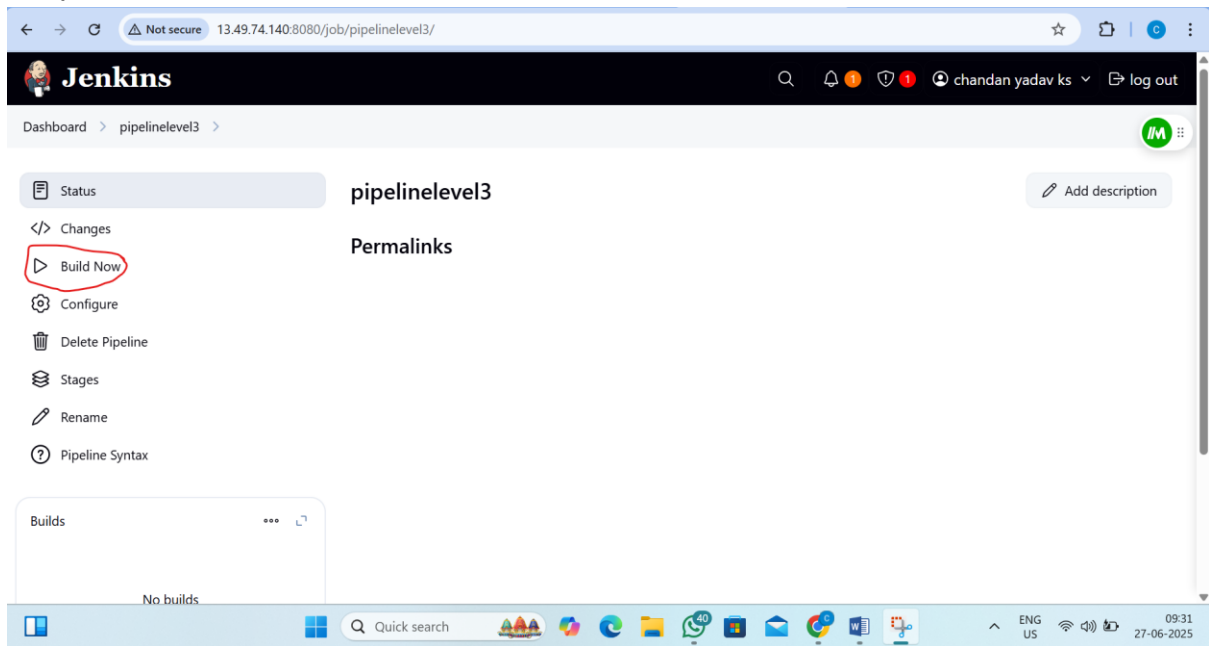
try sample Pipeline...

☒ Use Groovy Sandbox ?

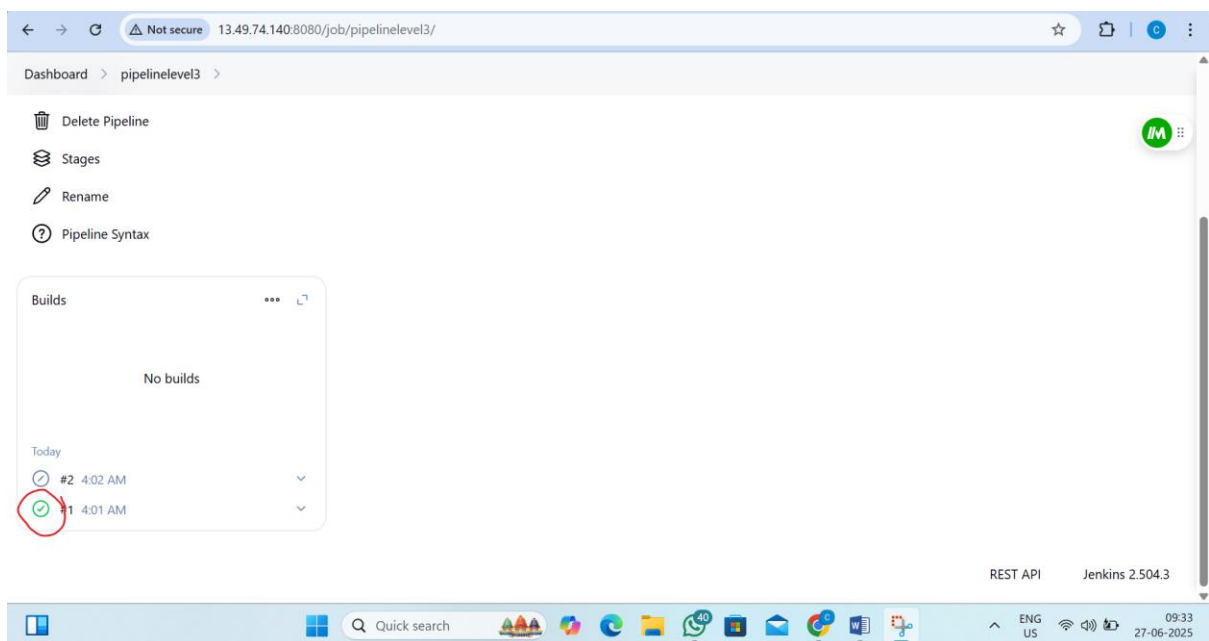
Save Apply

Quick search ENG US 09:30 27-06-2025

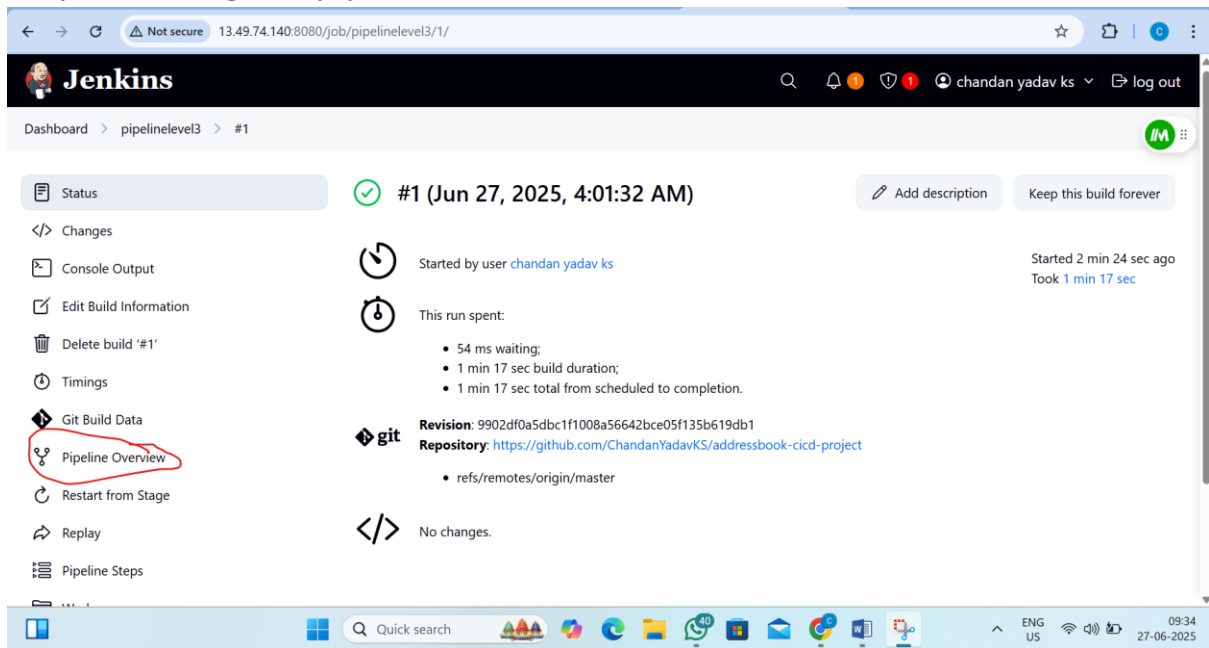
Step10 :-now build now



now click on 1

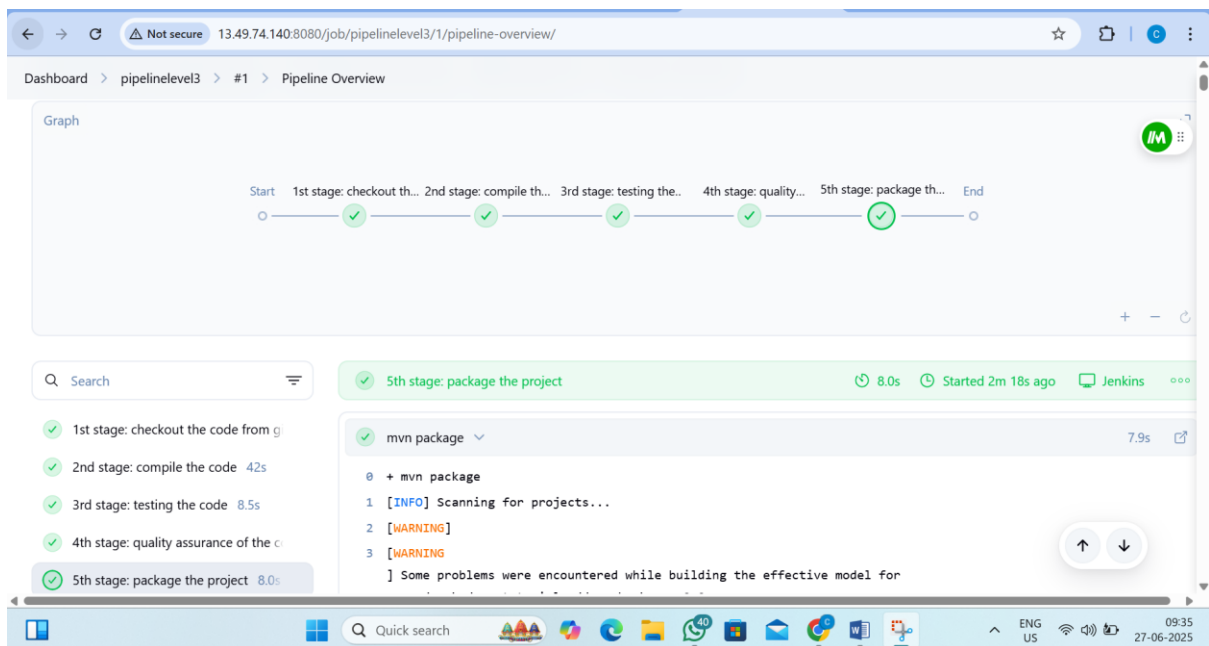


Step11:-now go to pipeline overview



The screenshot shows the Jenkins web interface for a pipeline named 'pipelinelevel3' at job '#1'. The left sidebar contains a list of actions: Status, Changes, Console Output, Edit Build Information, Delete build '#1', Timings, Git Build Data, Pipeline Overview (highlighted with a red circle), Restart from Stage, Replay, and Pipeline Steps. The main content area displays the pipeline's status as a green checkmark and the build number '#1 (Jun 27, 2025, 4:01:32 AM)'. It includes buttons for 'Add description' and 'Keep this build forever'. The build was started by user 'chandan yadav ks' and took 1 min 17 sec. The console output shows the build duration breakdown: 54 ms waiting, 1 min 17 sec build duration, and 1 min 17 sec total from scheduled to completion. The Git repository is 'https://github.com/ChandanYadavKS/addressbook-cicd-project' at revision '9902df0a5dbc1f1008a56642bce05f135b619db1'. The pipeline is currently 'No changes'.

Step12:we will see 5 stages is running

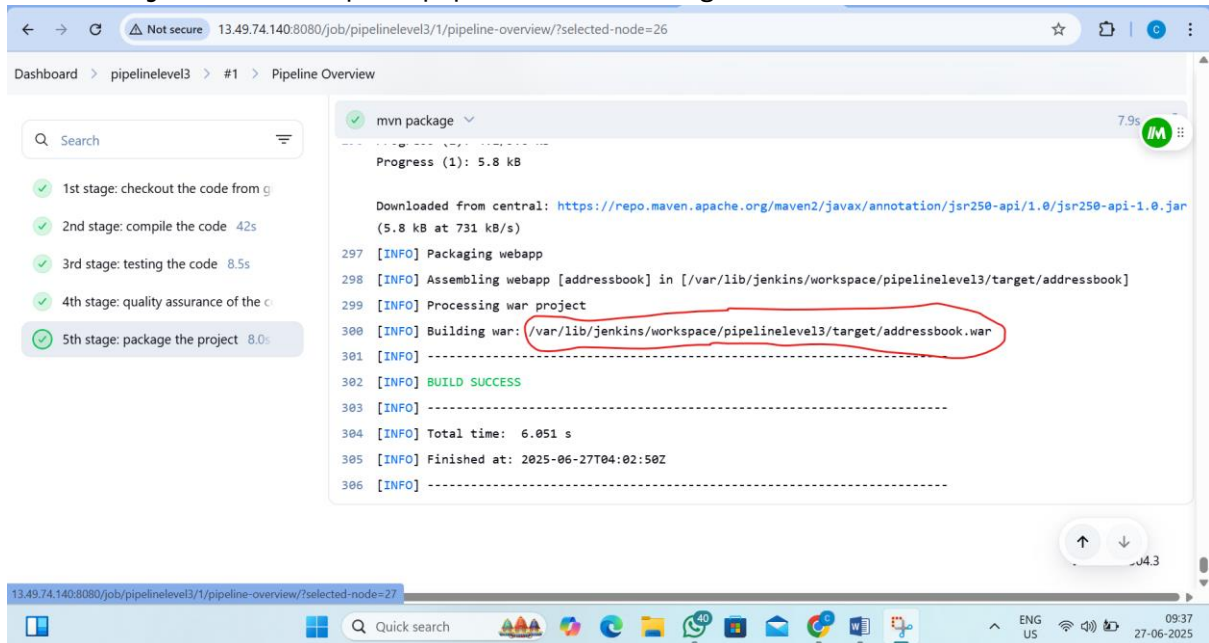


The screenshot shows the Jenkins web interface for the 'Pipeline Overview' of job '#1'. The top section displays a graph of the pipeline stages: Start, 1st stage: checkout th..., 2nd stage: compile th..., 3rd stage: testing the..., 4th stage: quality..., 5th stage: package th..., and End. All stages are marked with green checkmarks, indicating successful completion. Below the graph, a search bar is visible. The 5th stage, 'package the project', is selected and expanded, showing a duration of 8.0s and a start time of 2m 18s ago. The console output for this stage shows the command 'mvn package' and the output: '[INFO] Scanning for projects...', '[WARNING]', and '[WARNING] Some problems were encountered while building the effective model for...'. The bottom of the screen shows the Windows taskbar with various application icons and the system clock.

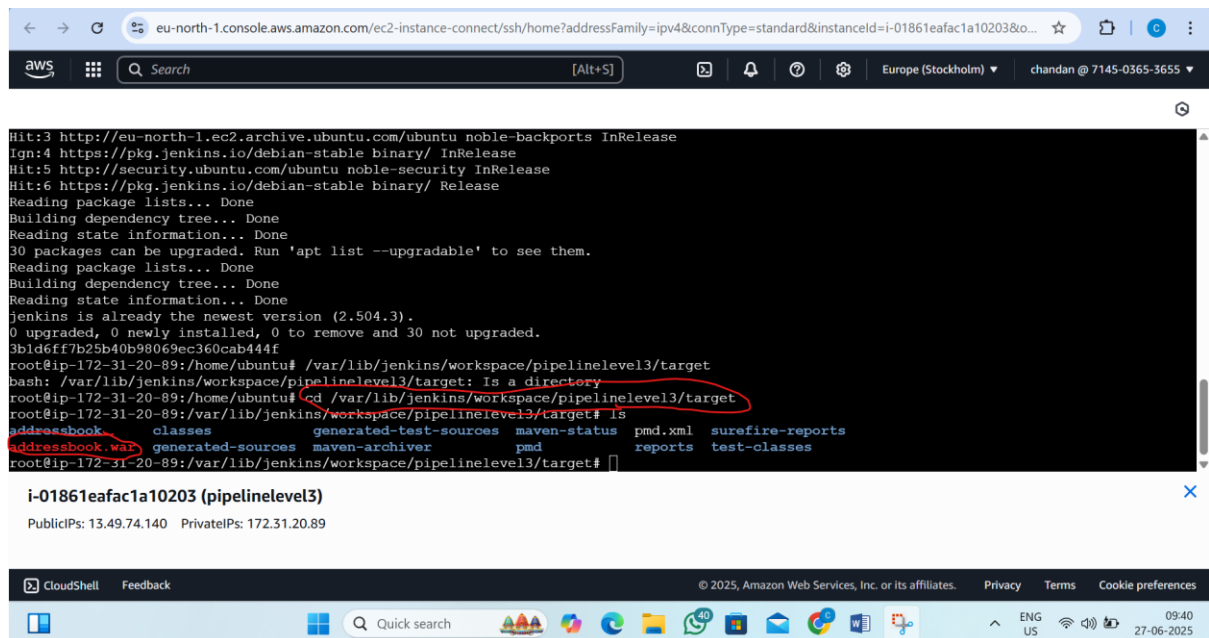
Step13:scroll down and copy the war file

this is war file:-

/var/lib/jenkins/workspace/pipelinelevel3/target/addressbook.war



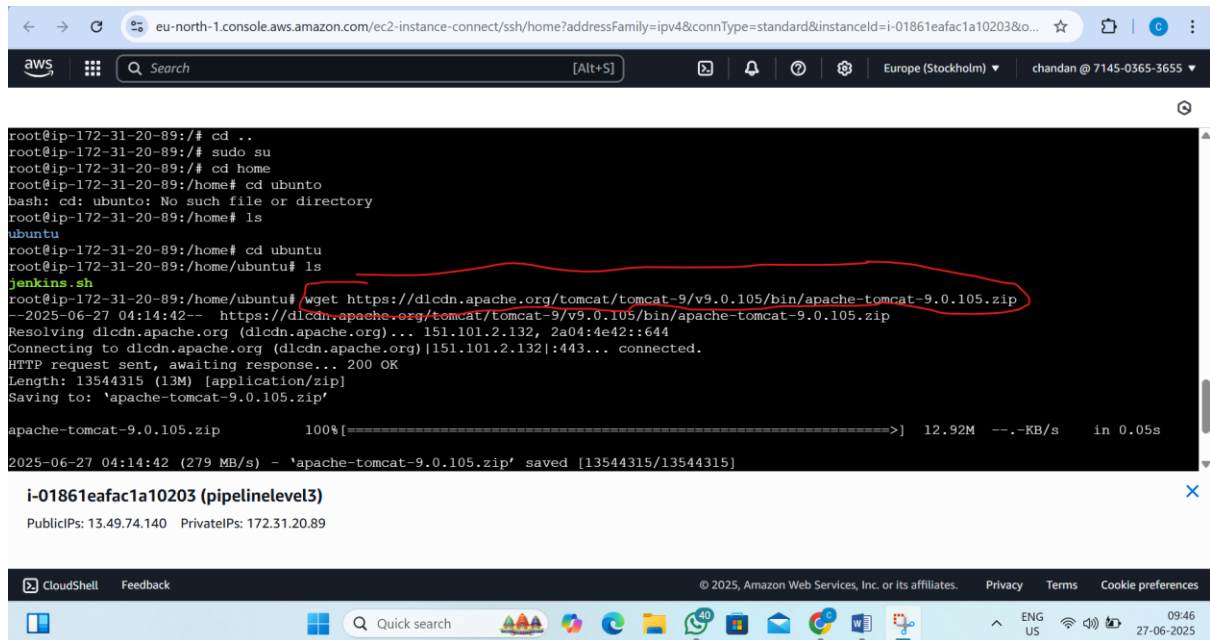
Step14:we can see the location of war file in ubuntu also



Step15:now deploy the tomcat by following the steps

i)install the tomcat in the machine

by the command:- wget <https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.105/bin/apache-tomcat-9.0.105.zip>



```
root@ip-172-31-20-89:/# cd ..
root@ip-172-31-20-89:/# sudo su
root@ip-172-31-20-89:/# cd home
root@ip-172-31-20-89:/home# cd ubuntu
bash: cd: ubuntu: No such file or directory
root@ip-172-31-20-89:/home# ls
ubuntu
root@ip-172-31-20-89:/home# cd ubuntu
root@ip-172-31-20-89:/home/ubuntu# ls
jenkins.sh
root@ip-172-31-20-89:/home/ubuntu# wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.105/bin/apache-tomcat-9.0.105.zip
--2025-06-27 04:14:42-- https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.105/bin/apache-tomcat-9.0.105.zip
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13544315 (13M) [application/zip]
Saving to: 'apache-tomcat-9.0.105.zip'

apache-tomcat-9.0.105.zip      100%[=====>] 12.92M  --.-KB/s  in 0.05s

2025-06-27 04:14:42 (279 MB/s) - 'apache-tomcat-9.0.105.zip' saved [13544315/13544315]
```

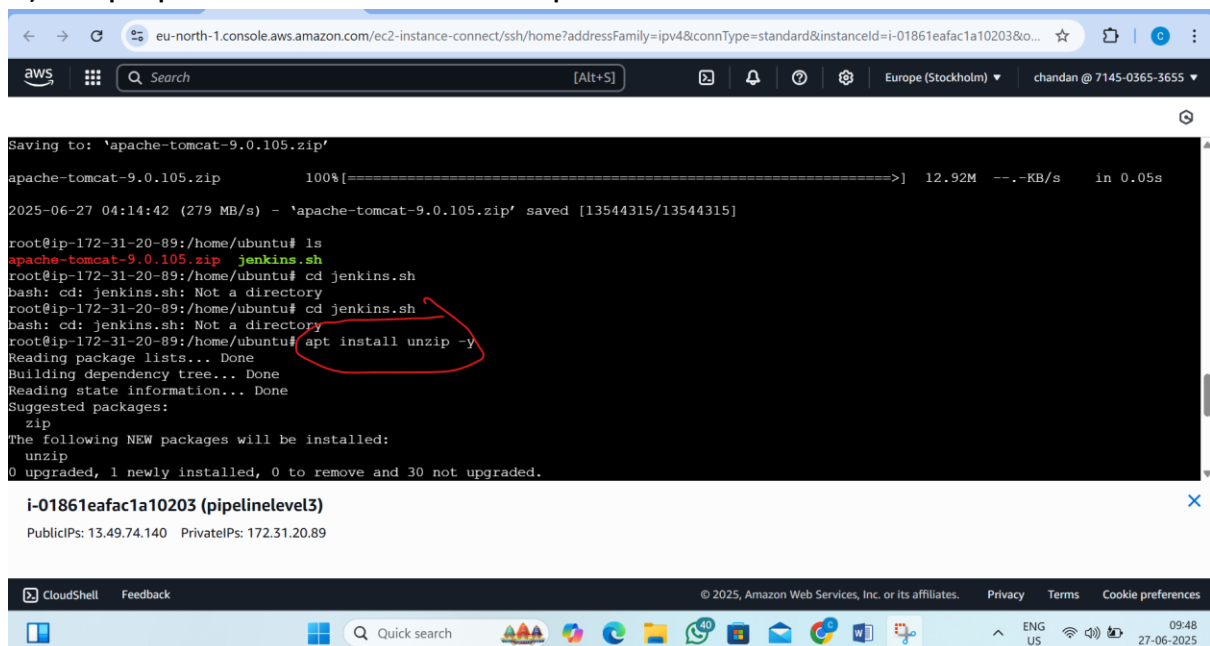
i-01861eafac1a10203 (pipelinelevel3)

PublicIPs: 13.49.74.140 PrivateIPs: 172.31.20.89

Step16:now install the unzip file and start

by the command:- 1)apt install unzip -y

2)unzip apache-tomcat-9.0.105.zip



```
Saving to: 'apache-tomcat-9.0.105.zip'

apache-tomcat-9.0.105.zip      100%[=====>] 12.92M  --.-KB/s  in 0.05s

2025-06-27 04:14:42 (279 MB/s) - 'apache-tomcat-9.0.105.zip' saved [13544315/13544315]

root@ip-172-31-20-89:/home/ubuntu# ls
apache-tomcat-9.0.105.zip jenkins.sh
root@ip-172-31-20-89:/home/ubuntu# cd jenkins.sh
bash: cd: jenkins.sh: Not a directory
root@ip-172-31-20-89:/home/ubuntu# cd jenkins.sh
bash: cd: jenkins.sh: Not a directory
root@ip-172-31-20-89:/home/ubuntu# apt install unzip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  zip
The following NEW packages will be installed:
  unzip
0 upgraded, 1 newly installed, 0 to remove and 30 not upgraded.
```

i-01861eafac1a10203 (pipelinelevel3)

PublicIPs: 13.49.74.140 PrivateIPs: 172.31.20.89

```
aws
Search [Alt+S]
Europe (Stockholm) chandan @ 7145-0365-3655

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-20-89:/home/ubuntu# ls
apache-tomcat-9.0.105.zip jenkins.sh
root@ip-172-31-20-89:/home/ubuntu# unzip apache-tomcat-9.0.105.zip
Archive:  apache-tomcat-9.0.105.zip
  creating: apache-tomcat-9.0.105/
  creating: apache-tomcat-9.0.105/bin/
  creating: apache-tomcat-9.0.105/conf/
  creating: apache-tomcat-9.0.105/lib/
  creating: apache-tomcat-9.0.105/logs/
  creating: apache-tomcat-9.0.105/temp/
  creating: apache-tomcat-9.0.105/webapps/
  creating: apache-tomcat-9.0.105/webapps/ROOT/
  creating: apache-tomcat-9.0.105/webapps/ROOT/WEB-INF/
  creating: apache-tomcat-9.0.105/webapps/docs/
  creating: apache-tomcat-9.0.105/webapps/docs/META-INF/
  creating: apache-tomcat-9.0.105/webapps/docs/WEB-INF/
  creating: apache-tomcat-9.0.105/webapps/docs/WEB-INF/jsp/
  creating: apache-tomcat-9.0.105/webapps/docs/annotationapi/
  creating: apache-tomcat-9.0.105/webapps/docs/api/

i-01861eafac1a10203 (pipelinelevel3)
PublicIPs: 13.49.74.140 PrivateIPs: 172.31.20.89

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```

step17:-now problem is Jenkins and tocat is working in same port that is 8080 so we want to change the port number of tomcat by following step

1)cd apache-tomcat-9.0.105

```
eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-01861eafac1a10203&co...
aws
Search [Alt+S]
Europe (Stockholm) chandan @ 7145-0365-3655

inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/401.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/403.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/404.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/connectorCerts.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/connectorCiphers.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/connectorTrustedCerts.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/sessionDetail.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/sessionsList.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/web.xml
inflating: apache-tomcat-9.0.105/webapps/manager/css/manager.css
inflating: apache-tomcat-9.0.105/webapps/manager/images/asf-logo.svg
inflating: apache-tomcat-9.0.105/webapps/manager/images/tomcat.svg
inflating: apache-tomcat-9.0.105/webapps/manager/index.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/status.xsd
inflating: apache-tomcat-9.0.105/webapps/manager/xform.xsl
root@ip-172-31-20-89:/home/ubuntu# ls
apache-tomcat-9.0.105 apache-tomcat-9.0.105.zip jenkins.sh
root@ip-172-31-20-89:/home/ubuntu# cd apache-tomcat-9.0.105
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105# ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105#

i-01861eafac1a10203 (pipelinelevel3)
PublicIPs: 13.49.74.140 PrivateIPs: 172.31.20.89

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Quick search
ENG US 09:52 27-06-2025
```


2)cd conf

The screenshot shows the AWS CloudShell interface. The terminal output displays the following commands and results:

```
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/connectorCiphers.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/connectorTrustedCerts.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/sessionDetail.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/sessionsList.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/web.xml
inflating: apache-tomcat-9.0.105/webapps/manager/css/manager.css
inflating: apache-tomcat-9.0.105/webapps/manager/images/asf-logo.svg
inflating: apache-tomcat-9.0.105/webapps/manager/images/tomcat.svg
inflating: apache-tomcat-9.0.105/webapps/manager/index.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/status.xsd
inflating: apache-tomcat-9.0.105/webapps/manager/xform.xsl
root@ip-172-31-20-89:/home/ubuntu# ls
apache-tomcat-9.0.105  apache-tomcat-9.0.105.zip  jenkins.sh
root@ip-172-31-20-89:/home/ubuntu# cd apache-tomcat-9.0.105
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105# cd conf
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/conf# ls
catalina.policy  context.xml  jaspic-providers.xsd  server.xml  tomcat-users.xsd
catalina.properties  jaspic-providers.xml  logging.properties  tomcat-users.xml  web.xml
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/conf#
```

Below the terminal output, the instance ID **i-01861eafac1a10203 (pipelinelevel3)** is shown, along with public and private IP addresses.

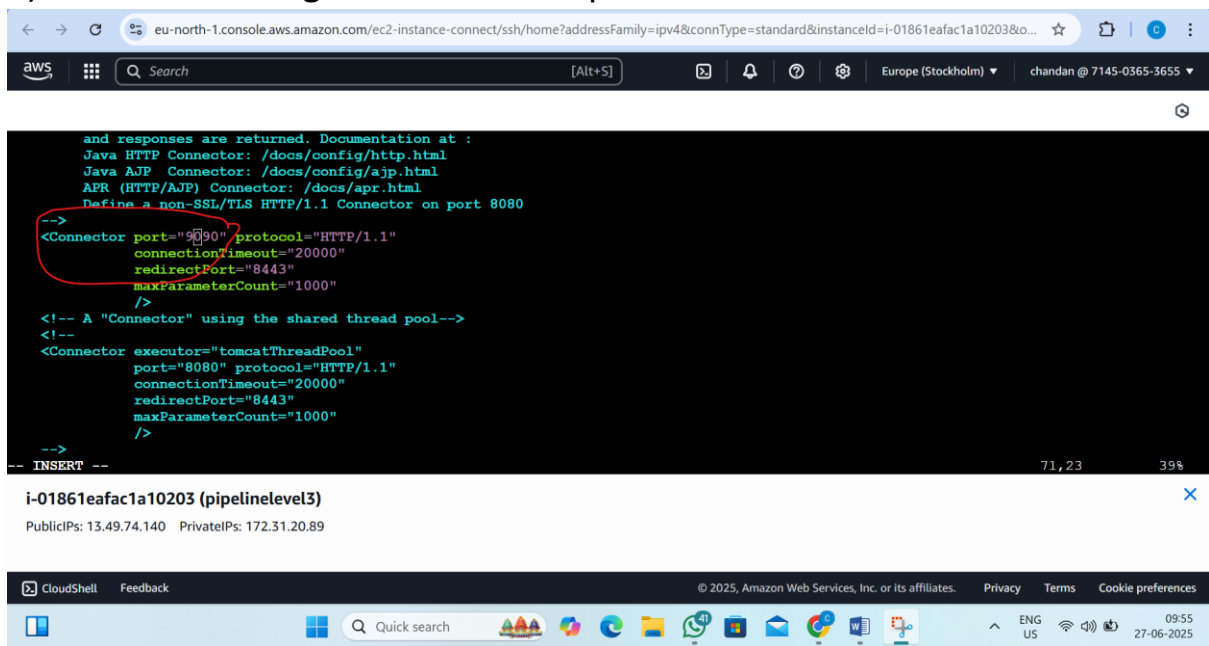
3)now edit the server.xml by the command:-vi server.xml

The screenshot shows the AWS CloudShell interface. The terminal output displays the following commands and results:

```
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/connectorTrustedCerts.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/sessionDetail.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/jsp/sessionsList.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/WEB-INF/web.xml
inflating: apache-tomcat-9.0.105/webapps/manager/css/manager.css
inflating: apache-tomcat-9.0.105/webapps/manager/images/asf-logo.svg
inflating: apache-tomcat-9.0.105/webapps/manager/images/tomcat.svg
inflating: apache-tomcat-9.0.105/webapps/manager/index.jsp
inflating: apache-tomcat-9.0.105/webapps/manager/status.xsd
inflating: apache-tomcat-9.0.105/webapps/manager/xform.xsl
root@ip-172-31-20-89:/home/ubuntu# ls
apache-tomcat-9.0.105  apache-tomcat-9.0.105.zip  jenkins.sh
root@ip-172-31-20-89:/home/ubuntu# cd apache-tomcat-9.0.105
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105# cd conf
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/conf# ls
catalina.policy  context.xml  jaspic-providers.xsd  server.xml  tomcat-users.xsd
catalina.properties  jaspic-providers.xml  logging.properties  tomcat-users.xml  web.xml
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/conf# vi server.xml
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/conf#
```

Below the terminal output, the instance ID **i-01861eafac1a10203 (pipelinelevel3)** is shown, along with public and private IP addresses.

4)now inside change the connector port 8080 to 9090



```
and responses are returned. Documentation at :
Java HTTP Connector: /docs/config/http.html
Java AJP Connector: /docs/config/ajp.html
APR (HTTP/AJP) Connector: /docs/apr.html
Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
-->
<Connector port="9090" protocol="HTTP/1.1"
  connectionTimeout="20000"
  redirectPort="8443"
  maxParameterCount="1000"
/>
<!-- A "Connector" using the shared thread pool-->
<!--
<Connector executor="tomcatThreadPool"
  port="8080" protocol="HTTP/1.1"
  connectionTimeout="20000"
  redirectPort="8443"
  maxParameterCount="1000"
/>
-->
-- INSERT --
```

i-01861eafac1a10203 (pipelinelevel3)

PublicIPs: 13.49.74.140 PrivateIPs: 172.31.20.89

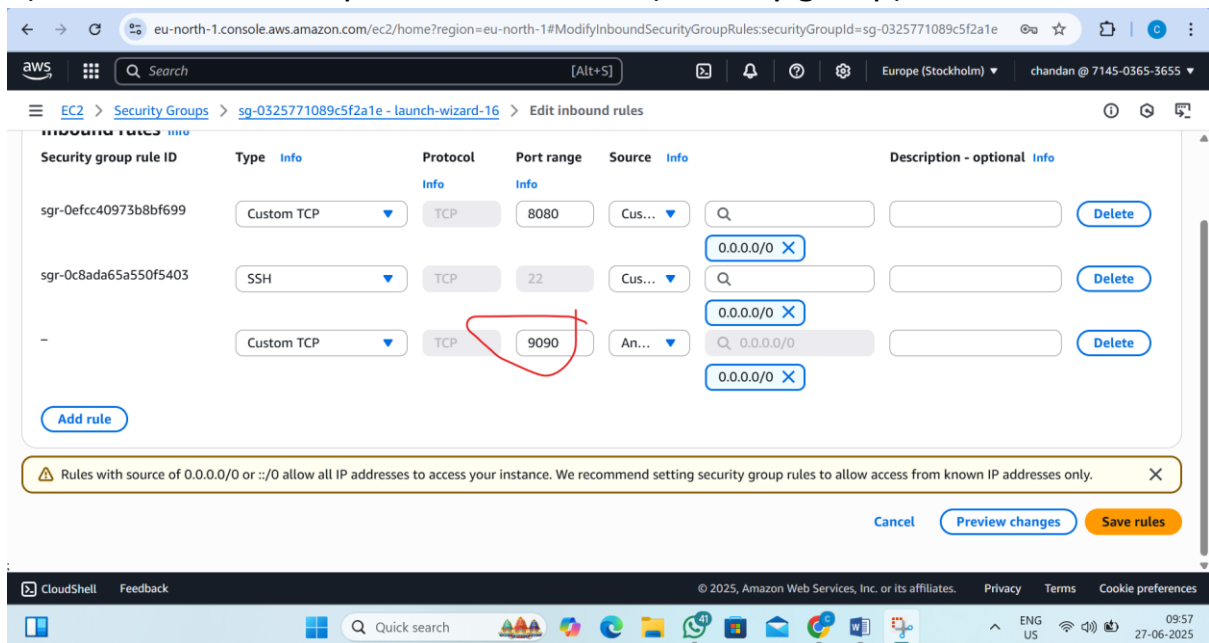
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5)now add the 9090 port in inbound rule(security group)



Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-0efcc40973b8bf699	Custom TCP	TCP	8080	Cus...	
sgr-0c8ada65a550f5403	SSH	TCP	22	Cus...	
-	Custom TCP	TCP	9090	An...	

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes Save rules

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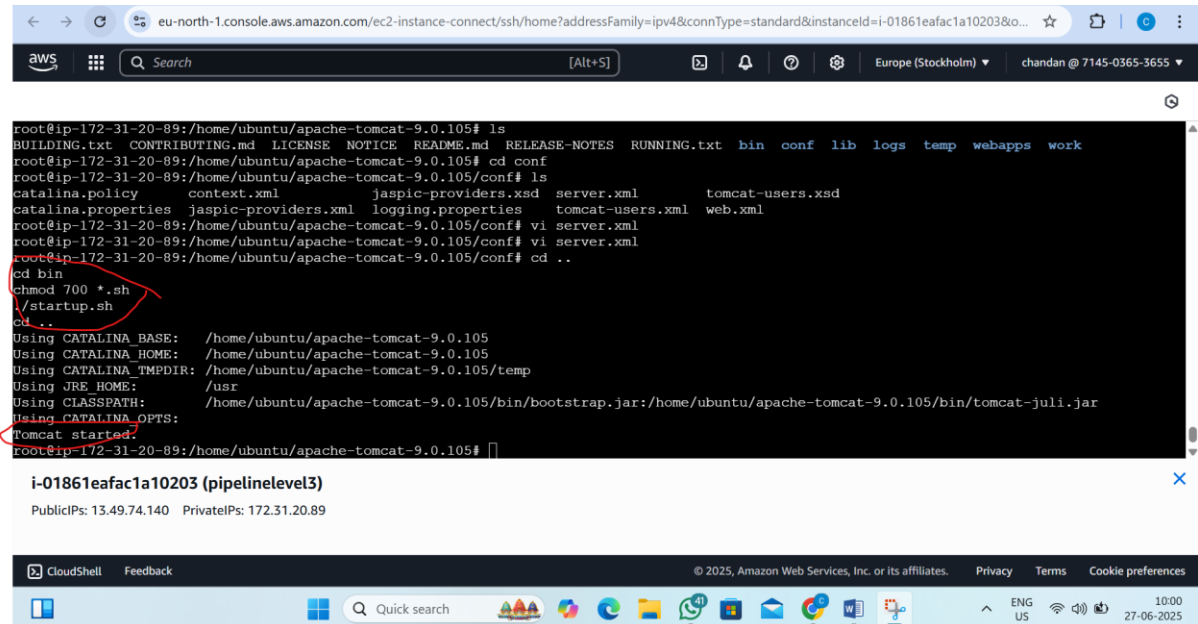
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6) now we will run the tomcat on 9090 port
first pass the command in ubuntu to start the tomcat

```
cd ..  
cd bin  
chmod 700 *.sh  
./startup.sh  
cd ..
```

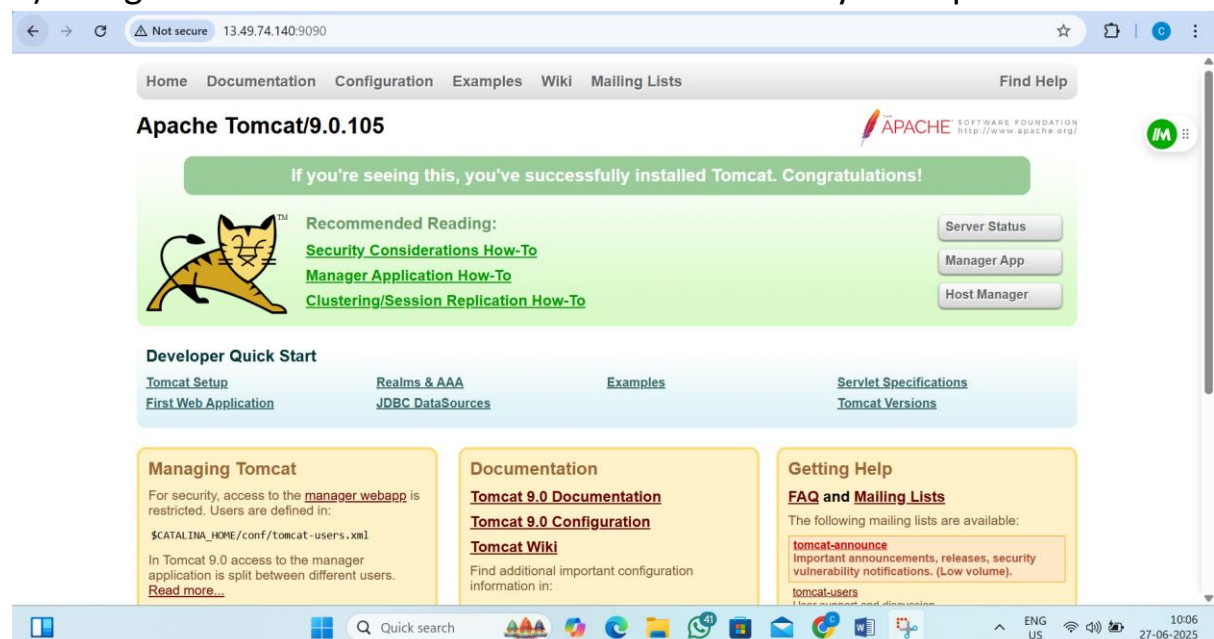


The screenshot shows the AWS CloudShell interface. The terminal window displays the following commands and output:

```
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105# ls  
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work  
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105# cd conf  
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/conf# ls  
catalina.policy context.xml jaspic-providers.xsd server.xml tomcat-users.xsd  
catalina.properties jaspic-providers.xml logging.properties tomcat-users.xml web.xml  
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/conf# vi server.xml  
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/conf# cd ..  
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105# cd bin  
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/bin# chmod 700 *.sh  
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/bin# ./startup.sh  
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/bin# cd ..  
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105#  
Using CATALINA_BASE:   /home/ubuntu/apache-tomcat-9.0.105  
Using CATALINA_HOME:   /home/ubuntu/apache-tomcat-9.0.105  
Using CATALINA_TMPDIR: /home/ubuntu/apache-tomcat-9.0.105/temp  
Using JRE_HOME:        /usr  
Using CLASSPATH:        /home/ubuntu/apache-tomcat-9.0.105/bin/bootstrap.jar:/home/ubuntu/apache-tomcat-9.0.105/bin/tomcat-juli.jar  
Tomcat started.  
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105#
```

Below the terminal, the instance ID **i-01861eafac1a10203 (pipelinelevel3)** is shown, along with Public IPs (13.49.74.140) and Private IPs (172.31.20.89).

7) now go to browser and start the tomcat server by 9090 port



Step18:now go to Jenkins job ->configure and update the code
code is :-

```
pipeline {
    agent any

    stages {
        stage("1st stage: checkout the code from github") {
            steps {
                git url: "https://github.com/ChandanYadavKS/addressbook-cicd-project"
            }
        }

        stage("2nd stage: compile the code") {
            steps {
                sh "mvn compile"
            }
        }

        stage("3rd stage: testing the code") {
            steps {
                sh "mvn test"
            }
        }

        stage("4th stage: quality assurance of the code") {
            steps {
                sh "mvn pmd:pmd"
            }
        }

        stage("5th stage: package the project") {
            steps {
                sh "mvn package"
            }
        }

        stage("6th stage: deploy the project") {
            steps {
                sh "sudo cp /var/lib/jenkins/workspace/pipelinelevel3/target/addressbook.war /home/ubuntu/apache-tomcat-9.0.105/webapps"
            }
        }
    }
}
```

Dashboard > pipelinelevel3 > Configuration

Configure

- General
- Triggers
- Pipeline
- Advanced

```
21 sh "mvn pmd:pmd"
22 }
23 }
24 stage("5th stage: package the project"){
25 steps{
26 sh "mvn package"
27 }
28 }
29 stage("6th stage :depoly the project"){
30 sh "sudo cp /var/lib/jenkins/workspace/pipelinelevel3/target/addressbook.war /home/ubuntu/s
31 }
32 }
33 }
34 }
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Save Apply

step19:now give the sudo permission to a user in linux
by the command:-visudo

eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-01861eafac1a10203&o...

Search [Alt+S] Europe (Stockholm) chandan @ 7145-0365-3655

```
catalina.bat commons-daemon-native.tar.gz daemon.sh makebase.sh shutdown.sh tomcat-native.tar.gz version.sh
catalina.sh commons-daemon.jar digest.bat setclasspath.bat startup.bat tool-wrapper.bat
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/bin# sh shutdown.sh
Using CATALINA_BASE: /home/ubuntu/apache-tomcat-9.0.105
Using CATALINA_HOME: /home/ubuntu/apache-tomcat-9.0.105
Using CATALINA_TMPDIR: /home/ubuntu/apache-tomcat-9.0.105/temp
Using JRE_HOME: /usr
Using CLASSPATH: /home/ubuntu/apache-tomcat-9.0.105/bin/bootstrap.jar:/home/ubuntu/apache-tomcat-9.0.105/bin/tomcat-juli.jar
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS: --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=
s=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=
java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/bin# sh startup.sh
Using CATALINA_BASE: /home/ubuntu/apache-tomcat-9.0.105
Using CATALINA_HOME: /home/ubuntu/apache-tomcat-9.0.105
Using CATALINA_TMPDIR: /home/ubuntu/apache-tomcat-9.0.105/temp
Using JRE_HOME: /usr
Using CLASSPATH: /home/ubuntu/apache-tomcat-9.0.105/bin/bootstrap.jar:/home/ubuntu/apache-tomcat-9.0.105/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@ip-172-31-20-89:/home/ubuntu/apache-tomcat-9.0.105/bin# visudo
```

i-01861eafac1a10203 (pipelinelevel3)

PublicIPs: 13.49.74.140 PrivateIPs: 172.31.20.89

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eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-01861eafac1a10203&o...

```
GNU nano 7.2 /etc/sudoers.tmp *
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"
# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"
# Host alias specification
# User alias specification
# Cmnd alias specification
# User privilege specification
root ALL=(ALL:ALL) ALL
jenkins ALL=(ALL:ALL) NOPASSWD:ALL
#Members of the admin group may gain root privileges
admin ALL=(ALL) ALL

Help Write Out Where Is Cut Execute Location Undo Set Mark To Bracket
Exit Read File Replace Paste Justify Go To Line Redo Copy Where Was
```

i-01861eafac1a10203 (pipelinelevel3)
PublicIPs: 13.49.74.140 PrivateIPs: 172.31.20.89

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Step20:now restart the Jenkins and build the job by the command:./Jenkins.sh

Not secure 13.49.74.140:8080/job/pipelinelevel3/

Dashboard > pipelinelevel3 >

Status

Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

pipelinelevel3

Add description

Permalinks

- Last build (#5), 4 min 16 sec ago
- Last stable build (#5), 4 min 16 sec ago
- Last successful build (#5), 4 min 16 sec ago
- Last failed build (#3), 8 min 12 sec ago
- Last unsuccessful build (#3), 8 min 12 sec ago
- Last completed build (#5), 4 min 16 sec ago

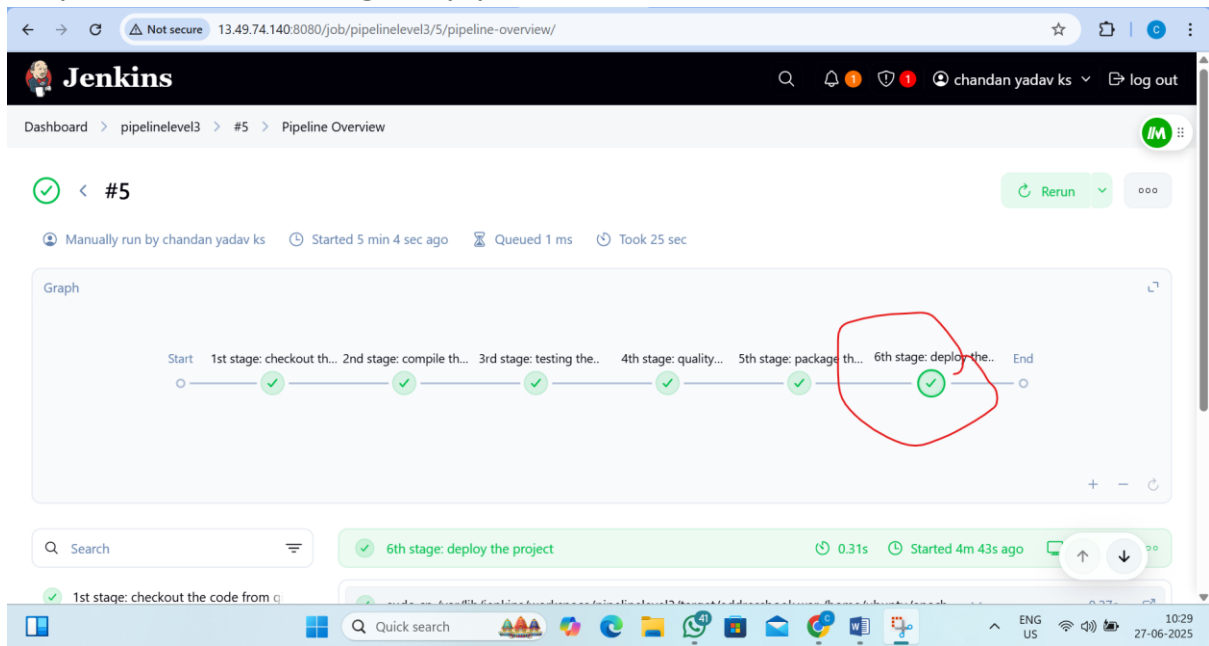
Builds

Filter

Today

- #5 4:54 AM
- #4 4:52 AM

Step21:see the 6th stage in pipeline overview



The screenshot shows the Jenkins Pipeline Overview for a pipeline named 'pipelinelevel3'. The pipeline is currently at stage #5. The stages are: 1st stage: checkout th..., 2nd stage: compile th..., 3rd stage: testing the..., 4th stage: quality..., 5th stage: package th..., and 6th stage: deploy the... The 6th stage is highlighted with a red circle. The pipeline status is 'Completed' (indicated by a green checkmark). The pipeline was manually run by 'chandan yadav ks' and started 5 min 4 sec ago. The 6th stage '6th stage: deploy the project' is shown as completed with a duration of 0.31s and started 4m 43s ago.

Dashboard > pipelinelevel3 > #5 > Pipeline Overview

✓ < #5

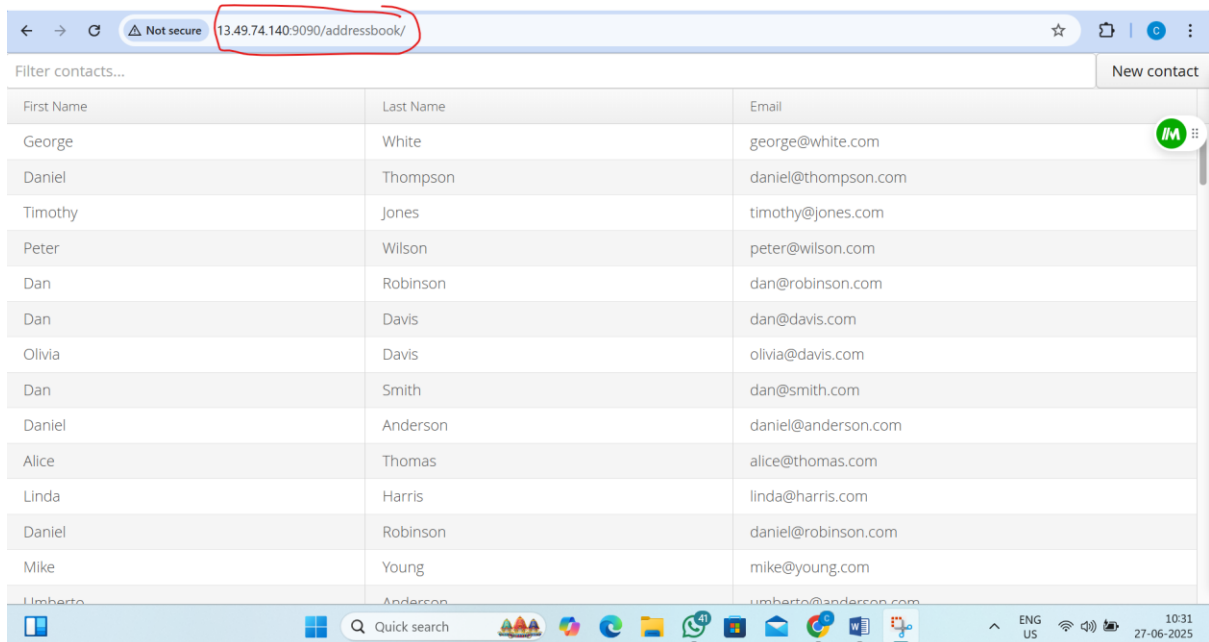
Manually run by chandan yadav ks Started 5 min 4 sec ago Queued 1 ms Took 25 sec

Graph

Start 1st stage: checkout th... 2nd stage: compile th... 3rd stage: testing the... 4th stage: quality... 5th stage: package th... 6th stage: deploy the... End

6th stage: deploy the project 0.31s Started 4m 43s ago

Step22:now see the output in browser



The screenshot shows a web browser displaying an addressbook application. The URL is '13.49.74.140:9090/addressbook/'. The application has a search bar and a 'New contact' button. The contact list is displayed in a table with columns: First Name, Last Name, and Email. The contacts are listed in descending order of email address.

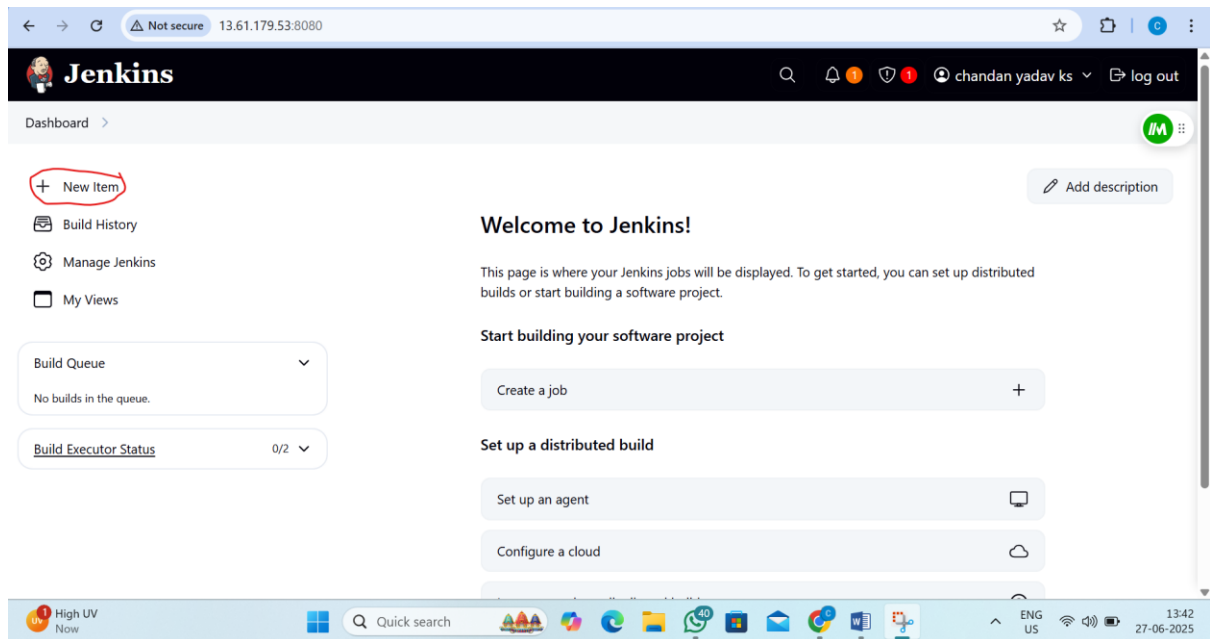
Filter contacts... New contact

First Name	Last Name	Email
George	White	george@white.com
Daniel	Thompson	daniel@thompson.com
Timothy	Jones	timothy@jones.com
Peter	Wilson	peter@wilson.com
Dan	Robinson	dan@robinson.com
Dan	Davis	dan@davis.com
Olivia	Davis	olivia@davis.com
Dan	Smith	dan@smith.com
Daniel	Anderson	daniel@anderson.com
Alice	Thomas	alice@thomas.com
Linda	Harris	linda@harris.com
Daniel	Robinson	daniel@robinson.com
Mike	Young	mike@young.com
Umberto	Anderson	umberto@anderson.com

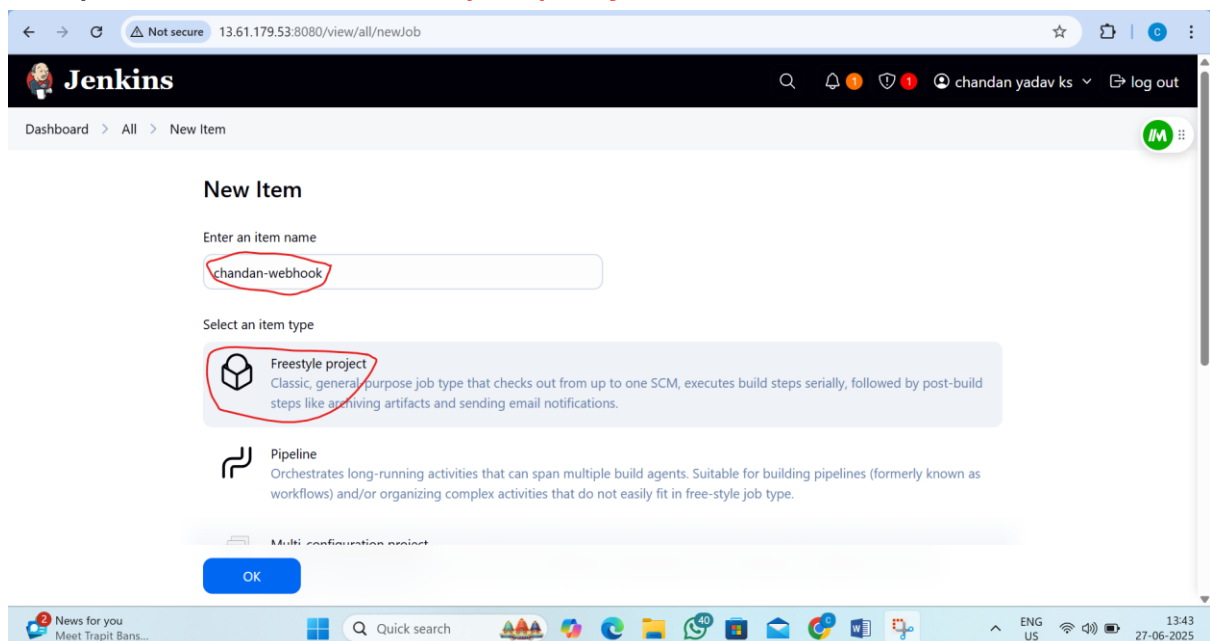
L4 - Automate the CI/CD Pipeline using GitHub webhook and Poll-SCM

GitHub web hook

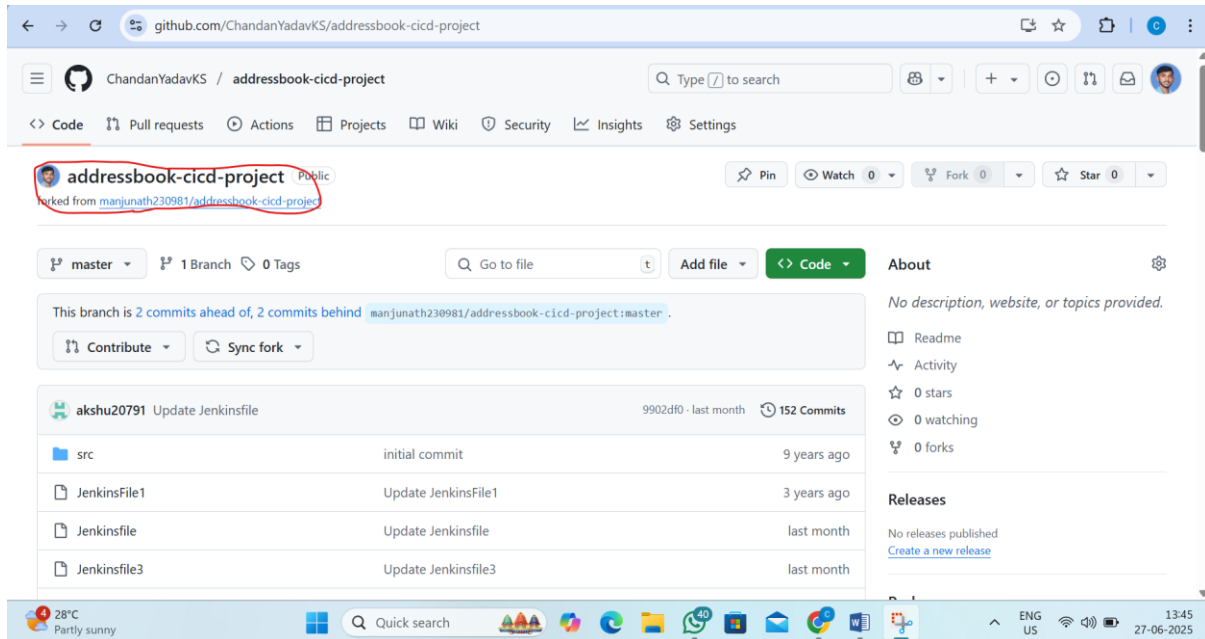
Step1:create a new item



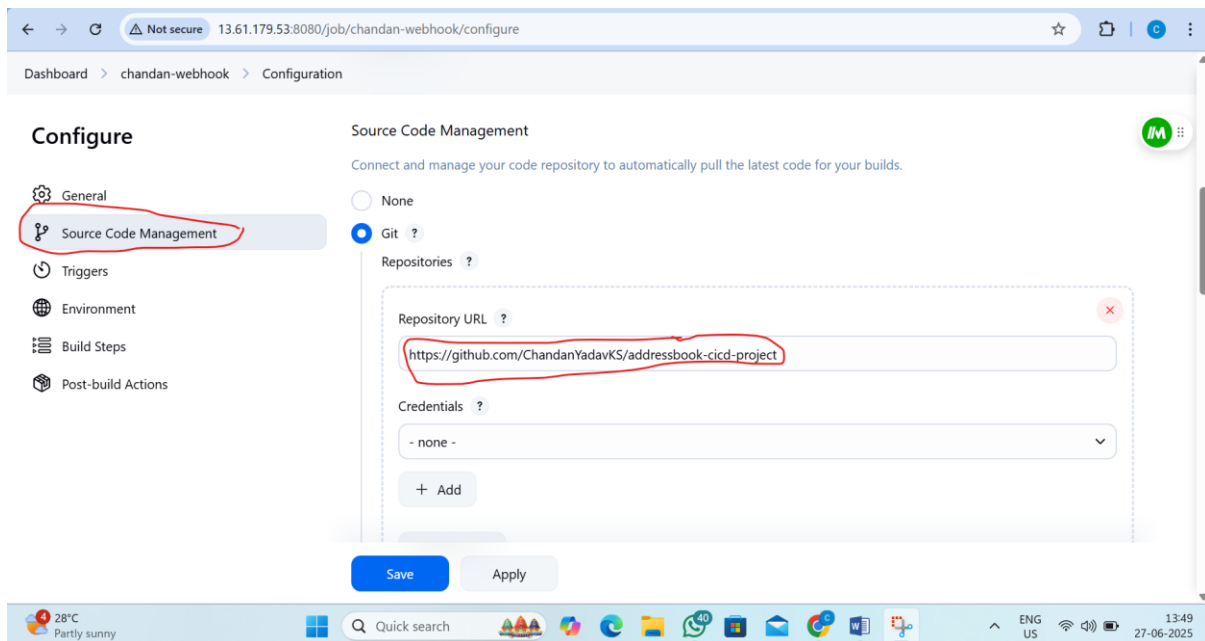
Step2:create a freestyle project



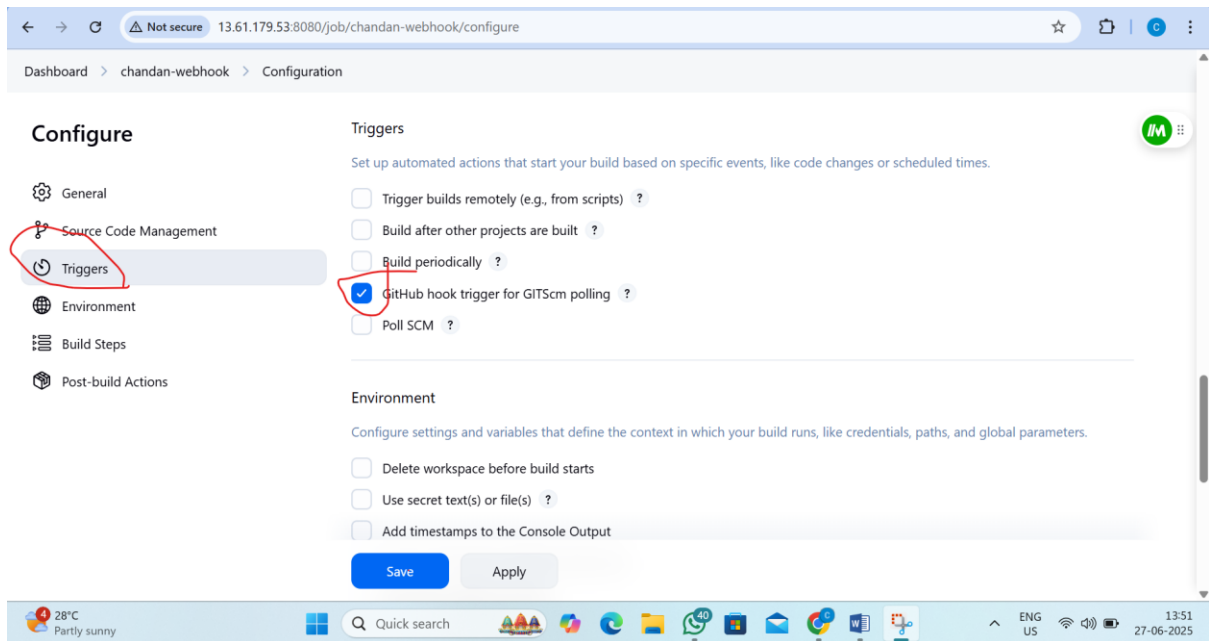
Step3 :now go to github repository and copy the repository



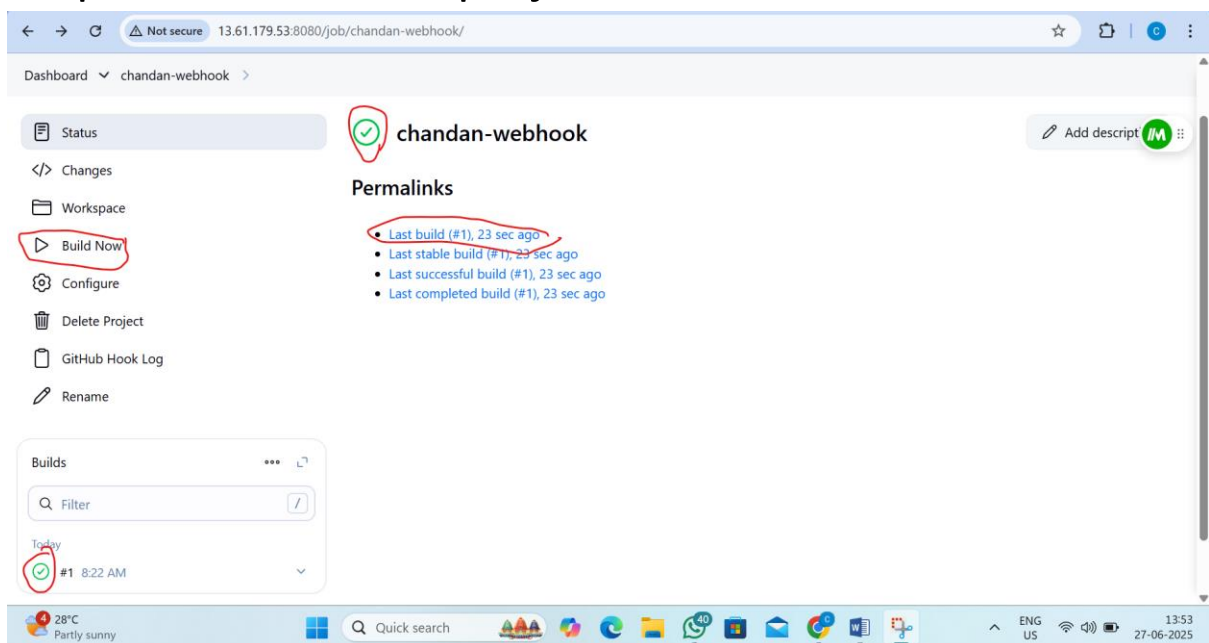
Step4:go to Jenkins



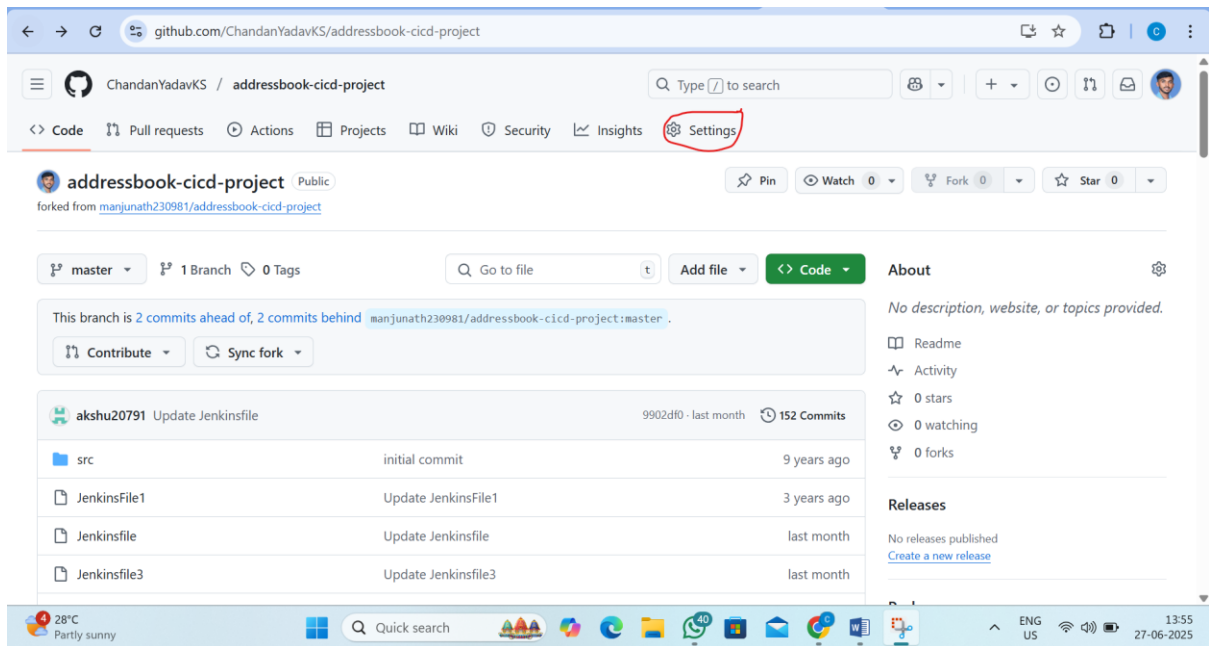
Step5:now go to **triggers** and select the **webhook**



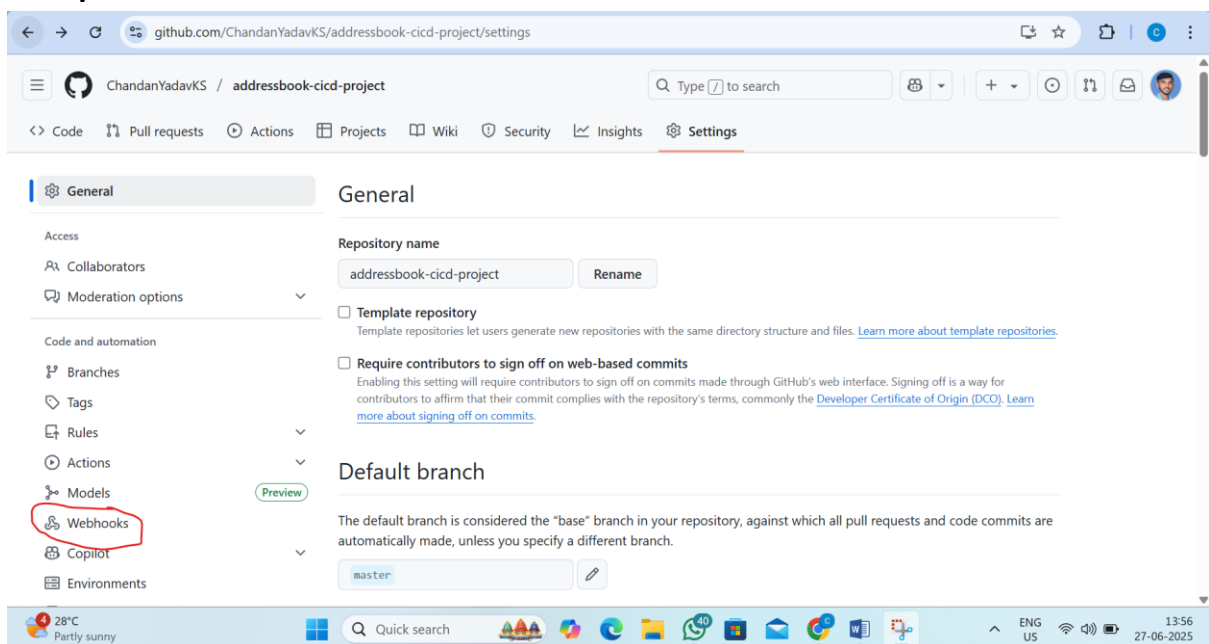
Step6:now build the project



Step7:now go to **github repository** and click on **setting**



Step8:now click on **webhook**



Step9:now add webhook

The screenshot shows the 'Add webhook' form in the GitHub settings for the 'addressbook-cicd-project'. The 'Webhooks' tab is selected. The form includes the following fields and options:

- Payload URL ***: `http://13.61.179.53:8080/github-webhook/`
- Content type ***: `application/json`
- Secret**: (Empty text input field)
- SSL verification**: ☒ By default, we verify SSL certificates when delivering payloads.

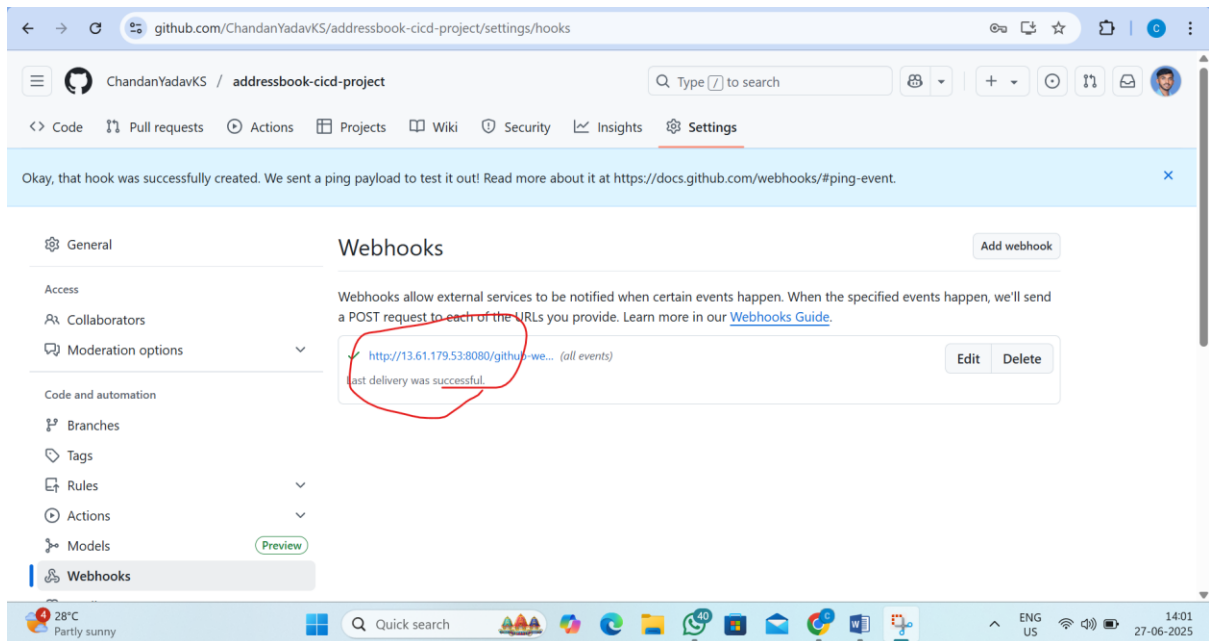
The 'Add webhook' button is located at the bottom of the form.

Step10:And select send everything

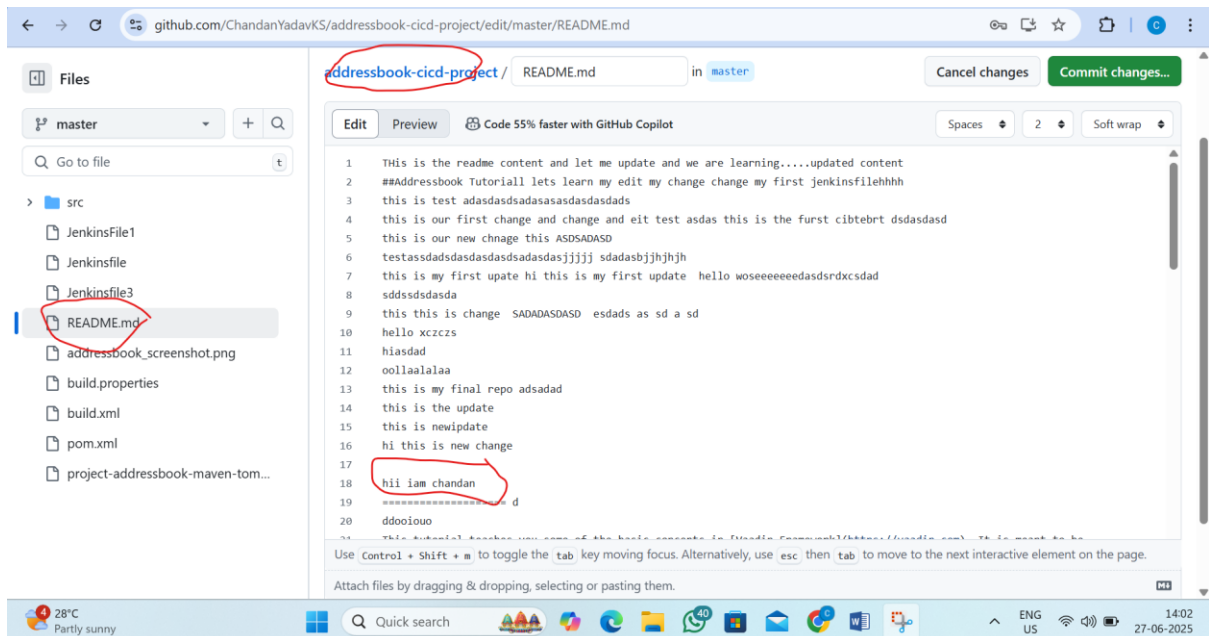
The screenshot shows the 'Add webhook' form with the 'Webhooks' tab selected. The 'Which events would you like to trigger this webhook?' section is expanded, showing the following options:

- ☒ **Send me everything**
- ☐ Just the push event.
- ☐ Let me select individual events.

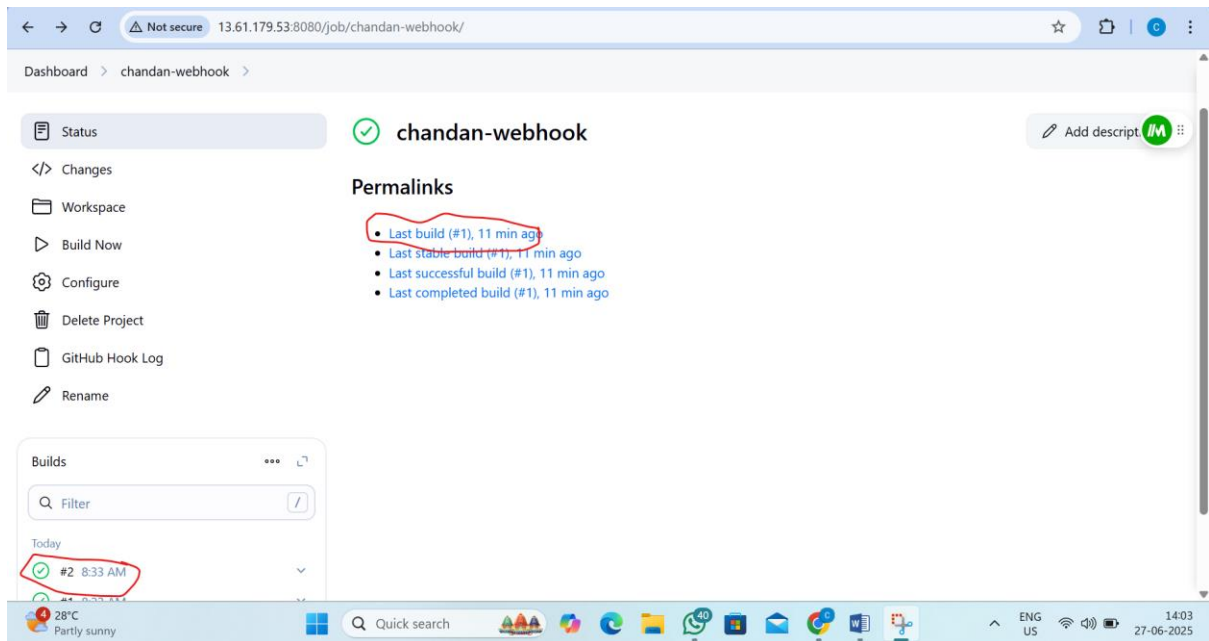
The 'Active' checkbox is checked, and the 'Add webhook' button is highlighted at the bottom of the form.



Step11:now go to repository change any thing in repository it will automatically build

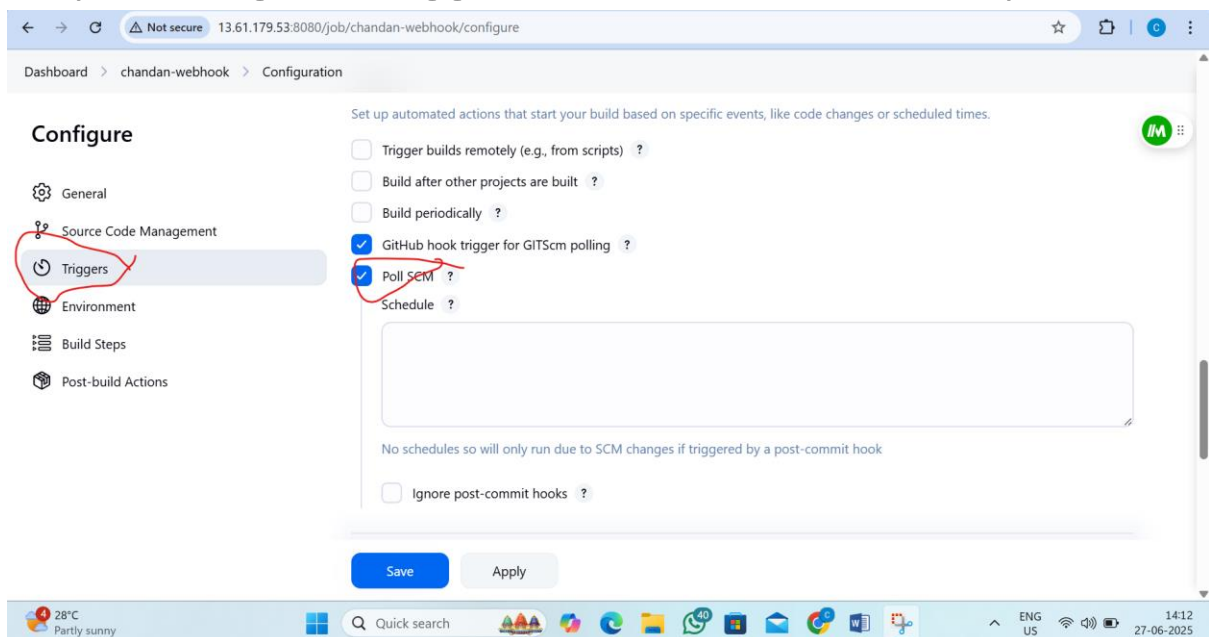


Step12:now see in the Jenkins atomatically it is bulid

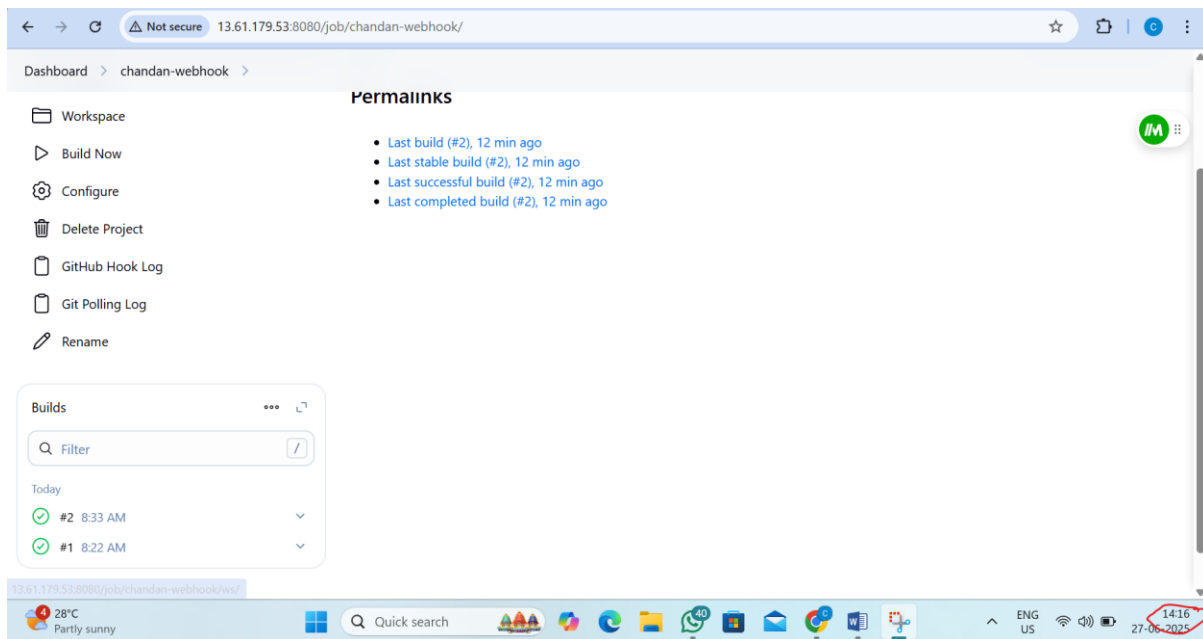


Poll-SCM

step1:now go to triggers in Jenkins and select poll SCM



Step2:now schedule the time[H/1****]->it means every minute it automatically build



After 1 minute it automatically build

